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JAMB Chemistry: The Complete Course

Principles, worked examples and exercises for the UTME

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Chapter 1: Matter, Separation Techniques, Purification and Chemical Laws

Chemistry is the branch of science that studies matter: what it is made of, how it behaves, how it changes and the energy that accompanies those changes. Everything around you, from the water in a Lagos borehole to the groundnut oil in a Kano market, from the air in your lungs to the phone in your hand, is made of matter. This opening chapter builds the vocabulary on which the rest of the course depends. It describes the states of matter and the kinetic theory that explains them, distinguishes elements, compounds and mixtures, shows how mixtures are separated and substances purified, and ends with the fundamental laws of chemical combination, which historically convinced scientists that matter is made of atoms.

1.1 Matter and its states

Matter is anything that has mass and occupies space. It exists mainly in three physical states: solid, liquid and gas. A fourth state, plasma, occurs at very high temperatures (in lightning and in the sun), but the UTME syllabus concentrates on the first three.

Property	Solid	Liquid	Gas
Shape	Fixed	Takes the shape of the container	Fills the container completely
Volume	Fixed	Fixed	Not fixed; expands to fill space
Compressibility	Almost incompressible	Almost incompressible	Easily compressed
Particle arrangement	Closely packed in a regular pattern	Close but disordered	Far apart and random
Particle motion	Vibration about fixed positions	Sliding past one another	Rapid random motion in straight lines
Attractive forces	Strong	Moderate	Very weak

The kinetic theory of matter

The kinetic theory states that all matter consists of tiny particles (atoms, molecules or ions) which are in constant motion. The average kinetic energy of the particles is proportional to the absolute temperature. In a solid the particles can only vibrate; in a liquid they have enough energy to move past each other; in a gas they have enough energy to overcome nearly all attractive forces and move freely. Two familiar observations support the theory.

- **Diffusion** is the spreading of one substance through another without stirring, as when the smell of frying onions reaches the next room. It is fastest in gases, slower in liquids and almost absent in solids. Lighter gas molecules diffuse faster than heavier ones, so ammonia (M = 17) diffuses faster than hydrogen chloride (M = 36.5).
- **Brownian motion** is the continuous, random zigzag movement of tiny solid particles (such as pollen grains or smoke) suspended in a liquid or gas. It is caused by the unequal bombardment of the visible particle by the invisible, fast-moving molecules of the medium.

Changes of state

When a solid is heated its particles vibrate more strongly until, at the **melting point**, the regular pattern collapses and the solid becomes a liquid. Further heating gives the particles enough energy to escape from the liquid entirely at the **boiling point**. During a change of state the temperature stays constant even though heat is still being supplied, because the energy is used to overcome the forces between particles (latent heat) rather than to raise their average kinetic energy. The direct change from solid to gas is called **sublimation**.

> **Key point.** A pure substance melts and boils at sharp, fixed temperatures. An impure substance melts over a range of temperatures and at a lower temperature than the pure substance, and it boils at a higher temperature. Melting and boiling points are therefore used as tests of purity.

1.2 Elements, compounds and mixtures

All matter can be classified as either a pure substance or a mixture. Pure substances are divided into elements and compounds.

An **element** is a substance that cannot be broken down into simpler substances by ordinary chemical means; all its atoms have the same atomic number. Examples are hydrogen, iron, sulphur and chlorine. More than 110 elements are known, and each has a symbol of one or two letters, such as Na for sodium (from the Latin *natrium*) and K for potassium (from *kalium*).

A **compound** is a pure substance formed when two or more elements combine chemically in a fixed proportion by mass. Water (H_2O), sodium chloride (NaCl) and carbon dioxide (CO_2) are compounds. The properties of a compound are quite different from those of the elements that form it: sodium is a soft, reactive metal and chlorine is a poisonous green gas, yet together they form the harmless table salt used in every Nigerian kitchen.

A **mixture** contains two or more substances that are not chemically combined. The following table compares mixtures and compounds.

A mixture may be **homogeneous** (uniform throughout, like salt dissolved in water or the air, and called a solution) or **heterogeneous** (with visibly different parts, like sand in water or palm oil in water). A **suspension** is a heterogeneous mixture whose particles settle on standing; a **colloid** (such as milk, fog, starch paste or smoke) has particles intermediate in size that do not settle and scatter a beam of light, an effect called the Tyndall effect.

Physical and chemical changes

A **physical change** does not produce a new substance and is usually easily reversed: melting wax, dissolving sugar, boiling water. A **chemical change** produces one or more new substances and is generally not easily reversed: burning wood, rusting of iron, souring of milk, frying an egg. Signs of chemical change include a colour change, evolution of a gas, formation of a precipitate, and a change in temperature or emission of light.

1.3 Separation of mixtures

Because the components of a mixture are not chemically combined, they can be separated using physical differences between them: in particle size, solubility, boiling point, magnetic behaviour, density or the tendency to sublime. The method chosen depends on the type of mixture.

Filtration and decantation

Filtration separates an insoluble solid from a liquid by passing the mixture through filter paper. The liquid that passes through is the **filtrate** and the solid retained is the **residue**. **Decantation** is the gentle pouring off of the liquid after the solid has settled; it is simple but less complete.

Evaporation and crystallisation

To recover a dissolved solid from its solution, the solvent can be removed by **evaporation**. When the solid decomposes on strong heating, or when well-formed crystals are wanted (as with copper(II) sulphate), the solution is heated only until it is saturated and then left to cool; this is **crystallisation**. Solubility generally rises with temperature, so crystals form as the hot saturated solution cools. The crystals are filtered off, washed with a little cold distilled water and dried between filter papers.

Simple distillation

Simple distillation separates a pure liquid from a solution containing a dissolved non-volatile solid, for instance obtaining distilled water from salty water. The solution is boiled in a flask, the vapour passes into a Liebig condenser, where cold water circulates in the outer jacket, and the condensed liquid (the **distillate**) is collected. The cooling water enters at the bottom and leaves at the top so that the jacket stays full.

Fractional distillation

Fractional distillation separates a mixture of miscible liquids whose boiling points differ, such as ethanol (boiling point 78 °C) and water (100 °C). A fractionating column, packed with glass beads or rings, is placed between the flask and the condenser. Vapour rising up the column repeatedly condenses and re-evaporates on the packing. Each cycle enriches the vapour in the more volatile component, so ethanol, which has the lower boiling point, reaches the top first and distils over at about 78 °C while the thermometer reads a steady value. Once it has all gone, the temperature rises towards 100 °C and water follows.

Sublimation

If one component of a solid mixture sublimes, the mixture can be separated by heating. Ammonium chloride sublimes on heating but sodium chloride does not, so a mixture of the two is heated in an evaporating dish covered with an inverted funnel; ammonium chloride vapour condenses as a white solid on the cool funnel walls, leaving sodium chloride behind.

Magnetic separation

A magnet removes iron, cobalt or nickel from a non-magnetic substance. This method is used to remove iron filings from sulphur or sand and to pick out iron scrap from rubbish at recycling plants.

Chromatography

Chromatography separates small amounts of substances that differ in how strongly they are held by a stationary phase (such as paper) and how soluble they are in the moving phase (the solvent). In paper chromatography a spot of the mixture, for example a mixture of dyes in ink, is placed above the solvent level on a pencil base line near the bottom of the paper. As the solvent rises by capillary action, each component travels at its own rate. Components that are more soluble in the solvent and less strongly adsorbed on the paper travel further. The base line is drawn in pencil because graphite does not dissolve.

The retention factor is defined as

> **Retention factor.** R_f = distance moved by the substance ÷ distance moved by the solvent front.

Its value lies between 0 and 1 and is constant for a given substance in a given solvent, so it is used to identify unknown compounds. A single spot on the chromatogram, even when different solvents are tried, suggests that the substance is

pure.

WORKED EXAMPLE 1.1

In a chromatography experiment, a red dye moved 3.0 cm while the solvent front moved 7.5 cm. A yellow dye in the same ink moved 6.0 cm. Calculate the R_f value of each dye and say which is more soluble in the solvent.

Solution: R_f (red) = 3.0 ÷ 7.5 = 0.40. R_f (yellow) = 6.0 ÷ 7.5 = 0.80. The yellow dye has the larger R_f value, so it is more soluble in the solvent (or less strongly held by the paper).

1.4 Purification and tests for purity

A substance that has been separated from a mixture is rarely perfectly pure. Several techniques refine it further.

- **Recrystallisation** purifies a solid. The impure solid is dissolved in the minimum amount of hot solvent, the hot solution is filtered to remove insoluble impurities, and the filtrate is cooled slowly. The pure compound crystallises while soluble impurities, present in small quantities, stay in the cold solution. The crystals are filtered, washed and dried.
- **Distillation** purifies liquids, as described above. Tap water is distilled to give distilled water, which is used in laboratories and in car batteries.

WORKED EXAMPLE 1.2

The solubility of potassium nitrate is 110 g per 100 g of water at 60 °C and 32 g per 100 g of water at 20 °C. A saturated solution containing 200 g of water at 60 °C is cooled to 20 °C. Calculate the mass of crystals that separate.

Solution: At 60 °C, 200 g of water dissolves 2 × 110 = 220 g of the salt. At 20 °C, 200 g of water can hold only 2 × 32 = 64 g. Mass crystallised = 220 − 64 = 156 g.

1.5 Chemical laws of combination

Between about 1770 and 1810 chemists made careful measurements of the masses and volumes of substances in chemical reactions. Their results were summarised in several laws that any theory of matter had to explain. Dalton's atomic theory (Chapter 2) was proposed to explain them.

The law of conservation of mass

Formulated by Antoine Lavoisier, the law states that **matter can neither be created nor destroyed in a chemical reaction; the total mass of the reactants equals the total mass of the products**. When magnesium burns in air, the white ash weighs more than the magnesium ribbon, but only because oxygen from the air has combined with the metal. If the reaction is done in a closed vessel, the total mass before and after is identical. Chemical equations are balanced because of this law: the atoms in the reactants are merely rearranged into products.

The law of constant (definite) composition

Proposed by Joseph Proust, this law states that **all pure samples of a particular compound contain the same elements combined in the same proportion by mass, regardless of the source or method of preparation**. Water from a river in

Ibadan, from rain, or made by burning hydrogen always contains 1 part by mass of hydrogen to 8 parts by mass of oxygen.

The law of multiple proportions

Proposed by John Dalton, it states that **when two elements combine to form more than one compound, the different masses of one element that combine with a fixed mass of the other are in a simple whole-number ratio**. Carbon forms two oxides. In carbon monoxide, 12 g of carbon combines with 16 g of oxygen; in carbon dioxide, 12 g of carbon combines with 32 g of oxygen. The masses of oxygen that combine with 12 g of carbon are in the ratio 16 : 32 = 1 : 2, a simple ratio.

WORKED EXAMPLE 1.3

Two oxides of nitrogen were analysed. Oxide A contains 7.0 g of nitrogen and 4.0 g of oxygen. Oxide B contains 7.0 g of nitrogen and 16.0 g of oxygen. Show that the data illustrate the law of multiple proportions and suggest formulae.

Solution: The mass of nitrogen is fixed at 7.0 g in both. The masses of oxygen are 4.0 g and 16.0 g, in the ratio 4 : 16 = 1 : 4, a simple whole-number ratio. Moles of atoms in A: N = 7.0 ÷ 14 = 0.5, O = 4.0 ÷ 16 = 0.25, so the ratio N : O = 2 : 1 and A is N_2O. In B: N = 0.5, O = 16.0 ÷ 16 = 1.0, so N : O = 1 : 2 and B is NO_2.

The law of reciprocal proportions

Proposed by Jeremias Richter, the law states that **if two elements A and B each combine with a fixed mass of a third element C, then the masses of A and B that combine with C are the same, or simple multiples of, the masses in which A and B combine with each other**. Hydrogen combines with oxygen in the ratio 1 : 8 by mass in water, and carbon combines with oxygen in the ratio 12 : 32 in

carbon dioxide. Fixing the oxygen at 8 g, the mass of carbon is 12 × 8 ÷ 32 = 3 g. So 1 g of hydrogen and 3 g of carbon combine with the same 8 g of oxygen, and the law predicts that carbon and hydrogen will combine in the ratio 3 : 1 by mass. In methane, CH_4, the ratio is 12 : 4 = 3 : 1, as predicted.

Gay-Lussac's law of combining volumes

Joseph Gay-Lussac found in 1808 that **when gases react, they do so in volumes that bear a simple whole-number ratio to one another and to the volumes of any gaseous products, provided all volumes are measured at the same temperature and pressure**. Two volumes of hydrogen react with one volume of oxygen to give two volumes of steam. Hydrogen and chlorine react in equal volumes to give twice that volume of hydrogen chloride.

Avogadro's law

Gay-Lussac's results were explained in 1811 by Amedeo Avogadro, who proposed that **equal volumes of all gases, under the same conditions of temperature and pressure, contain the same number of molecules**. It follows that the ratio of the volumes of reacting gases equals the ratio of the numbers of molecules in the balanced equation. This idea leads directly to the mole and the molar volume of a gas (22.4 dm³ at s.t.p.), which are treated in Chapters 5 and 6.

WORKED EXAMPLE 1.4

Using Gay-Lussac's law, calculate the volume of oxygen needed to burn completely 40 cm³ of hydrogen to steam, and the volume of steam formed, all measured at the same temperature and pressure.

Solution: $2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$. The volume ratio is 2 : 1 : 2. Oxygen needed = 40 × 1/2 = 20 cm³. Steam formed = 40 × 2/2 = 40 cm³.

Exercise 1.1

1. Which of the following is a heterogeneous mixture? (A) air (B) brass (C) muddy water (D) sea water filtered through paper.
2. State two pieces of evidence that a liquid is pure.
3. Name the method you would use to separate (a) a mixture of two miscible liquids with boiling points of 65 °C and 98 °C, and (b) iodine from sand.
4. A dye travels 4.5 cm on a chromatogram when the solvent front has moved 9.0 cm. Calculate its R_f value.
5. A solution saturated at 80 °C contains 90 g of salt in 150 g of water. It is cooled to 20 °C, where the solubility is 36 g per 100 g of water. Calculate the mass of crystals formed.
6. When 4.8 g of magnesium burns completely in oxygen, 8.0 g of magnesium oxide is formed. Calculate the mass of oxygen used and the ratio Mg : O by mass. Which law does this illustrate?
7. Sulphur forms two oxides. In the first, 16 g of sulphur combines with 16 g of oxygen. In the second, 16 g of sulphur combines with 24 g of oxygen. Show that the masses of oxygen are in a simple ratio and name the law.
8. What volume of nitrogen reacts with 60 cm³ of hydrogen to form ammonia, and what volume of ammonia forms? ($N_2 + 3H_2 \rightarrow 2NH_3$; all gases at the same temperature and pressure.)

Answers to Exercise 1.1

1. (C) muddy water has visible, undissolved particles; the others are uniform.
2. It has a sharp, fixed boiling point (or melting point) equal to the tabulated value; it gives a single spot on a chromatogram.
3. (a) Fractional distillation; (b) sublimation (heat the mixture and condense the purple iodine vapour on a cool surface).

4. $R_f = 4.5 \div 9.0 = 0.50$.
5. At 20 °C, 150 g of water holds $1.5 \times 36 = 54$ g. Crystals = $90 - 54 = 36$ g.
6. Mass of oxygen = $8.0 - 4.8 = 3.2$ g; Mg : O = 4.8 : 3.2 = 3 : 2. Law of conservation of mass (and constant composition: 24 : 16 = 3 : 2).
7. Oxygen masses 16 : 24 = 2 : 3, a simple whole-number ratio; law of multiple proportions (the oxides are SO_2 and SO_3).
8. Ratio $N_2 : H_2 : NH_3 = 1 : 3 : 2$. Nitrogen = $60 \div 3 = 20$ cm³; ammonia = $60 \times 2/3 = 40$ cm³.

Chapter summary

- Matter has mass and occupies space; its three common states differ in the arrangement, motion and attraction of particles, as described by the kinetic theory. Diffusion and Brownian motion are evidence for particle motion.
- Elements contain one kind of atom; compounds contain elements combined chemically in fixed proportions; mixtures can be separated by physical means.
- Separation methods include filtration, decantation, evaporation, crystallisation, simple and fractional distillation, sublimation, magnetic separation and chromatography, each exploiting a particular physical difference.
- R_f = distance moved by substance ÷ distance moved by solvent front, and it is always less than 1.
- Purity is judged by sharp melting and boiling points; impurities lower the melting point and raise the boiling point. Recrystallisation and distillation are standard purification methods.
- The laws of chemical combination are conservation of mass, constant composition, multiple proportions, reciprocal proportions, Gay-Lussac's

law of combining volumes, and Avogadro's law. Dalton's atomic theory explains them.

Chapter 2: Atomic Structure, Isotopes and Electronic Configuration

Chapter 1 ended with the laws of chemical combination and the suggestion that matter is made of atoms. This chapter opens the atom itself. You will see how experiments revealed its internal parts, how the numbers that describe an atom are defined, why atoms of the same element can differ in mass, how electrons are arranged around the nucleus, and how unstable nuclei change by radioactive decay. The electronic arrangement is the key to almost everything that follows in the course: the periodic table, bonding, reactivity and the properties of compounds all depend on it.

2.1 From Dalton to the nuclear atom

In 1803 John Dalton proposed that all matter consists of tiny indivisible particles called atoms; that atoms of one element are identical in mass and properties and differ from those of other elements; that atoms are neither created nor destroyed in a chemical reaction but are rearranged; and that compounds form when atoms of different elements combine in simple whole-number ratios. These ideas explained the laws of conservation of mass, constant composition and multiple proportions. Later discoveries showed that two statements were wrong. Atoms are divisible, being made of smaller particles, and atoms of the same element are not all identical, because isotopes exist.

Thomson, Rutherford and Bohr

- **J. J. Thomson (1897)** passed electricity through gases at low pressure and discovered cathode rays, streams of negatively charged particles now called **electrons**. Because the same particles came from every metal tested, he concluded that electrons are present in all atoms. He pictured the atom as a

sphere of positive charge with electrons embedded in it, like seeds in a watermelon.

- **Ernest Rutherford (1911)** directed a beam of alpha particles at very thin gold foil. Most passed straight through, a few were deflected through large angles and very few bounced back. He concluded that the atom is mostly empty space, that nearly all its mass and all its positive charge are concentrated in a tiny, dense **nucleus**, and that the electrons move around it.
- **Niels Bohr (1913)** proposed that electrons move in fixed circular paths called energy levels or shells. An electron in a shell has a definite energy and does not radiate it. When an electron drops from a higher level to a lower one, it emits energy as light of a definite frequency, which explains why heated elements give **line spectra** rather than continuous colour. The yellow of sodium street lamps and the red of strontium in fireworks arise in this way.
- **James Chadwick (1932)** discovered the **neutron**, a neutral particle in the nucleus with nearly the same mass as the proton.

Modern quantum mechanics replaces Bohr's circular paths with **orbitals**, regions of space around the nucleus where there is a high probability of finding an electron. Bohr's shells are still a useful first approximation.

2.2 The sub-atomic particles

Particle	Symbol	Relative charge	Relative mass	Location
Proton	p	+1	1	Nucleus
Neutron	n	0	1	Nucleus
Electron	e^-	−1	about 1/1840	Outside the nucleus

The mass of an atom is therefore concentrated in the nucleus. An atom is electrically neutral because the number of protons equals the number of electrons. Protons and neutrons are collectively called **nucleons**.

2.3 Atomic number, mass number and isotopes

The **atomic number** (Z) is the number of protons in the nucleus of an atom. It identifies the element: every atom with 11 protons is sodium. The **mass number** (A) is the total number of protons and neutrons. The number of neutrons is therefore A − Z. An atom of an element X is written $^{A}_{Z}\mathrm{X}$, so $^{23}_{11}\mathrm{Na}$ has 11 protons, 11 electrons and 23 − 11 = 12 neutrons.

An **ion** forms when an atom gains or loses electrons. The number of protons does not change. A cation is positive because electrons have been lost; an anion is negative because electrons have been gained.

WORKED EXAMPLE 2.1

State the numbers of protons, neutrons and electrons in (a) $^{56}_{26}\mathrm{Fe}^{3+}$ and (b) $^{32}_{16}\mathrm{S}^{2-}$.

Solution: (a) Protons = Z = 26. Neutrons = 56 − 26 = 30. The ion has lost three electrons, so electrons = 26 − 3 = 23. (b) Protons = 16. Neutrons = 32 − 16 = 16. The ion has gained two electrons, so electrons = 16 + 2 = 18.

Isotopes

Isotopes are atoms of the same element with the same atomic number but different mass numbers; they have the same number of protons but different numbers of neutrons. Hydrogen has three isotopes: protium ($^{1}\mathrm{H}$, no neutron), deuterium ($^{2}\mathrm{H}$, one neutron) and tritium ($^{3}\mathrm{H}$, two neutrons). Chlorine has two common isotopes, $^{35}\mathrm{Cl}$ and $^{37}\mathrm{Cl}$. Isotopes of an element have identical chemical properties, because chemical behaviour depends on the electrons, but slightly different physical properties such as density, melting point and rate of diffusion, because their masses differ.

Key point. Isotopes: same Z, different A. Different neutron number. Same chemistry, different mass. Do not confuse isotopes with allotropes (different structural forms of the same element, such as diamond and graphite).

Relative atomic mass

An atom is far too light to weigh in grams, so masses are compared with a standard. The **relative atomic mass** (A_r) of an element is the weighted average mass of the atoms of its naturally occurring isotopes, relative to 1/12 of the mass of one atom of carbon-12. Because it is an average over isotopes, it is often not a whole number. If the isotopes have masses m_1, m_2, … with abundances p_1, p_2, … per cent, then

Formula. $A_r = (m_1p_1 + m_2p_2 + \ldots) \div 100$

WORKED EXAMPLE 2.2

Chlorine consists of 75% ^{35}Cl and 25% ^{37}Cl. Calculate its relative atomic mass.

Solution: $A_r = (35 \times 75 + 37 \times 25) \div 100 = (2625 + 925) \div 100 = 3550 \div 100 = 35.5$.

WORKED EXAMPLE 2.3

Boron has two isotopes, ^{10}B and ^{11}B. Its relative atomic mass is 10.8. Calculate the percentage abundance of each isotope.

Solution: Let the abundance of ^{11}B be x per cent, so that of ^{10}B is $(100 - x)$. Then $10.8 \times 100 = 10(100 - x) + 11x$, so $1080 = 1000 - 10x + 11x = 1000 + x$, giving $x = 80$. Thus ^{11}B is 80% and ^{10}B is 20%.

WORKED EXAMPLE 2.4

Magnesium has three isotopes: ^{24}Mg (78.6%), ^{25}Mg (10.1%) and ^{26}Mg (11.3%). Calculate the relative atomic mass to one decimal place.

Solution: 24 × 78.6 = 1886.4; 25 × 10.1 = 252.5; 26 × 11.3 = 293.8. The sum is 1886.4 + 252.5 + 293.8 = 2432.7. A_r = 2432.7 ÷ 100 = 24.327, which is 24.3 to one decimal place.

Isotopes and their abundances are measured with a **mass spectrometer**. A vaporised sample is ionised, the ions are accelerated and deflected by a magnetic field, and lighter ions are deflected more than heavier ones. A detector records a peak for each isotope, and the height of each peak is proportional to its abundance.

2.4 Electronic configuration

Electrons occupy **shells** numbered n = 1, 2, 3, … outward from the nucleus. The maximum number of electrons in a shell is $2n^2$. The first shell holds up to 2 electrons, the second 8, the third 18 and the fourth 32. Electrons fill the lowest available energy level first. For the first twenty elements this gives a simple arrangement, for example Na (Z = 11) is 2, 8, 1 and Cl (Z = 17) is 2, 8, 7, with the outermost shell never holding more than 8 electrons.

Each shell is divided into **sub-shells** labelled s, p, d and f, made of orbitals. An s sub-shell has 1 orbital (holding 2 electrons), p has 3 orbitals (6 electrons), d has 5 orbitals (10 electrons) and f has 7 orbitals (14 electrons). An s orbital is spherical; a p orbital is dumbbell-shaped and the three p orbitals point along the x, y and z axes. The shell with n = 1 has only 1s; n = 2 has 2s and 2p; n = 3 has 3s, 3p and 3d; and so on.

The three rules for filling orbitals

1. **Aufbau (building-up) principle.** Electrons enter the orbital of lowest energy available. The order of increasing energy is 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s. Note that 4s fills before 3d.
2. **Pauli exclusion principle.** An orbital holds at most two electrons, and they must have opposite spins.
3. **Hund's rule.** Electrons occupy orbitals of the same energy singly, with parallel spins, before any orbital receives a second electron. Nitrogen ($2p^3$) therefore has three unpaired electrons in its three 2p orbitals.

The configuration is written by listing each sub-shell with its number of electrons as a superscript.

Element	Z	Configuration	Shell arrangement
Carbon	6	$1s^2\ 2s^2\ 2p^2$	2, 4
Oxygen	8	$1s^2\ 2s^2\ 2p^4$	2, 6
Sodium	11	$1s^2\ 2s^2\ 2p^6\ 3s^1$	2, 8, 1
Chlorine	17	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$	2, 8, 7
Calcium	20	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$	2, 8, 8, 2
Iron	26	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ 4s^2$	2, 8, 14, 2

Exceptions: chromium and copper

A half-filled (d^5) or completely filled (d^{10}) d sub-shell is unusually stable. Chromium (Z = 24) is therefore [Ar] $3d^5\ 4s^1$ rather than $3d^4\ 4s^2$, and copper (Z = 29) is [Ar] $3d^{10}\ 4s^1$ rather than $3d^9\ 4s^2$. Zinc (Z = 30) is the regular [Ar] $3d^{10}\ 4s^2$. Here [Ar] stands for $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$, the argon core.

Ions and the order of removal

When atoms of the transition elements form positive ions they lose their 4s electrons first, even though 4s fills before 3d. Thus Fe^{2+} is [Ar] $3d^6$ and Fe^{3+} is [Ar] $3d^5$, which has the stable half-filled d sub-shell and explains why iron(III) is a common oxidation state. Non-metal atoms gain electrons to complete the octet: S^{2-} is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$, the same as argon. Species with the same electronic configuration are said to be **isoelectronic**; K^+, Ca^{2+}, Cl^-, S^{2-} and Ar are all isoelectronic with 18 electrons.

WORKED EXAMPLE 2.5

Write the electronic configuration of (a) the phosphorus atom (Z = 15), (b) the manganese atom (Z = 25) and (c) the Mn^{2+} ion. State the number of unpaired electrons in each.

Solution: (a) P: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$. The three 3p electrons occupy separate orbitals by Hund's rule, so there are 3 unpaired electrons. (b) Mn: 25 electrons. After [Ar] (18 electrons), 7 remain: [Ar] $3d^5\ 4s^2$. The five 3d electrons are in five separate orbitals, so there are 5 unpaired electrons. (c) Mn^{2+} loses the two 4s electrons: [Ar] $3d^5$, with 5 unpaired electrons.

Valence electrons and chemical behaviour

The electrons in the outermost shell are the **valence electrons**. Elements with the same number of valence electrons behave similarly and sit in the same group of the periodic table. Atoms with an outer shell of 8 electrons (or 2 for helium) are very stable, which is why the noble gases are almost unreactive. Other atoms react by losing, gaining or sharing electrons to reach this arrangement.

Sodium (2, 8, 1) readily loses one electron to form Na^+ (2, 8). Chlorine (2, 8, 7) readily gains one to form Cl^- (2, 8, 8). Aluminium (2, 8, 3) forms Al^{3+} and oxygen (2, 6) forms O^{2-}. The number of electrons lost or gained is the **valency** in these simple cases.

2.5 Radioactivity

The nuclei of some isotopes are unstable and break up spontaneously, emitting radiation. This is **radioactivity**, discovered by Henri Becquerel in 1896 and investigated by Marie and Pierre Curie. The change is a nuclear process, so it is not affected by temperature, pressure or chemical combination.

Radiation	Nature	Charge	Penetration
Alpha (α)	Helium nucleus, $^{4}_{2}\mathrm{He}$	+2	Low; stopped by paper or skin
Beta (β)	Fast electron, $^{0}_{-1}\mathrm{e}$	−1	Medium; stopped by thin aluminium
Gamma (γ)	High-energy electromagnetic radiation	0	High; reduced by thick lead or concrete

In an alpha emission the mass number falls by 4 and the atomic number by 2. In a beta emission a neutron changes into a proton and an electron, so the mass number is unchanged and the atomic number rises by 1. Gamma emission changes neither number. Nuclear equations are balanced by making the sums of mass numbers and of atomic numbers equal on both sides.

WORKED EXAMPLE 2.6

Uranium-238 (Z = 92) emits an alpha particle and the product then emits a beta particle. Identify the final nucleus.

Solution: $^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$ (238 − 4 = 234; 92 − 2 = 90, thorium). Then $^{234}_{90}\mathrm{Th} \rightarrow {}^{234}_{91}\mathrm{Pa} + {}^{0}_{-1}\mathrm{e}$ (mass number unchanged, atomic number 91, protactinium). The final nucleus is $^{234}_{91}\mathrm{Pa}$.

Half-life

Radioactive decay is random for any single nucleus, but a large sample decays at a steady statistical rate. The **half-life** ($t_{½}$) is the time taken for half of the radioactive nuclei in a sample to decay. After n half-lives the fraction remaining is $(½)^n$, and n = total time ÷ half-life.

WORKED EXAMPLE 2.7

Phosphorus-32 has a half-life of 14 days. What mass of a 64 g sample remains after 56 days?

Solution: Number of half-lives = 56 ÷ 14 = 4. Mass remaining = 64 × $(½)^4$ = 64 ÷ 16 = 4 g.

WORKED EXAMPLE 2.8

A sample of a radioisotope decays from 100 mg to 12.5 mg in 15 hours. Find its half-life.

Solution: 100 → 50 → 25 → 12.5 shows that three half-lives have passed. Half-life = 15 ÷ 3 = 5 hours.

Radioisotopes have many uses. **Carbon-14** dating estimates the age of wooden or bone objects from archaeological sites, such as Nok terracotta context material, from the fraction of carbon-14 remaining. Cobalt-60 gamma rays are used in cancer therapy and to sterilise medical equipment. Iodine-131 is used to treat thyroid disorders, and radioisotopes serve as tracers for finding leaks in pipelines. Radiation is also dangerous because it ionises living tissue and can cause burns, cancer and genetic damage, so radioactive waste is stored in shielded containers. In **nuclear fission** a heavy nucleus such as uranium-235 splits into smaller nuclei, releasing great energy; in **nuclear fusion** light nuclei join to form a heavier one, as in the sun.

Exercise 2.1

1. How many neutrons are there in an atom of $^{27}_{13}\text{Al}$?
2. Write the electronic configuration of the sulphide ion S^{2-} (Z = 16) and state how many electrons it has.
3. Copper exists as ^{63}Cu and ^{65}Cu. If its relative atomic mass is 63.5, find the percentage of ^{65}Cu.
4. An element has the shell arrangement 2, 8, 3. Give its atomic number and the formula of its ion.
5. How many electrons are in the Ca^{2+} ion (Z = 20)? Name a negative ion isoelectronic with it.
6. A radioisotope has a half-life of 6 hours. What fraction of a sample remains after 24 hours?
7. Radium-226 (Z = 88) decays by alpha emission. Write the symbol of the daughter nucleus.
8. What is the maximum number of electrons that can occupy the third shell?
9. Write the configuration of the Cu atom and explain why it is not $3d^9\,4s^2$.
10. Explain why the relative atomic mass of chlorine is not a whole number.
11. How many unpaired electrons does an iron(III) ion (Z = 26) have?
12. Rutherford found that a few alpha particles were deflected through large angles. What conclusion did this lead to?

Answers to Exercise 2.1

1. 27 − 13 = 14 neutrons.
2. $1s^2\,2s^2\,2p^6\,3s^2\,3p^6$; 16 + 2 = 18 electrons.
3. Let x be the percentage of ^{65}Cu: 63(100 − x) + 65x = 6350, so 6300 + 2x = 6350 and x = 25%.
4. Z = 2 + 8 + 3 = 13 (aluminium); the ion is Al^{3+}.

5. 20 − 2 = 18 electrons. Isoelectronic anions include Cl^- and S^{2-} (also P^{3-}).
6. 24 ÷ 6 = 4 half-lives, so the fraction remaining is $(\tfrac{1}{2})^4 = 1/16$.
7. $^{222}_{86}Rn$ (mass number 226 − 4 = 222; atomic number 88 − 2 = 86, radon).
8. $2n^2 = 2 \times 9 = 18$ electrons.
9. [Ar] $3d^{10}$ $4s^1$. A completely filled 3d sub-shell is especially stable, so one 4s electron moves into the 3d sub-shell.
10. It is a weighted average of the masses of its isotopes, ^{35}Cl and ^{37}Cl, in the ratio about 3 : 1.
11. Fe^{3+} is [Ar] $3d^5$, with five d electrons in five separate orbitals, so 5 unpaired electrons.
12. The positive charge and most of the mass of the atom are concentrated in a very small, dense nucleus, since only a direct approach to such a nucleus would repel an alpha particle strongly.

Chapter summary

- Thomson discovered the electron, Rutherford the nucleus, Bohr the energy levels and Chadwick the neutron. Dalton's theory had to be modified because atoms are divisible and isotopes exist.
- The atomic number is the number of protons; the mass number is protons plus neutrons; neutrons = A − Z. Ions are formed by gain or loss of electrons, never of protons.
- Isotopes have the same atomic number but different mass numbers and share the same chemical properties.
- Relative atomic mass is the weighted average mass of isotopes compared with 1/12 of a carbon-12 atom: $A_r = \Sigma(\text{mass} \times \%\ \text{abundance}) \div 100$.
- Shells hold $2n^2$ electrons. Orbitals fill by the Aufbau principle, the Pauli exclusion principle and Hund's rule; 4s fills before 3d, and Cr and Cu are exceptions with $4s^1$.

- Alpha emission lowers A by 4 and Z by 2; beta emission raises Z by 1; gamma emission changes neither. After n half-lives the fraction left is $(½)^{n}$.

Chapter 3: The Periodic Table and Periodicity

By the middle of the nineteenth century about sixty elements were known, and chemists were looking for order among them. The periodic table is the most successful of all such attempts. It arranges the elements so that those with similar properties fall in the same vertical column, and it allows a student who knows the position of an element to predict its electronic structure, its valency, the formulae of its compounds and the way it behaves in reactions. This chapter explains how the table is built, why it takes the form it does, and how properties such as atomic size, ionisation energy and electronegativity change across a period and down a group. Chapters 4, 11 and 12 will rely heavily on these ideas.

3.1 Development of the periodic table

Early attempts at classification used atomic mass. In 1829 Johann Döbereiner noticed **triads**, groups of three elements with similar properties in which the atomic mass of the middle element is roughly the average of the other two. Calcium (40), strontium (87.6) and barium (137) form such a triad: (40 + 137)/2 = 88.5, which is close to 87.6. In 1864 John Newlands arranged the elements in order of increasing atomic mass and observed that every eighth element resembled the first, his **law of octaves**. The pattern broke down after calcium because it did not allow for undiscovered elements.

In 1869 Dmitri Mendeleev published a table in which elements were arranged by increasing atomic mass but placed in columns according to their chemical properties. Where an element seemed not to fit he left a gap and boldly predicted that an unknown element would be found to fill it. He predicted the properties of the element he called eka-silicon, and when germanium was discovered in 1886 its density, atomic mass and the formula of its oxide matched his predictions closely.

This success persuaded chemists of the value of the table. Mendeleev's table had weaknesses: tellurium (A_r = 127.6) had to be placed before iodine (126.9) to keep both in the correct columns, so ordering by mass was not always consistent with chemical behaviour.

The difficulty was resolved by Henry Moseley in 1913. From the X-ray spectra of elements he showed that the fundamental quantity is the **atomic number** (the number of protons), not the atomic mass. The modern statement of the **periodic law** is therefore:

> **Periodic law.** The properties of the elements are a periodic function of their atomic numbers. When elements are arranged in order of increasing atomic number, elements with similar properties recur at regular intervals.

3.2 Structure of the modern periodic table

The modern table contains horizontal rows called **periods** and vertical columns called **groups**. There are seven periods and eighteen groups (numbered 1 to 18 by the current convention; older books number the main groups I to VII and 0).

- **Period number** equals the number of occupied electron shells in the atom. Sodium (2,8,1) has three shells and is in Period 3.
- **Group number** for the main-group elements relates to the number of outer-shell electrons. Elements of Group 1 have one outer electron, Group 2 have two, and Groups 13 to 17 have three to seven. In the older numbering these are Groups I to VII. The noble gases (Group 18, old Group 0) have a full outer shell: two electrons for helium and eight for the others.
- Atoms in the same group have the same number of outer electrons and therefore similar chemical properties.

The periods differ in length because shells hold different numbers of electrons and because sub-shells fill in a definite order (Chapter 2). Period 1 contains only two elements (H and He). Periods 2 and 3 contain eight each, as the 2s, 2p and 3s, 3p sub-shells fill. Period 4 contains eighteen elements because the 3d sub-shell is filled between calcium and gallium.

Blocks of the table

The table can be divided into blocks according to the sub-shell receiving the last electron.

Block	Groups	Last electron enters	Examples
s-block	1 and 2 (and He)	s sub-shell	Na, Mg, Ca
p-block	13 to 18	p sub-shell	Al, S, Cl, Ne
d-block	3 to 12	d sub-shell	Fe, Cu, Zn
f-block	lanthanides and actinides	f sub-shell	Ce, U

WORKED EXAMPLE 3.1

An element X has the electronic configuration 2,8,6. State its period, group, valency, and the formula of its compound with magnesium.

Solution: There are three occupied shells, so X is in Period 3. There are six outer electrons, so it is in Group 16 (old Group VI). Atomic number = 2 + 8 + 6 = 16, so X is sulphur. To fill its outer shell it gains 2 electrons, forming X^{2-}, so its valency is 2. Magnesium (2,8,2) loses two electrons to form Mg^{2+}. One Mg^{2+} balances one X^{2-}, so the formula is MgS.

3.3 Families of elements

Group 1: the alkali metals

Lithium, sodium, potassium, rubidium and caesium each have one outer electron, which is easily lost to give a unipositive ion M^+. They are soft metals with low densities and low melting points, kept under oil because they react rapidly with air and water. Reaction with water gives the hydroxide and hydrogen: $2Na + 2H_2O \rightarrow 2NaOH + H_2$. Reactivity increases down the group because the outer electron is further from the nucleus and better shielded, so it is lost more easily.

Group 2: the alkaline-earth metals

Beryllium, magnesium, calcium, strontium and barium have two outer electrons and form M^{2+} ions. They are harder, denser and have higher melting points than the Group 1 metals in the same period. Reactivity also increases down the group; magnesium reacts slowly with cold water but burns vigorously in steam, whereas calcium reacts readily with cold water.

Group 17: the halogens

Fluorine, chlorine, bromine, iodine and astatine have seven outer electrons and exist as diatomic molecules X_2. They gain one electron to form X^-. Unlike the metals, **reactivity decreases down the group**: fluorine is the most reactive and iodine is the least, because the incoming electron is added further from the nucleus and is more shielded. Physical state at room temperature changes from gas (F_2, Cl_2) to liquid (Br_2) to solid (I_2), and colour deepens from pale yellow, to greenish-yellow, to red-brown, to grey-black. A more reactive halogen displaces a less reactive one from solution of its salt: $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$.

Group 18: the noble gases

Helium, neon, argon, krypton, xenon and radon have complete outer shells and are monatomic and chemically almost inert. Helium is used to fill balloons and in breathing mixtures, neon in advertising signs and argon in electric light bulbs.

The transition elements

The d-block elements (such as iron, copper, zinc, nickel and chromium) are usually called **transition elements**, although strictly zinc is excluded because Zn^{2+} has a full 3d sub-shell ($3d^{10}$). Transition metals share four characteristic properties:

- they show variable oxidation states (iron forms Fe^{2+} and Fe^{3+}; manganese ranges from +2 to +7);
- they form coloured compounds (Cu^{2+} is blue, Fe^{2+} is pale green, Fe^{3+} is yellow-brown, MnO_4^- is purple);
- they form complex ions such as $[Cu(NH_3)_4]^{2+}$;
- they and their compounds are frequently good catalysts (iron in the Haber process, vanadium(V) oxide in the Contact process, nickel in the hydrogenation of oils).

WORKED EXAMPLE 3.2

Strontium lies between calcium and barium in Group 2. Predict the formula of its chloride and its oxide, and say how it reacts with cold water.

Solution: Group 2 elements form M^{2+} ions. Chloride is Cl^-, so the chloride must be $SrCl_2$; oxide is O^{2-}, so the oxide is SrO. Strontium is below calcium, so it is more reactive than calcium and reacts readily with cold water giving the hydroxide and hydrogen: $Sr + 2H_2O \rightarrow Sr(OH)_2 + H_2$.

3.4 Periodic trends in atomic and ionic size

The size of an atom is described by its **atomic radius**, usually half the distance between the nuclei of two bonded atoms of the same element. Two opposing factors control size: the number of occupied shells and the effective nuclear charge felt by the outer electrons.

Across a period

Moving from left to right across a period the atomic number increases by one at each step, but the added electron enters the same shell. Inner electrons shield the outer electrons only weakly from other electrons in the same shell, so the effective nuclear charge increases and the outer electrons are pulled closer. **Atomic radius therefore decreases across a period.** In Period 3 the radius falls from sodium (about 0.186 nm) to chlorine (about 0.099 nm).

Down a group

Each new period adds a shell, so the outer electrons are further from the nucleus and are shielded by more inner shells. **Atomic radius increases down a group.** The radius of lithium is smaller than that of sodium, which is smaller than that of potassium.

Ionic radius

- A **positive ion is smaller than its parent atom**. The metal atom loses its outer shell (or at least has the same number of protons holding fewer electrons), so Na^+ (2,8) is much smaller than Na (2,8,1).
- A **negative ion is larger than its parent atom**. The extra electrons repel one another while the nuclear charge is unchanged, so Cl^- is larger than Cl.
- Among **isoelectronic ions** (ions with the same number of electrons, such as N^{3-}, O^{2-}, F^-, Na^+, Mg^{2+} and Al^{3+}, all with 10 electrons) the radius decreases as the nuclear charge increases: $N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$.

WORKED EXAMPLE 3.3

Arrange S^{2-}, Cl^{-}, K^{+} and Ca^{2+} in order of increasing ionic radius and explain the order.

Solution: Each ion has 18 electrons (2,8,8), so they are isoelectronic. The nuclear charges are +16 (S), +17 (Cl), +19 (K) and +20 (Ca). The greater the nuclear charge the more strongly the same 18 electrons are attracted and the smaller the ion. Increasing radius: $Ca^{2+} < K^{+} < Cl^{-} < S^{2-}$.

3.5 Ionisation energy

The **first ionisation energy** is the minimum energy required to remove one mole of electrons from one mole of gaseous atoms of an element to form one mole of gaseous unipositive ions: $M(g) \rightarrow M^{+}(g) + e^{-}$. It is measured in kJ mol^{-1}. The second ionisation energy is the energy needed to remove a second mole of electrons from the resulting M^{+} ions, and so on.

Ionisation energy is always positive (endothermic), because energy must be supplied to pull an electron away from the attraction of the nucleus. It depends on the nuclear charge, the atomic radius, the shielding by inner electrons and, to a lesser extent, the type of sub-shell the electron comes from.

Trend down a group

The first ionisation energy **decreases down a group**. The outer electron is further from the nucleus and is shielded by more inner shells, so it is held less firmly. For the alkali metals the values fall from about 520 kJ mol^{-1} (Li) to about 496 (Na) and 419 (K).

Trend across a period

Generally the first ionisation energy **increases across a period** because the nuclear charge rises while the electrons are added to the same shell, so the atom becomes smaller and holds its outer electron more tightly. The noble gas at the end of the period has the highest value, and the alkali metal at the start has the lowest. There are two small irregularities in Periods 2 and 3.

Element (Period 3)	Na	Mg	Al	Si	P	S	Cl	Ar
First ionisation energy (kJ mol^{-1})	496	738	578	789	1012	1000	1251	1521

- **Mg to Al drops.** The outer electron of aluminium ($3p^1$) is in a 3p sub-shell, which is higher in energy and slightly better shielded than the 3s sub-shell holding the outer electrons of magnesium.
- **P to S drops.** In phosphorus ($3p^3$) the three 3p electrons are unpaired in separate orbitals. In sulphur ($3p^4$) one orbital holds a pair, and the repulsion between the paired electrons makes one of them easier to remove.

Successive ionisation energies

Successive ionisation energies of one element increase steadily because each electron is removed from an ion with a larger positive charge. A very large jump occurs when removal begins from a new inner shell, which is much closer to the nucleus. The position of the jump shows the number of outer electrons and therefore the group of the element.

WORKED EXAMPLE 3.4

The successive ionisation energies (kJ mol^{-1}) of an element Y are 590, 1150, 4900, 6500 and 8100. State the group of Y and the formula of its oxide.

Solution: The values rise gently from the first to the second (590 to 1150, roughly a factor of 2), but there is a very large jump from the second to the third (1150 to 4900, a factor of more than 4). The third electron must therefore come from an inner shell, so Y has two outer electrons and is in Group 2. It forms Y^{2+}, so the oxide is YO. (Calcium, with a first value of 590, fits.)

3.6 Electron affinity and electronegativity

The **electron affinity** of an element is the energy change when one mole of gaseous atoms each gains one electron to form one mole of gaseous unipositive negative ions: $X(g) + e^- \rightarrow X^-(g)$. For most elements the first electron affinity is exothermic (negative); the halogens have the most exothermic values, and chlorine's is slightly more exothermic than fluorine's, because the very small fluorine atom has strong repulsion between the incoming electron and those already present. Adding a second electron (O^- to O^{2-}) is endothermic, since a negative ion repels the incoming electron.

Electronegativity is the tendency of an atom to attract the shared pair of electrons in a covalent bond toward itself. It has no units. On the Pauling scale fluorine, the most electronegative element, has a value of 4.0, followed by oxygen (3.5), nitrogen and chlorine (both about 3.0). Electronegativity increases across a period and decreases down a group, so the most electronegative elements lie at the top right of the table (excluding the noble gases) and the least electronegative at the bottom left (caesium and francium). Differences in electronegativity decide the type of bond formed, as Chapter 4 explains.

3.7 Metallic and non-metallic character

Metals form positive ions by loss of electrons and their oxides are usually basic. Non-metals tend to gain electrons or share them, and their oxides are usually acidic. Because electrons are lost more easily towards the bottom left of the table, **metallic character increases down a group and decreases across a period**. A diagonal staircase running from boron to astatine separates the metals from the non-metals. Elements along it, such as silicon, germanium and arsenic, are **metalloids** with intermediate properties; silicon is a semiconductor, which is why it is used in the chips of phones and computers.

Within a group the same trend is seen in the change from non-metal to metal. In Group 14 carbon is a non-metal, silicon and germanium are metalloids, and tin and lead are metals. In Group 15 nitrogen is a gas whose oxides are acidic, while bismuth is a metal.

3.8 Properties of the elements and compounds of Period 3

Period 3 illustrates every trend well.

Element	Na	Mg	Al	Si	P	S	Cl	Ar
Outer electrons	1	2	3	4	5	6	7	8
Structure of element	Giant metallic			Giant covalent	Simple molecules (P_4, S_8)		Cl_2	Single atoms
Typical oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2, SO_3	Cl_2O_7	None
Nature of oxide	Basic		Amphoteric	Acidic				

The oxides change from strongly basic (Na_2O, which dissolves in water to give NaOH), through amphoteric (Al_2O_3), to acidic (SO_3 gives H_2SO_4). An **amphoteric oxide** reacts with both acids and strong alkalis to form salts:

- $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$
- $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$ (sodium aluminate)

Zinc oxide and lead(II) oxide are also amphoteric. The melting point rises from sodium to silicon (reaching a maximum for the giant covalent silicon) and then falls sharply for the simple molecules of phosphorus, sulphur and chlorine, which are held together by weak forces between molecules.

> **Exam tip.** If a question asks which element in Period 3 has the highest melting point, the answer is silicon (a giant covalent structure). If it asks which oxide is amphoteric, the answer is aluminium oxide.

3.9 Diagonal relationships and anomalies

The first member of each group often differs from the others because its atom is very small. Lithium behaves in some respects like magnesium (its neighbour diagonally down and to the right): both form normal oxides on burning in air (Li_2O and MgO) whereas the other Group 1 metals form peroxides or superoxides, and the carbonates and nitrates of both decompose on heating. Similarly beryllium resembles aluminium: both have amphoteric oxides and both form covalent chlorides. This is called a **diagonal relationship**. It arises because the increase in charge on moving across the table is compensated by an increase in size on moving down it, giving similar charge density.

WORKED EXAMPLE 3.5

Element Z has atomic number 19 and element W has atomic number 35. State the group and period of each, which has the larger atomic radius, and the formula of the compound they form.

Solution: Z: 2,8,8,1 has four shells and one outer electron, so Period 4, Group 1 (potassium). W: 2,8,18,7 has four shells and seven outer electrons, so Period 4, Group 17 (bromine). Both are in the same period, and atomic radius decreases across a period, so potassium has the larger radius. K loses one electron and Br gains one, so the compound is KBr, an ionic compound.

3.10 Uses of the periodic table in prediction

Once the position of an element is known, much can be deduced. Suppose we are told that element R is in Group 14, Period 3. Its configuration is 2,8,4, so it is silicon; its valency is 4; its oxide RO_2 is acidic and has a giant structure; its chloride RCl_4 is a covalent liquid. Likewise, an element in Group 1 must be a soft, reactive metal whose hydroxide is a strong base. Chemists also use the table to estimate unknown quantities. For example, if the boiling points of the hydrides of Group 17 are known for HCl, HBr and HI, a rough value for HAt can be extrapolated.

Finally, remember that the arrangement is a summary of the electronic structure of atoms: position in the table shows the outer-shell structure, outer-shell structure controls bonding, and bonding determines properties. That is why Chapter 4, on chemical bonding, follows directly from this one.

Exercise 3.1

1. State the modern periodic law and name the scientist who established that atomic number, not atomic mass, is the basis of the table.

2. An element has the electronic configuration 2,8,8,2. Give its period, group and the formula of its oxide.
3. Which of the following has the largest radius: Na^{+}, Mg^{2+}, F^{-}, O^{2-}? Explain.
4. Why does the first ionisation energy of potassium (419 kJ mol^{-1}) fall below that of sodium (496 kJ mol^{-1})?
5. The successive ionisation energies (kJ mol^{-1}) of an element are 800, 2400, 3700, 25000, 32800. State its group.
6. Write balanced equations for the reaction of chlorine with (a) potassium iodide solution and (b) sodium bromide solution. Why does iodine not displace chlorine from potassium chloride?
7. Give three characteristic properties of transition elements and an example of each.
8. Explain why the first ionisation energy of sulphur is lower than that of phosphorus.
9. Aluminium oxide is amphoteric. Write equations to show it reacting with (a) hydrochloric acid and (b) sodium hydroxide solution.
10. Calcium (40), strontium (87.6) and barium (137) form a Döbereiner triad. Using the same rule, estimate the atomic mass of the middle element of the triad lithium (7), sodium and potassium (39), and compare it with the actual value of 23.
11. Explain the trend in electronegativity across Period 2 from lithium to fluorine.

Answers to Exercise 3.1

1. The properties of the elements are a periodic function of their atomic numbers; Henry Moseley.

2. Four occupied shells, so Period 4; two outer electrons, so Group 2 (calcium); oxide CaO.
3. O^{2-}. The four ions are isoelectronic (10 electrons), so size falls as nuclear charge rises: O^{2-} (+8) > F^{-} (+9) > Na^{+} (+11) > Mg^{2+} (+12).
4. The outer electron of potassium is in a fourth shell, further from the nucleus and more shielded, so less energy is needed to remove it.
5. Group 13 (old Group III). The big jump comes after the third value (3700 to 25000), so three electrons are in the outer shell.
6. (a) $Cl_2 + 2KI \rightarrow 2KCl + I_2$; (b) $Cl_2 + 2NaBr \rightarrow 2NaCl + Br_2$. Iodine is less reactive (a weaker oxidising agent) than chlorine, so it cannot take an electron from Cl^{-}.
7. Variable oxidation states (Fe^{2+}/Fe^{3+}); coloured compounds ($CuSO_4$ is blue); catalytic activity (iron in the Haber process). Complex ion formation is also acceptable.
8. In P ($3p^3$) each p electron is alone in an orbital; in S ($3p^4$) two electrons are paired in one orbital and repel each other, so one is removed more easily.
9. (a) $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$; (b) $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$.
10. (7 + 39)/2 = 46/2 = 23, which agrees exactly with the actual value.
11. Electronegativity increases from Li to F because the nuclear charge increases while the atomic radius decreases, so the nucleus attracts the bonding pair more strongly.

Chapter summary

- Döbereiner (triads), Newlands (octaves), Mendeleev (gaps and predictions) and Moseley (atomic number) developed the table. The periodic law states that properties are a periodic function of atomic number.

- Period number equals the number of occupied shells; for main-group elements the group relates to the number of outer electrons. The table is divided into s-, p-, d- and f-blocks.
- Alkali metals and alkaline-earth metals become more reactive down the group; halogens become less reactive down the group. Noble gases are inert because their outer shells are complete.
- Transition elements show variable oxidation states, coloured compounds, complex ions and catalytic activity.
- Atomic radius decreases across a period and increases down a group. Positive ions are smaller, and negative ions larger, than their atoms.
- First ionisation energy generally increases across a period and decreases down a group, with dips at Al and S in Period 3. A large jump in successive ionisation energies shows the number of outer electrons.
- Electronegativity increases across a period and decreases down a group; fluorine is the most electronegative element.
- Across Period 3 oxides change from basic to amphoteric (Al_2O_3) to acidic, and metallic character falls.

Chapter 4: Chemical Bonding and Structure

Very few elements exist as free atoms. Apart from the noble gases, atoms join together to form molecules, crystals and metals, and the forces that hold them together are called chemical bonds. Why atoms bond, which type of bond forms, and how the bond type decides properties such as melting point, hardness and electrical conductivity, are among the most important questions in chemistry. The explanation rests on the electronic structures studied in Chapters 2 and 3: atoms bond in order to reach a more stable arrangement of outer electrons, usually that of a noble gas. This chapter treats ionic, covalent, dative and metallic bonding, the shapes of molecules, the weaker forces between molecules, and the way structure determines properties.

4.1 Why atoms bond

The noble gases have complete outer shells (2 electrons for helium, 8 for the others) and are exceptionally stable. Other atoms achieve a similar arrangement by losing, gaining or sharing electrons. This is the **octet rule** (and the **duplet rule** for small atoms such as hydrogen and lithium). Energy is released when a stable bond forms, so the bonded system has lower energy than the separate atoms. There are three principal kinds of strong bond: ionic (transfer of electrons), covalent (sharing of electrons) and metallic (a lattice of positive ions in a sea of delocalised electrons). The octet rule has exceptions, such as BF_3 (six electrons around boron) and PCl_5 (ten around phosphorus), but it explains most of the compounds at UTME level.

4.2 Ionic (electrovalent) bonding

An **ionic bond** is the electrostatic attraction between oppositely charged ions formed by the transfer of electrons from one atom to another. It usually forms between a metal (low ionisation energy, little attraction for electrons) and a non-metal (high electronegativity).

In sodium chloride, sodium (2,8,1) loses one electron to give Na^+ (2,8) and chlorine (2,8,7) gains it to give Cl^- (2,8,8). In magnesium oxide, Mg (2,8,2) transfers two electrons to O (2,6), giving Mg^{2+} and O^{2-}. The number of electrons gained or lost is the **electrovalency**. In calcium chloride one Ca^{2+} ion needs two Cl^- ions to balance charge, giving $CaCl_2$.

Ionic solids do not consist of separate molecules. Each ion is surrounded by ions of the opposite charge in a regular three-dimensional **giant lattice**. In sodium chloride each Na^+ is surrounded by six Cl^- and each Cl^- by six Na^+. The formula NaCl simply gives the ratio of ions.

Properties of ionic compounds

- They are hard, crystalline solids with **high melting and boiling points**, because a great deal of energy is needed to overcome the strong attractions throughout the lattice (NaCl melts at 801 °C).
- They are brittle: a blow displaces layers so that ions of the same charge come together and repel, splitting the crystal.
- They **do not conduct electricity when solid** because the ions are fixed in position, but they conduct when molten or dissolved in water, when the ions are free to move.
- Many are soluble in water, which is a polar solvent, but insoluble in non-polar solvents such as kerosene and benzene.

WORKED EXAMPLE 4.1

Element A has atomic number 13 and element B has atomic number 8. Deduce the formula of the compound formed and the type of bond.

Solution: A is 2,8,3 and loses three electrons to form A^{3+}. B is 2,6 and gains two electrons to form B^{2-}. To balance charges the lowest common multiple of 3 and 2 is 6, so two A^{3+} (+6) are needed for three B^{2-} (−6). The formula is A_2B_3, that is Al_2O_3, and the bond is ionic (metal plus non-metal).

4.3 Covalent bonding

A **covalent bond** is formed when two atoms share one or more pairs of electrons. It occurs mainly between non-metal atoms, which both need electrons and neither of which can easily lose them. The shared pair is attracted by both nuclei and holds the atoms together.

- In a **single bond** one pair is shared: H–H in hydrogen, Cl–Cl in chlorine, H–Cl.
- In a **double bond** two pairs are shared: O=O in oxygen, and C=C in ethene.
- In a **triple bond** three pairs are shared: N≡N in nitrogen, and C≡C in ethyne.

In water, oxygen (2,6) shares one pair with each of two hydrogen atoms, so each hydrogen has two electrons and oxygen has eight; two further pairs on oxygen do not take part in bonding and are called **lone pairs**. In ammonia, nitrogen shares three pairs with three hydrogen atoms and keeps one lone pair. In methane, carbon shares four pairs with four hydrogens. In carbon dioxide, each oxygen is doubly bonded to carbon: O=C=O. In nitrogen, the triple bond N≡N is very strong, which explains why nitrogen is so unreactive.

Bond strength and length are related. For carbon–carbon bonds the order of strength is C≡C > C=C > C–C, while the order of length is the reverse: a multiple bond is shorter and stronger than a single bond between the same atoms.

Properties of simple covalent substances

Substances made of discrete molecules (water, methane, iodine, sugar) have **low melting and boiling points** because only the weak forces between molecules, not the strong bonds inside them, are overcome on melting. They are often gases, liquids or soft solids, are poor conductors of electricity (there are no free ions or electrons), and are generally insoluble in water but soluble in organic solvents. Exceptions are molecules able to interact with water, such as sugar, ethanol and ammonia.

4.4 Polar covalent bonds

When two identical atoms bond, the electrons are shared equally. When the atoms differ in electronegativity the shared pair is pulled toward the more electronegative atom, which acquires a small negative charge (δ−) while the other acquires a small positive charge (δ+). Such a bond is **polar**. In H–Cl, chlorine (3.0) is more electronegative than hydrogen (2.1), so the bond is $H^{\delta+}$–$Cl^{\delta-}$.

A rough guide is that a difference in electronegativity greater than about 1.7 gives a predominantly ionic bond, a difference between about 0.4 and 1.7 gives a polar covalent bond, and a smaller difference gives a non-polar covalent bond. Thus the bond in NaCl (difference about 2.1) is ionic, that in HCl (0.9) is polar covalent, and that in Cl_2 (0) is non-polar.

A molecule can contain polar bonds and still be non-polar overall if the bond dipoles cancel by symmetry. Carbon dioxide (linear, O=C=O) and tetrachloromethane (tetrahedral) are non-polar, but water (bent) and ammonia (pyramidal) are polar. Polar molecules dissolve ionic substances and other polar substances well, which is why water is such a good solvent.

4.5 Dative (coordinate) covalent bonds

In a **dative covalent bond** both electrons of the shared pair come from one atom, the donor, which must have a lone pair; the other atom, the acceptor, must have an empty orbital. Once formed, a dative bond is identical to any other covalent bond. It is shown as an arrow from donor to acceptor.

- **Ammonium ion:** $NH_3 + H^+ \rightarrow NH_4^+$. The lone pair on nitrogen is donated to the hydrogen ion, which has an empty 1s orbital. All four N–H bonds in NH_4^+ are then equivalent.
- **Hydroxonium (oxonium) ion:** $H_2O + H^+ \rightarrow H_3O^+$.
- **Ammonia–boron trifluoride:** $NH_3 \rightarrow BF_3$, where boron has only six electrons and accepts the lone pair of nitrogen.
- **Carbon monoxide:** C≡O has two covalent bonds plus a dative bond from oxygen to carbon.

4.6 Shapes of molecules

The shape of a molecule is explained by the **valence shell electron pair repulsion (VSEPR) theory**: pairs of electrons in the outer shell of the central atom, both bonding pairs and lone pairs, repel each other and arrange themselves as far apart as possible. A lone pair repels more strongly than a bonding pair, so lone pairs compress bond angles.

Bonding pairs	Lone pairs	Shape	Bond angle	Example
2	0	Linear	180°	$BeCl_2$, CO_2
3	0	Trigonal planar	120°	BF_3
4	0	Tetrahedral	109.5°	CH_4, NH_4^+
3	1	Pyramidal	about 107°	NH_3
2	2	Bent (V-shaped)	about 104.5°	H_2O

The decreasing angles in the series CH_4 (109.5°), NH_3 (107°) and H_2O (104.5°) show the growing effect of lone pairs. A double bond is treated as one region of electron density, so CO_2 is linear.

WORKED EXAMPLE 4.2

Predict the shape of the molecule of phosphine, PH_3, and of the ion H_3O^+.

Solution: Phosphorus (2,8,5) has five outer electrons; three form bonds with hydrogen, leaving one lone pair. With three bonding pairs and one lone pair PH_3 is pyramidal. In H_3O^+ oxygen has three bonding pairs (one of them dative) and one lone pair, so it is also pyramidal, like NH_3.

4.7 Giant covalent structures

Some covalent substances form a continuous network of atoms joined by covalent bonds throughout the crystal. These **giant covalent** (macromolecular) structures have very high melting points and are hard, because to melt them many strong covalent bonds must be broken.

- **Diamond:** each carbon is bonded to four others tetrahedrally. It is the hardest natural substance, has a melting point above 3500 °C, and does not conduct electricity because all four outer electrons are used in bonding. It is used in cutting tools and drill bits.
- **Graphite:** each carbon is bonded to three others in flat hexagonal layers; the fourth electron is delocalised along the layers. Layers are held by weak forces and slide over each other, so graphite is soft and slippery (used in pencils and as a lubricant) and conducts electricity (used for electrodes in electrolysis and dry cells).
- **Silicon(IV) oxide (silica, sand, quartz):** each silicon is bonded to four oxygens and each oxygen to two silicons, giving a hard solid of very high melting point (about 1700 °C). Silica is the main ingredient of glass.

Diamond and graphite are **allotropes** of carbon: different forms of the same element in the same physical state, differing in the arrangement of atoms. Other allotropes are fullerenes (such as C_{60}) and, in other elements, the rhombic and monoclinic forms of sulphur and the white and red forms of phosphorus.

4.8 Metallic bonding

A metal consists of a regular lattice of positive ions surrounded by a "sea" of delocalised outer electrons that are free to move throughout the structure. The **metallic bond** is the electrostatic attraction between the positive ions and the delocalised electrons. This explains the properties of metals:

- **High electrical and thermal conductivity**: delocalised electrons carry charge and energy through the metal.
- **Malleability and ductility**: layers of ions can slide over each other without breaking the bond, since the electron sea simply readjusts.
- **High melting points and density** in most metals, particularly where each atom contributes several electrons (magnesium melts at a higher temperature than sodium; the strength of the bond depends on ionic charge, ionic size and the number of delocalised electrons).
- A shiny lustre, because mobile electrons reflect light.

4.9 Forces between molecules

The forces *between* molecules are much weaker than bonds within molecules, but they decide the physical state and boiling point of covalent substances.

Van der Waals (dispersion) forces

Electrons move constantly, so at any instant the charge distribution in a molecule is uneven, producing a temporary dipole that induces a dipole in neighbouring molecules. The resulting attractions exist between all molecules and **increase with**

the number of electrons, that is, with molecular size. This explains why the boiling points of the halogens rise down the group (F_2 is a gas and I_2 a solid) and why the noble gases liquefy at lower temperatures for the lighter members. Alkanes show the same pattern: methane is a gas, hexane a liquid and candle wax a solid.

Dipole–dipole attractions

Polar molecules such as HCl attract each other because the δ+ end of one is attracted to the δ− end of another. These forces add to the van der Waals forces.

Hydrogen bonding

A **hydrogen bond** is a relatively strong attraction between a hydrogen atom covalently bonded to a very electronegative atom (N, O or F) and a lone pair on another N, O or F atom of a neighbouring molecule. It is much stronger than the other intermolecular forces, though still about ten times weaker than a covalent bond.

- Water has an unusually high boiling point (100 °C) compared with H_2S (−60 °C), because of hydrogen bonding. HF, NH_3, ethanol and ethanoic acid show similar effects.
- When water freezes, hydrogen bonds hold the molecules in an open lattice that is less dense than the liquid, so ice floats; this protects aquatic life in cold regions because lakes freeze from the top.
- Hydrogen bonding explains the solubility of ethanol and sugar in water and the high surface tension of water.

WORKED EXAMPLE 4.3

Explain why ethanol (C_2H_5OH) boils at 78 °C while methoxymethane (CH_3OCH_3), which has the same molecular formula, boils at −25 °C.

Solution: Ethanol contains an O–H group, so its molecules form hydrogen bonds with each other, and these require much extra energy to overcome. Methoxymethane has no hydrogen attached to oxygen (no O–H or N–H), so it cannot form hydrogen bonds with itself, and only weak dipole and van der Waals forces act between its molecules. Hence ethanol has the much higher boiling point.

4.10 Structure and properties: a summary

Type of structure	Particles	Melting point	Conductivity	Examples
Giant ionic	Ions	High	Conducts when molten or dissolved, not solid	NaCl, MgO
Giant covalent	Atoms	Very high	None (except graphite)	Diamond, SiO_2
Simple molecular	Molecules	Low	None	H_2O, CO_2, I_2
Giant metallic	Positive ions and electrons	Usually high	Good in solid and liquid	Cu, Fe, Na

WORKED EXAMPLE 4.4

The table shows properties of four substances. Identify the type of structure of each.

- P: melting point 1085 °C, conducts when solid.
- Q: melting point 801 °C, does not conduct as a solid but conducts when molten.
- R: melting point −7 °C, non-conductor in all states.
- S: melting point above 1700 °C, hard, non-conductor.

Solution: P conducts as a solid and melts at a high temperature, so it is a metal (copper melts at 1085 °C). Q conducts only when molten, so it is ionic (sodium chloride, 801 °C). R has a low melting point and does not conduct, so it is a simple molecular substance (for example, bromine). S is hard, non-conducting and has a very high melting point, so it is a giant covalent structure such as silicon(IV) oxide.

4.11 Other types of bonding to note

Many compounds contain more than one type of bond. In ammonium chloride, NH_4Cl, there are covalent bonds (three ordinary and one dative) inside the NH_4^+ ion and an ionic bond between NH_4^+ and Cl^-. In sodium hydroxide, the O–H bond is covalent but the attraction between Na^+ and OH^- is ionic. In the sulphate ion the S–O bonds are covalent while the bond between a metal ion and SO_4^{2-} is ionic.

Some ionic compounds have a covalent character. According to **Fajans' rules**, covalent character increases when the cation is small and highly charged (it polarises the anion), and when the anion is large and easily polarised. Aluminium chloride, with the small, triply charged Al^{3+}, is largely covalent, which is why it sublimes easily and is a poor conductor when molten.

Exercise 4.1

1. Describe, using electron configurations, how an ionic bond forms in magnesium chloride and give its formula.
2. Why does solid sodium chloride not conduct electricity while molten sodium chloride does?
3. Draw (in words) the dot-and-cross arrangement for ammonia and state its shape and bond angle.
4. Explain what is meant by a dative covalent bond, using the formation of NH_4^+ as an example.
5. Give two reasons why graphite is a good lubricant and conducts electricity although diamond does not.
6. Explain why the boiling point of water (100 °C) is much higher than that of hydrogen sulphide (−60 °C).
7. Predict the shape of (a) BF_3 and (b) CH_4, and state the bond angle in each.
8. Explain why metals are malleable and good conductors of heat.
9. Carbon tetrachloride, CCl_4, has polar C–Cl bonds but is a non-polar molecule. Explain.
10. Element X (atomic number 11) reacts with element Y (atomic number 17). Give the formula, the bond type and two physical properties of the product.
11. Arrange F_2, Cl_2, Br_2 and I_2 in order of increasing boiling point and explain.

Answers to Exercise 4.1

1. Mg (2,8,2) loses two electrons to give Mg^{2+} (2,8); each Cl (2,8,7) gains one to give Cl^- (2,8,8); two chloride ions are needed, so the formula is $MgCl_2$.

2. In the solid the ions are fixed in the lattice; in the melt they are free to move and carry charge.
3. Nitrogen shares three pairs with three hydrogen atoms (each H has 2, N has 8) and has one lone pair. The shape is pyramidal with a bond angle of about 107°.
4. It is a covalent bond in which both shared electrons come from one atom. In NH_4^+ the lone pair of nitrogen in NH_3 is donated to an H^+ ion, which has an empty 1s orbital.
5. Graphite has layers held by weak forces which slide over each other, giving lubrication; each carbon uses only three electrons in bonding, and the fourth is delocalised and carries current. In diamond all four electrons are used in bonding.
6. Water molecules form hydrogen bonds (O–H···O), which need much energy to break; H_2S cannot form hydrogen bonds because sulphur is much less electronegative.
7. (a) Trigonal planar, 120°; (b) tetrahedral, 109.5°.
8. Layers of positive ions can slide over each other while the electron sea holds the structure together; the mobile electrons transfer kinetic energy rapidly through the metal.
9. The four bonds are arranged tetrahedrally and symmetrically, so the individual bond dipoles cancel.
10. X is sodium (2,8,1), Y is chlorine (2,8,7). The product NaCl is ionic; it has a high melting point (801 °C) and conducts electricity when molten or dissolved (also soluble in water).
11. $F_2 < Cl_2 < Br_2 < I_2$. Van der Waals forces increase with the number of electrons and the size of the molecule.

Chapter summary

- Atoms bond to attain a stable noble gas arrangement of electrons by transferring (ionic), sharing (covalent) or delocalising (metallic) electrons.

- Ionic compounds form giant lattices with high melting points, conduct only when molten or dissolved, and are brittle and often water-soluble.
- Covalent bonds involve shared pairs; single, double and triple bonds increase in strength and decrease in length. Simple molecular substances have low melting and boiling points and do not conduct.
- A bond is polar when the atoms differ in electronegativity; a molecule is polar only if its bond dipoles do not cancel.
- A dative bond is a covalent bond in which one atom supplies both electrons (NH_4^+, H_3O^+, CO).
- VSEPR theory: linear (180°), trigonal planar (120°), tetrahedral (109.5°), pyramidal (about 107°), bent (about 104.5°).
- Diamond, graphite and silica are giant covalent structures; diamond and graphite are allotropes of carbon.
- Metallic bonding is the attraction between positive ions and delocalised electrons and explains conductivity and malleability.
- Intermolecular forces (van der Waals, dipole–dipole and hydrogen bonds) determine boiling points and solubility of molecular substances.

Chapter 5: The Mole Concept, Chemical Formulae and Stoichiometry

Chemists constantly need to answer practical questions. How much iron can be obtained from a tonne of ore? What volume of carbon dioxide is released when limestone is heated at a cement works in Ewekoro? How much sodium hydroxide must a soap maker in Kano weigh out to saponify a given quantity of palm oil? Atoms and molecules are far too small to count or weigh one at a time, so chemistry uses a counting unit of its own, the mole, which links the invisible world of particles to the masses and volumes measured in the laboratory. This chapter develops the mole concept, shows how formulae are found from experimental data, explains how chemical equations are used to predict quantities, and treats concentration of solutions and volumetric analysis. Calculations of this kind make up a large part of the UTME Chemistry paper, so every worked example here deserves careful study.

5.1 Relative atomic mass and relative molecular mass

The mass of a single atom is of the order of 10^{-23} g, an inconvenient figure. Chemists therefore compare masses on a relative scale. The **relative atomic mass** (A_r) of an element is the average mass of one atom of the element, taking account of its naturally occurring isotopes, divided by one-twelfth of the mass of one atom of carbon-12. Because it is a ratio, A_r has no units. The **relative molecular mass** (M_r) of a substance is the sum of the relative atomic masses of all atoms in its formula. For ionic compounds the term *relative formula mass* is often used, but the calculation is identical.

Throughout this book the following approximate values are used: H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, Al = 27, S = 32, Cl = 35.5, K = 39, Ca = 40, Fe = 56, Cu = 64, Zn = 65, Ag = 108, Pb = 207. These are the values supplied or

assumed in UTME questions.

Because most elements occur as mixtures of isotopes, relative atomic masses are often not whole numbers. Chlorine occurs as about 75% chlorine-35 and 25% chlorine-37, so

$A_r(\text{Cl}) = (75 \times 35 + 25 \times 37)/100 = (2625 + 925)/100 = 3550/100 = 35.5$.

To find the M_r of calcium nitrate, $\text{Ca(NO}_3)_2$, add: $40 + 2 \times (14 + 3 \times 16) = 40 + 2 \times 62 = 164$. For hydrated copper(II) sulphate, $\text{CuSO}_4\cdot 5\text{H}_2\text{O}$, the water of crystallisation is included: $64 + 32 + 64 + 5 \times 18 = 160 + 90 = 250$.

5.2 The mole and the Avogadro constant

The **mole** is the amount of substance that contains as many elementary particles (atoms, molecules, ions or electrons) as there are atoms in exactly 12 g of carbon-12. That number is the **Avogadro constant**, $L = 6.02 \times 10^{23}\ \text{mol}^{-1}$. One mole of any substance therefore contains 6.02×10^{23} particles, in the same way that a dozen always means twelve whatever is being counted.

The mass of one mole of a substance, expressed in grams, is its **molar mass**, M, with units g/mol. It is numerically equal to the relative atomic or molecular mass. Thus the molar mass of water is 18 g/mol, of carbon dioxide 44 g/mol and of sodium chloride 58.5 g/mol. The three basic relationships are:

- Amount of substance (mol) = mass (g) ÷ molar mass (g/mol)
- Number of particles = amount (mol) × 6.02×10^{23}
- For gases at standard temperature and pressure (s.t.p., 0 °C and 101.3 kPa), the **molar volume** is 22.4 dm^3/mol, so amount (mol) = volume (dm^3) ÷ 22.4.

Key point. Always state the particle you are counting. One mole of oxygen *molecules* (O_2) is 32 g and contains 6.02×10^{23} molecules but 1.204×10^{24} oxygen atoms. Conversions among mass, moles, particles and gas volume all pass through the mole.

WORKED EXAMPLE 5.1

Calculate (a) the mass of 0.25 mol of calcium carbonate, $CaCO_3$; (b) the number of water molecules in 3.6 g of water.

Solution: (a) $M(CaCO_3) = 40 + 12 + 3 \times 16 = 100$ g/mol. Mass $= 0.25 \times 100 =$ **25 g**.

(b) $M(H_2O) = 18$ g/mol. Amount $= 3.6/18 = 0.20$ mol. Number of molecules $= 0.20 \times 6.02 \times 10^{23} =$ **1.204×10^{23}**.

WORKED EXAMPLE 5.2

What volume at s.t.p. is occupied by 8.0 g of oxygen gas, and how many atoms does it contain?

Solution: $M(O_2) = 32$ g/mol. Amount $= 8.0/32 = 0.25$ mol. Volume $= 0.25 \times 22.4 =$ **5.6 dm^3**. Molecules $= 0.25 \times 6.02 \times 10^{23} = 1.505 \times 10^{23}$; each molecule has two atoms, so atoms = **3.01×10^{23}**.

5.3 Percentage composition and formulae

The **percentage composition** of a compound is the percentage by mass of each element in it:

percentage of element = (mass of the element in one mole of compound ÷ molar mass of compound) × 100.

For iron(III) oxide, Fe_2O_3, $M_r = 2 \times 56 + 3 \times 16 = 160$, so the iron content is $112/160 \times 100 = 70\%$, and oxygen makes up the remaining 30%. A farmer comparing two fertilisers can use the same idea: urea, $CO(NH_2)_2$ ($M_r = 60$), contains $28/60 \times 100 = 46.7\%$ nitrogen, while ammonium nitrate, NH_4NO_3 ($M_r = 80$), contains $28/80 \times 100 = 35\%$ nitrogen.

Empirical and molecular formulae

The **empirical formula** shows the simplest whole-number ratio of atoms of each element in a compound. The **molecular formula** shows the actual number of atoms in one molecule and is always a whole-number multiple of the empirical formula: molecular formula = $(\text{empirical formula})_n$, where $n = M_r \div$ empirical formula mass.

To find an empirical formula from analysis data, follow this routine: (1) write the mass or percentage of each element; (2) divide each by its relative atomic mass to obtain moles; (3) divide all by the smallest number; (4) if the ratios are not whole numbers (for example 1.5 or 2.33), multiply through by the smallest integer that makes them whole.

WORKED EXAMPLE 5.3

A compound contains 40.0% carbon, 6.7% hydrogen and 53.3% oxygen by mass, and its relative molecular mass is 180. Find its empirical and molecular formulae.

Solution: Moles: C = 40.0/12 = 3.33; H = 6.7/1 = 6.7; O = 53.3/16 = 3.33. Dividing by 3.33 gives C : H : O = 1 : 2 : 1. Empirical formula = $\mathbf{CH_2O}$, with mass 12 + 2 + 16 = 30. n = 180/30 = 6, so the molecular formula is $\mathbf{C_6H_{12}O_6}$ (glucose).

WORKED EXAMPLE 5.4

Analysis of an iron oxide shows that 5.6 g of iron combine with 2.4 g of oxygen. Determine the empirical formula.

Solution: Moles of Fe = 5.6/56 = 0.10; moles of O = 2.4/16 = 0.15. Ratio Fe : O = 0.10 : 0.15 = 1 : 1.5. Multiplying by 2 gives 2 : 3. The formula is **Fe_2O_3**.

Water of crystallisation

Many salts crystallise from solution with a fixed number of water molecules, for example $Na_2CO_3 \cdot 10H_2O$ (washing soda) and $CuSO_4 \cdot 5H_2O$. Heating drives off the water, leaving the anhydrous salt, and the loss in mass gives the value of x.

WORKED EXAMPLE 5.5

A sample of hydrated copper(II) sulphate, $CuSO_4 \cdot xH_2O$, of mass 2.50 g loses 0.90 g on heating to constant mass. Find x.

Solution: Mass of anhydrous salt = 2.50 − 0.90 = 1.60 g. Moles $CuSO_4$ = 1.60/160 = 0.010. Moles H_2O = 0.90/18 = 0.050. Ratio = 0.050/0.010 = 5. So x = **5**.

5.4 Chemical equations

A chemical equation represents a reaction using formulae. By the law of conservation of mass (Chapter 1), the number of atoms of each element must be the same on both sides, so equations must be **balanced** by placing coefficients in front of formulae, never by changing the formulae themselves. State symbols, (s), (l), (g) and (aq), are often added. A balanced equation tells us the ratio in which substances react in moles; this mole ratio is the key to all stoichiometric calculations.

Consider the burning of propane: $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$. Balance carbon first (3 CO_2), then hydrogen (4 H_2O), and finally oxygen (6 + 4 = 10 atoms, hence 5 O_2).

Ionic equations show only the species that actually change. When silver nitrate solution is added to sodium chloride solution, the full equation is $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$. Sodium and nitrate ions are *spectator ions*, present before and after, so the ionic equation is $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$. An ionic equation must balance in charge as well as in atoms.

5.5 Calculations from equations

The method has four steps: (1) write the balanced equation; (2) convert the given quantity to moles; (3) use the coefficients to find the moles of the required substance; (4) convert to mass, volume or number as requested.

WORKED EXAMPLE 5.6

Magnesium burns in oxygen: $2Mg + O_2 \rightarrow 2MgO$. What mass of magnesium oxide forms when 6.0 g of magnesium is burnt completely?

Solution: Moles Mg = 6.0/24 = 0.25. From the equation 2 mol Mg gives 2 mol MgO, so moles MgO = 0.25. M(MgO) = 40 g/mol. Mass = 0.25 × 40 = **10 g**.

WORKED EXAMPLE 5.7

Calcium carbonate reacts with hydrochloric acid: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$. What volume of carbon dioxide at s.t.p. is produced from 10.0 g of calcium carbonate?

Solution: Moles $CaCO_3$ = 10.0/100 = 0.100. The ratio $CaCO_3$: CO_2 is 1 : 1, so moles CO_2 = 0.100. Volume = 0.100 × 22.4 = **2.24 dm^3**.

Gas volumes in reactions

By Gay-Lussac's law and Avogadro's law, gases at the same temperature and pressure react in simple volume ratios equal to their mole ratios, so volumes may be used directly without conversion to moles. For example, in $2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$, 100 cm^3 of hydrogen reacts with 50 cm^3 of oxygen to give 100 cm^3 of steam, all measured under the same conditions.

Limiting reagent and percentage yield

When reactants are not supplied in the exact ratio of the equation, one of them is used up first. This is the **limiting reagent**, and it controls the amount of product. The other is in **excess**. To identify the limiting reagent, divide the moles of each reactant by its coefficient; the smaller value belongs to the limiting reagent.

WORKED EXAMPLE 5.8

2.4 g of magnesium is added to 100 cm^3 of 1.0 mol/dm^3 hydrochloric acid: $Mg + 2HCl \rightarrow MgCl_2 + H_2$. Which reagent is limiting, and what volume of hydrogen at s.t.p. is evolved?

Solution: Moles Mg = 2.4/24 = 0.10. Moles HCl = 1.0 × 0.100 = 0.10. Dividing by coefficients: Mg 0.10/1 = 0.10; HCl 0.10/2 = 0.05. HCl is smaller, so **HCl is limiting** and magnesium is in excess. Moles H_2 = 0.10/2 = 0.05. Volume = 0.05 × 22.4 = **1.12 dm^3**.

In practice the mass of product actually obtained is usually less than the calculated value, because of incomplete reactions, side reactions and losses during separation. The **percentage yield** is (actual yield ÷ theoretical yield) × 100. If 10.0 g of a product is expected and 8.5 g is collected, the yield is 8.5/10.0 × 100 = 85%. The **percentage purity** of a sample is (mass of pure substance ÷ mass of sample) × 100, and is used when an impure reactant such as limestone is involved.

WORKED EXAMPLE 5.9

A 20.0 g sample of impure limestone releases 3.36 dm^3 of carbon dioxide at s.t.p. when strongly heated: $CaCO_3 \rightarrow CaO + CO_2$. Calculate the percentage of $CaCO_3$ in the sample, assuming the impurities do not decompose.

Solution: Moles CO_2 = 3.36/22.4 = 0.150. Moles $CaCO_3$ = 0.150 (1 : 1). Mass = 0.150 × 100 = 15.0 g. Purity = 15.0/20.0 × 100 = **75%**.

5.6 Concentration of solutions

A **solution** consists of a solute dissolved in a solvent. Its concentration states how much solute is present in a given volume, and can be expressed in two common ways:

- **Molar concentration** (mol/dm^3, symbol M or mol dm^{-3}) = moles of solute ÷ volume in dm^3.
- **Mass concentration** (g/dm^3) = mass of solute ÷ volume in dm^3.

The two are connected by: mass concentration = molar concentration × molar mass. Remember that 1 dm^3 = 1000 cm^3, so a volume in cm^3 must be divided by 1000 before it is used. A **standard solution** is one whose concentration is accurately known.

WORKED EXAMPLE 5.10

4.0 g of sodium hydroxide is dissolved in water and made up to 500 cm^3. Calculate the concentration in (a) mol/dm^3 and (b) g/dm^3.

Solution: M(NaOH) = 23 + 16 + 1 = 40 g/mol. Moles = 4.0/40 = 0.10. Volume = 0.500 dm^3. (a) Concentration = 0.10/0.500 = **0.20 mol/dm^3**. (b) 4.0/0.500 = **8.0 g/dm^3**; check: 0.20 × 40 = 8.0.

Dilution

Adding solvent does not change the amount of solute, so moles before = moles after, giving $C_1V_1 = C_2V_2$. If 25.0 cm^3 of 2.0 mol/dm^3 solution is diluted to 100 cm^3, then $C_2 = (2.0 \times 25.0)/100 = 0.50$ mol/dm^3.

5.7 Volumetric analysis (titration)

Volumetric analysis determines the concentration of an unknown solution by reacting it with a standard solution. A **pipette** delivers a fixed volume (usually 25.0 cm^3) of one solution into a conical flask, and a **burette** adds the other solution gradually until the **end point**, shown by a colour change of an indicator, is reached. The titration is repeated until two or more readings agree within 0.10 cm^3, and the average of these concordant titres is used. (Indicators and acid–base theory are treated in Chapter 9.)

For the general reaction $aA + bB \rightarrow$ products, where A is the acid and B the base, the relation is

$$(C_A \times V_A)/(C_B \times V_B) = n_A/n_B,$$

in which n_A and n_B are the coefficients in the balanced equation. It is safer to work through moles explicitly, as below.

WORKED EXAMPLE 5.11

In a titration, 25.0 cm^3 of sodium hydroxide solution required 20.0 cm^3 of 0.100 mol/dm^3 hydrochloric acid for neutralisation. Calculate the concentration of the alkali.

Solution: $NaOH + HCl \rightarrow NaCl + H_2O$ (1 : 1). Moles HCl = $0.100 \times 20.0/1000 = 0.00200$. Moles NaOH = 0.00200. Concentration = 0.00200/0.0250 = **0.0800 mol/dm^3**.

WORKED EXAMPLE 5.12

25.0 cm^3 of sodium carbonate solution of concentration 0.100 mol/dm^3 was neutralised by 20.0 cm^3 of hydrochloric acid. Calculate the acid concentration. $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$.

Solution: Moles Na_2CO_3 = 0.100 × 0.0250 = 0.00250. Moles HCl = 2 × 0.00250 = 0.00500. Concentration = 0.00500/0.0200 = **0.250 mol/dm^3**.

Key point. In titration questions, always (1) write the balanced equation, (2) find moles of the substance whose concentration and volume are both known, (3) use the mole ratio to get moles of the other substance, (4) divide by its volume in dm^3. Skipping the equation is the commonest cause of wrong answers, especially with H_2SO_4 and Na_2CO_3 where the ratio is not 1 : 1.

Exercise 5.1

1. Calculate the mass of 0.50 mol of sodium carbonate, Na_2CO_3.
2. How many moles are present in 49 g of sulphuric acid, H_2SO_4?
3. How many molecules are present in 1.8 g of water?
4. A compound contains 32.4% sodium, 22.6% sulphur and 45.0% oxygen by mass. Find its empirical formula.
5. A hydrocarbon has empirical formula CH_2 and relative molecular mass 56. Find its molecular formula.
6. 4.6 g of sodium reacts with excess water: $2Na + 2H_2O \rightarrow 2NaOH + H_2$. What volume of hydrogen is released at s.t.p.?
7. What volume of carbon dioxide at s.t.p. is formed when 3.0 g of carbon burns completely in excess oxygen?

8. Calculate the molar concentration of a solution containing 5.3 g of Na_2CO_3 in 250 cm^3.
9. 25.0 cm^3 of 0.100 mol/dm^3 sodium carbonate required 20.0 cm^3 of hydrochloric acid for complete reaction. Calculate the concentration of the acid.
10. When 15.0 g of calcium carbonate is heated, 7.0 g of calcium oxide is obtained. Calculate the percentage yield.
11. 6.5 g of zinc is added to a solution containing 0.15 mol of hydrochloric acid: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$. Identify the limiting reagent and calculate the volume of hydrogen at s.t.p.
12. How many atoms are present in 12 g of magnesium?

Answers to Exercise 5.1

1. M = 106 g/mol; 0.50 × 106 = 53 g.
2. M = 98; 49/98 = 0.50 mol.
3. 1.8/18 = 0.10 mol; $0.10 \times 6.02 \times 10^{23} = 6.02 \times 10^{22}$ molecules.
4. Na 32.4/23 = 1.41; S 22.6/32 = 0.706; O 45.0/16 = 2.81. Dividing by 0.706 gives 2 : 1 : 4, so Na_2SO_4.
5. CH_2 has mass 14; 56/14 = 4; C_4H_8.
6. Na = 4.6/23 = 0.20 mol; H_2 = 0.10 mol; 0.10 × 22.4 = 2.24 dm^3.
7. $C + O_2 \rightarrow CO_2$; 3.0/12 = 0.25 mol; 0.25 × 22.4 = 5.6 dm^3.
8. 5.3/106 = 0.050 mol; 0.050/0.250 = 0.20 mol/dm^3.
9. Na_2CO_3 = 0.00250 mol; HCl = 0.00500 mol; 0.00500/0.0200 = 0.250 mol/dm^3.
10. $CaCO_3$ = 0.15 mol; theoretical CaO = 0.15 × 56 = 8.4 g; yield = 7.0/8.4 × 100 = 83.3%.
11. Zn = 6.5/65 = 0.10 mol, needing 0.20 mol HCl; only 0.15 mol is present, so HCl is limiting. H_2 = 0.075 mol; 0.075 × 22.4 = 1.68 dm^3.

12. 12/24 = 0.50 mol; $0.50 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$ atoms.

Chapter summary

- Relative atomic mass compares the average atom of an element with one-twelfth of a carbon-12 atom; M_r is the sum of the A_r values in the formula.
- One mole contains 6.02×10^{23} particles, has a mass equal to the molar mass in grams, and for a gas occupies 22.4 dm^3 at s.t.p.
- Empirical formulae come from mass or percentage data by dividing by A_r and finding the simplest ratio; the molecular formula is n times the empirical formula.
- Equations must balance in atoms (and charge, for ionic equations); the coefficients give the mole ratio used in all calculations.
- The limiting reagent decides the amount of product; percentage yield = actual/theoretical × 100.
- Concentration (mol/dm^3) = moles ÷ volume in dm^3; dilution obeys $C_1V_1 = C_2V_2$.
- Titration calculations use the balanced equation to link the moles of acid and base.

Chapter 6: Gases, Gas Laws, Solutions and Water

Every time a motorcyclist in Ibadan pumps up a tyre, every time a baker in Kano watches dough rise, and every time a bottle of soft drink fizzes when the cap is removed, the behaviour of gases is on display. Gases are the simplest of the three states of matter to describe mathematically, because their particles are so far apart that the nature of the particles matters much less than their number, the temperature and the space available. This chapter explains the kinetic theory of matter, develops the gas laws from Boyle to the ideal gas equation, and shows how to use them in calculations that the UTME examiners favour. It then turns to the second great theme of physical chemistry at this level, namely solutions: how solids and gases dissolve in water, how solubility changes with temperature, and why the water we drink, wash with and cook with can be hard or soft. The chapter closes with the properties of water itself and the causes and cures of hard water.

6.1 The kinetic theory of matter

The **kinetic theory** states that all matter consists of tiny particles (atoms, molecules or ions) that are in constant motion. The state of a substance depends on how much kinetic energy the particles possess compared with the strength of the attractions between them. In a solid the particles vibrate about fixed positions and are held in an orderly arrangement. In a liquid they are still close together but can slide past one another, so a liquid flows and takes the shape of its container while keeping a fixed volume. In a gas the particles are far apart, move rapidly in straight lines in all directions, collide with one another and with the walls of the container, and fill whatever space is available.

For gases the theory is made precise by a set of postulates which describe an *ideal gas*:

- A gas consists of a very large number of identical particles whose own volume is negligible compared with the volume of the container.
- The particles move randomly and continuously, and collisions between them and with the container walls are perfectly elastic, so no kinetic energy is lost.
- There are no attractive or repulsive forces between the particles except during collisions.
- The average kinetic energy of the particles is directly proportional to the absolute (kelvin) temperature of the gas.

The **pressure** of a gas is the force that the particles exert per unit area of the container wall as they strike it. It is measured in pascals (Pa), kilopascals (kPa), atmospheres (atm) or millimetres of mercury (mmHg). The relations to remember are 1 atm = 101.325 kPa = 760 mmHg, and for calculation purposes 101.3 kPa is accepted. The **kelvin temperature** is obtained from the Celsius temperature by adding 273: $T(\text{K}) = \theta(^\circ\text{C}) + 273$. The kelvin scale begins at absolute zero, the temperature (−273 °C) at which the kinetic energy of the particles would be at its minimum. Because gas laws relate quantities to the average kinetic energy of the particles, temperatures in gas calculations must *always* be converted to kelvin.

> **Key point.** Standard temperature and pressure (s.t.p.) means 273 K (0 °C) and 101.325 kPa (760 mmHg or 1 atm). At s.t.p. one mole of any ideal gas occupies 22.4 dm^3. Never substitute a Celsius temperature into a gas-law equation.

6.2 Boyle's law

Robert Boyle found that if the temperature of a fixed mass of gas is kept constant, the volume decreases as the pressure increases. **Boyle's law** states that *the volume of a fixed mass of gas at constant temperature is inversely proportional to its pressure*. In symbols, $V \propto 1/P$, or

PV = constant, so that $P_1V_1 = P_2V_2$.

The kinetic explanation is simple. If the same number of particles is squeezed into half the volume, they strike a given area of wall twice as often, so the pressure doubles. A graph of P against V is a curve (a hyperbola), whereas a graph of P against 1/V is a straight line through the origin.

WORKED EXAMPLE 6.1

A sample of gas occupies 600 cm^3 at a pressure of 100 kPa. What volume will it occupy at the same temperature if the pressure is raised to 250 kPa?

Solution: Temperature is constant, so $P_1V_1 = P_2V_2$. Then $V_2 = P_1V_1/P_2 = (100 \times 600)/250 = 60\,000/250 = 240$ cm^3. The pressure has been multiplied by 2.5, so the volume is divided by 2.5, which agrees: $600 \div 2.5 = 240$ cm^3.

6.3 Charles's law

Jacques Charles studied what happens to the volume of a gas when it is heated at constant pressure. **Charles's law** states that *the volume of a fixed mass of gas at constant pressure is directly proportional to its absolute temperature*:

$V \propto T$, or V/T = constant, so that $V_1/T_1 = V_2/T_2$.

A graph of volume against Celsius temperature is a straight line, which, if extended backwards, cuts the temperature axis at −273 °C. This is how the idea of absolute zero first arose.

WORKED EXAMPLE 6.2

A gas occupies 300 cm^3 at 27 °C. At constant pressure, what volume does it occupy at 87 °C?

Solution: Convert to kelvin: $T_1 = 27 + 273 = 300$ K and $T_2 = 87 + 273 = 360$ K. Then $V_2 = V_1 \times T_2/T_1 = 300 \times 360/300 = 360$ cm^3. A common error is to use 87/27, which would give nearly 967 cm^3, an impossible increase for a rise of only 60 degrees.

6.4 The pressure law and the combined gas law

The third relationship describes a gas heated in a container of fixed volume, such as an aerosol can or a gas cylinder. The **pressure law** states that *the pressure of a fixed mass of gas at constant volume is directly proportional to its absolute temperature*: $P_1/T_1 = P_2/T_2$. This is why a sealed can thrown into a fire may burst, and why tyre pressures rise after a long journey on the Lagos–Ibadan expressway.

The three laws combine into the **combined gas law**, which applies when all three variables change:

$P_1V_1/T_1 = P_2V_2/T_2$.

Any one of the three earlier laws is recovered by cancelling the quantity that is constant. The combined law is the tool most often used to convert a measured gas volume to its volume at s.t.p. so that it can be compared with the molar volume of 22.4 dm^3.

WORKED EXAMPLE 6.3

A gas occupies 500 cm^3 at 27 °C and 750 mmHg. Calculate its volume at s.t.p.

Solution: Initial conditions: P_1 = 750 mmHg, V_1 = 500 cm^3, T_1 = 300 K. Final conditions: P_2 = 760 mmHg, T_2 = 273 K. Then $V_2 = (P_1V_1/T_1) \times (T_2/P_2)$ = (750 × 500 × 273)/(300 × 760) = 102 375 000/228 000 ≈ 449 cm^3. The volume falls because the temperature is lowered (a factor 273/300 = 0.91) while the pressure is raised slightly (a factor 750/760 = 0.987).

6.5 Avogadro's law and the ideal gas equation

Avogadro's law states that *equal volumes of all gases, under the same conditions of temperature and pressure, contain the same number of molecules*. Equivalently, the volume of a gas at fixed temperature and pressure is proportional to the amount of gas in moles. It follows that at s.t.p. one mole of any gas, containing 6.02×10^{23} particles, occupies 22.4 dm^3, and that the gas densities at the same temperature and pressure are in the ratio of their molar masses. The law also explains Gay-Lussac's observation, met in Chapter 1, that gases react in simple whole-number ratios of volumes: the volume ratio in an equation is the same as the mole ratio.

Combining Boyle's, Charles's and Avogadro's laws gives the **ideal gas equation**:

$$PV = nRT$$

where P is the pressure, V the volume, n the amount of gas in moles, T the kelvin temperature and R the **molar gas constant**. Its value depends on the units: R = 8.314 J mol^{-1} K^{-1} when P is in pascals and V in cubic metres, which is the same as 8.314 kPa dm^3 mol^{-1} K^{-1} when P is in kPa and V in dm^3. In older units R = 0.0821 atm dm^3 mol^{-1} K^{-1}. You can check the value from s.t.p. data: R = PV/nT = (101.3 × 22.4)/(1 × 273) = 8.31.

Since n = m/M, where m is the mass and M the molar mass, the equation can be rearranged to find molar mass or density:

M = mRT/PV and density ρ = m/V = PM/RT.

WORKED EXAMPLE 6.4

A gas of mass 1.70 g occupies 1.25 dm^3 at 27 °C and 100 kPa. Calculate its molar mass and suggest the gas.

Solution: T = 300 K. n = PV/RT = (100 × 1.25)/(8.314 × 300) = 125/2494.2 = 0.0501 mol. Molar mass = m/n = 1.70/0.0501 = 33.9 g/mol, that is, about 34 g/mol. The gas is hydrogen sulphide, H_2S (2 × 1 + 32 = 34), the gas with the smell of rotten eggs.

6.6 Dalton's law of partial pressures

When several gases that do not react are placed in the same container, each gas behaves as though it were alone. **Dalton's law of partial pressures** states that *the total pressure of a mixture of gases is equal to the sum of the partial pressures of the individual gases*, the partial pressure of a gas being the pressure it would exert if it alone occupied the container at the same temperature. The partial pressure of a component equals its **mole fraction** multiplied by the total pressure:

$p_A = x_A \times P_{total}$, where $x_A = n_A/n_{total}$.

Dalton's law is essential when a gas is collected over water. Water evaporates into the gas, so the gas collected is a mixture of the gas itself and water vapour. The pressure of the gas alone is the atmospheric pressure minus the **saturated vapour pressure** (s.v.p.) of water at the temperature of the experiment. The s.v.p. of water rises with temperature: about 12 mmHg at 14 °C, 27 mmHg at 27 °C and 760 mmHg at 100 °C, when the water boils.

WORKED EXAMPLE 6.5

200 cm^3 of hydrogen is collected over water at 27 °C when the atmospheric pressure is 760 mmHg. The saturated vapour pressure of water at 27 °C is 27 mmHg. Calculate the volume of the dry hydrogen at s.t.p.

Solution: Pressure of hydrogen = 760 − 27 = 733 mmHg. Apply the combined gas law: $V_2 = (733 \times 200 \times 273)/(300 \times 760) = 40\,021\,800/228\,000 = 175.5\ cm^3$. If the vapour pressure were ignored the answer, 200 cm^3 × 273/300 = 182 cm^3, would be wrong by about 4%.

6.7 Diffusion and Graham's law

Diffusion is the spontaneous mixing of gases (or of any particles) as a result of their random motion, from a region of higher concentration to one of lower concentration. The smell of frying onions and pepper soup spreading from a kitchen through the house is a familiar example. **Effusion** is the escape of a gas through a very small hole. Thomas Graham showed that, at the same temperature and pressure, lighter gases diffuse faster. **Graham's law** states that *the rate of diffusion of a gas is inversely proportional to the square root of its density, or equivalently of its molar mass*:

$\text{Rate}_1/\text{Rate}_2 = \sqrt{(M_2/M_1)}$.

Since rate is inversely proportional to the time taken for a given volume to diffuse, the law can also be written $t_1/t_2 = \sqrt{(M_1/M_2)}$. The law is understood from the kinetic theory: at the same temperature all gas particles have the same average kinetic energy, $\tfrac{1}{2}mv^2$, so lighter particles must have greater average speeds. A classic demonstration places cotton wool soaked in concentrated ammonia at one end of a long glass tube and cotton wool soaked in concentrated hydrochloric acid at the other. A white ring of ammonium chloride forms nearer the hydrochloric acid end, because ammonia (M = 17) moves faster than hydrogen chloride (M = 36.5).

WORKED EXAMPLE 6.6

A gas X takes 40 seconds to diffuse through a porous plug, while the same volume of oxygen takes 20 seconds under the same conditions. Calculate the relative molecular mass of X.

Solution: $t_X/t_{O_2} = \sqrt{(M_X/M_{O_2})}$. Hence $40/20 = 2 = \sqrt{(M_X/32)}$. Squaring, $M_X/32 = 4$, so $M_X = 128$. (Compare the rates: oxygen diffuses twice as fast as X, and $\sqrt{(128/32)} = 2$.)

6.8 Solutions and solubility

A **solution** is a homogeneous mixture of two or more substances. The substance present in larger amount, usually a liquid, is the **solvent**, and the substance dissolved is the **solute**. A solution in which no more solute can dissolve at a given temperature, in the presence of undissolved solute, is **saturated**; one that holds less than the maximum is **unsaturated**. A **supersaturated** solution holds more than the saturated amount and is unstable: a single crystal dropped in makes the excess crystallise out.

The **solubility** of a solute at a given temperature is the maximum amount of it that will dissolve in a given quantity of solvent at that temperature to form a saturated solution. It is expressed either as grams of solute per 100 g of water or as mol/dm^3 (grams per dm^3 divided by the molar mass).

The general rule "like dissolves like" guides prediction. Polar and ionic substances, such as sodium chloride, sugar and ethanol, dissolve in polar solvents, such as water, because the solvent molecules attract the particles of the solute and pull them from the crystal; this attraction in water is called **hydration**. Non-polar substances, such as fats, candle wax, petrol and iodine, dissolve in non-polar solvents, such as hexane, tetrachloromethane and ether, but not in water. This is why palm oil stains on clothing are not removed by water alone but are lifted by soap or detergent.

Effect of temperature and pressure

For most solid solutes, such as potassium nitrate and sugar, solubility *increases* with temperature, because the dissolving process absorbs heat and is favoured by heating. A few, such as calcium hydroxide and anhydrous sodium sulphate above 32 °C, become less soluble as the temperature rises. The solubility of *gases* in water decreases with a rise in temperature, which is why a warm soft drink loses its fizz faster than a cold one and why fish can suffer in warm, still water that holds less dissolved oxygen. The solubility of a gas increases with pressure (Henry's law: the mass of gas dissolved is directly proportional to the pressure of the gas above the liquid); fizzy drinks are bottled under pressure of carbon dioxide, and the bubbles form when the pressure is released on opening the cap.

Solubility curves

A **solubility curve** is a graph of solubility (vertical axis) against temperature (horizontal axis). A point on the curve represents a saturated solution; a point below it, an unsaturated solution; a point above it, a supersaturated solution. A steep upward curve shows a salt whose solubility is very temperature-dependent, so that cooling a hot saturated solution crystallises a large amount of solute. Crystallisation is therefore a good purification method for such salts, and it is the principle used in the preparation of crystals of copper(II) sulphate.

WORKED EXAMPLE 6.7

The solubility of a salt is 45 g per 100 g of water at 60 °C and 15 g per 100 g of water at 20 °C. What mass of crystals separates when 250 g of the saturated solution at 60 °C is cooled to 20 °C? (Assume the crystals are anhydrous.)

Solution: At 60 °C, 145 g of saturated solution contains 45 g of salt and 100 g of water. In 250 g of solution the salt is 250 × 45/145 = 77.6 g and the water is 250 − 77.6 = 172.4 g. At 20 °C, 172.4 g of water can hold only 172.4 × 15/100 = 25.9 g of salt. Mass crystallised = 77.6 − 25.9 = 51.7 g. As a check, each 145 g of solution deposits 45 − 15 = 30 g on cooling, so 250 g deposits 250 × 30/145 = 51.7 g.

6.9 Water: composition, properties and uses

Water, H_2O, is a covalent compound whose bent molecules are polar and bound to each other by hydrogen bonds (Chapter 4). These attractions account for its high melting and boiling points relative to its size, its high specific heat capacity, and its power as a solvent. Pure water is a colourless, odourless, tasteless liquid that freezes at 0 °C and boils at 100 °C at standard pressure, with a maximum density of 1.00 g/cm^3 at 4 °C. Ice floats on water because, in ice, hydrogen bonding holds the molecules in an open lattice that is less dense than the liquid. Water reacts with sodium, potassium and calcium to give the hydroxide and hydrogen, combines with acidic oxides to form acids ($SO_3 + H_2O \rightarrow H_2SO_4$) and with basic oxides to form alkalis ($CaO + H_2O \rightarrow Ca(OH)_2$). It is detected by anhydrous copper(II) sulphate, which turns from white to blue.

6.10 Hardness of water

Hard water is water that does not readily form a lather with soap. It contains dissolved calcium and magnesium ions, which react with the soap (sodium stearate, say) to form an insoluble, greasy scum:

$$Ca^{2+}(aq) + 2C_{17}H_{35}COO^{-}(aq) \rightarrow Ca(C_{17}H_{35}COO)_2(s)$$

Only when all of these ions have been precipitated will the soap lather. **Soft water**, such as rainwater, lathers readily. Water becomes hard when rain, made slightly acidic by dissolved carbon dioxide, passes over limestone, chalk or dolomite: $CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$. Well water in limestone areas is therefore often hard.

Temporary hardness

Temporary hardness is caused by the hydrogencarbonates of calcium and magnesium, $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$. It is removed by *boiling*, which decomposes the hydrogencarbonate and precipitates the insoluble carbonate:

$$Ca(HCO_3)_2(aq) \rightarrow CaCO_3(s) + H_2O(l) + CO_2(g)$$

It can also be removed by adding the correct amount of calcium hydroxide (Clark's process): $Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 + 2H_2O$.

Permanent hardness

Permanent hardness is caused by the sulphates and chlorides of calcium and magnesium ($CaSO_4$, $MgSO_4$, $CaCl_2$, $MgCl_2$). These do not decompose on boiling. They are removed by adding sodium carbonate (washing soda), which precipitates the carbonate: $Ca^{2+}(aq) + CO_3^{2-}(aq) \rightarrow CaCO_3(s)$; by distillation; or by **ion exchange**, in which the water is passed through a column of a resin or an aluminosilicate (a zeolite, sometimes written Na_2Z) that swaps Ca^{2+} and Mg^{2+} for

Na^+: $Ca^{2+} + Na_2Z \rightarrow CaZ + 2Na^+$. Sodium carbonate and ion exchange remove *both* kinds of hardness. Detergents (soapless) do not form scum with hard water because their calcium salts are soluble.

Key point. Identify the type of hardness from the ions: hydrogencarbonates give temporary hardness (removed by boiling); sulphates and chlorides give permanent hardness (boiling does not help). Washing soda and ion-exchange resins remove both types. To find out whether a sample of water is hard, shake it with soap solution and see whether a lather forms.

Exercise 6.1

1. A gas occupies 1.5 dm^3 at 2.0 atm. What pressure is needed to compress it to 0.50 dm^3 at constant temperature?
2. 400 cm^3 of a gas at 7 °C is heated to 77 °C at constant pressure. Find the new volume.
3. The pressure of a gas in a sealed cylinder is 200 kPa at 27 °C. What is the pressure at 127 °C?
4. Calculate the volume at s.t.p. occupied by 11 g of carbon dioxide. (C = 12, O = 16)
5. A gas occupies 2.00 dm^3 at 273 °C and 202.6 kPa (about 2 atm). What volume does it occupy at s.t.p.?
6. How many times faster does methane (CH_4) diffuse than sulphur dioxide (SO_2)? (C = 12, H = 1, S = 32, O = 16)
7. A mixture of 0.4 mol of helium and 0.6 mol of neon exerts a total pressure of 500 kPa. What is the partial pressure of helium?
8. How many moles of gas are present in 5.0 dm^3 at 300 K and 249.4 kPa? ($R = 8.314$ kPa dm^3 mol^{-1} K^{-1})

9. The solubility of a salt is 30 g per 100 g of water at 50 °C and 12 g per 100 g at 10 °C. What mass of salt crystallises when 130 g of the saturated solution at 50 °C is cooled to 10 °C?
10. Calculate the density of nitrogen gas (N_2) at s.t.p. in g/dm^3. (N = 14)
11. A gas of mass 0.25 g occupies 100 cm^3 at s.t.p. Calculate its molar mass.
12. Explain why boiling removes temporary hardness but not permanent hardness, and name a reagent that removes both.

Answers to Exercise 6.1

1. $P_2 = P_1V_1/V_2 = (2.0 \times 1.5)/0.50 = 6.0$ atm.
2. $T_1 = 280$ K, $T_2 = 350$ K; $V = 400 \times 350/280 = 500$ cm^3.
3. $T_1 = 300$ K, $T_2 = 400$ K; $P = 200 \times 400/300 = 266.7$ kPa.
4. Moles = 11/44 = 0.25 mol; volume = $0.25 \times 22.4 = 5.6$ dm^3.
5. $T_1 = 546$ K; $V_2 = (202.6 \times 2.00 \times 273)/(546 \times 101.3) = 2.00 \times 2 \times 0.5 = 2.00$ dm^3 (pressure doubles the factor and temperature halves it, so they cancel).
6. Rate ratio = $\sqrt{(M(SO_2)/M(CH_4))} = \sqrt{(64/16)} = \sqrt{4} = 2$, so methane diffuses twice as fast.
7. Mole fraction of He = 0.4/1.0 = 0.4; partial pressure = $0.4 \times 500 = 200$ kPa.
8. $n = PV/RT = (249.4 \times 5.0)/(8.314 \times 300) = 1247/2494.2 = 0.50$ mol.
9. 130 g of saturated solution at 50 °C contains 30 g of salt and 100 g of water. At 10 °C the 100 g of water holds 12 g, so 30 − 12 = 18 g crystallises.
10. Density = 28/22.4 = 1.25 g/dm^3.
11. Moles = 0.100/22.4 = 0.004464 mol; M = 0.25/0.004464 = 56 g/mol.
12. Temporary hardness is due to hydrogencarbonates, which decompose on heating to insoluble carbonate; permanent hardness is due to sulphates and chlorides of calcium and magnesium, which are stable to heat. Washing soda (sodium carbonate), or an ion-exchange resin, removes both.

Chapter summary

- The kinetic theory describes particles in constant motion; for an ideal gas the average kinetic energy is proportional to the kelvin temperature. Always use $T(K) = \theta(°C) + 273$.
- Boyle's law: PV = constant at constant T. Charles's law: V/T = constant at constant P. Pressure law: P/T = constant at constant V. Combined: $P_1V_1/T_1 = P_2V_2/T_2$.
- Avogadro's law gives the molar volume of 22.4 dm^3 at s.t.p. and links volume ratios of gases with mole ratios; the ideal gas equation $PV = nRT$, with $R = 8.314$ J mol^{-1} K^{-1}, finds moles, molar mass and density.
- Dalton's law: total pressure is the sum of partial pressures; for a gas collected over water subtract the water's saturated vapour pressure.
- Graham's law: rate of diffusion is inversely proportional to $\sqrt{M}$; lighter gases diffuse faster.
- Solubility of most solids increases with temperature, that of gases decreases with temperature and increases with pressure; solubility curves show saturated, unsaturated and supersaturated regions.
- Hardness is caused by Ca^{2+} and Mg^{2+}: temporary (hydrogencarbonates) is removed by boiling; permanent (sulphates, chlorides) by washing soda, distillation or ion exchange.

Chapter 7: Acids, Bases, Salts and the pH Scale

The sour taste of lime and tamarind, the sting of ant bites, the soapy feel of wood-ash lye used by traditional soap makers, the tablets taken for indigestion, and the fertilisers spread on farms in Kaduna and Benue are all connected with acids, bases and salts. These substances occupy a central position in chemistry because so many reactions in the laboratory, in industry and in the living body are of the acid-base type. In this chapter you will study how acids and bases are defined, how their strength is distinguished from their concentration, how the pH scale measures acidity, how indicators and titrations work, and how salts are classified, prepared and recognised. The calculations on pH and titration are favourites in the UTME and reward careful practice.

7.1 Acids

The earliest definition describes an acid by its properties. An acid has a sour taste (never to be tested in the laboratory), turns blue litmus red, reacts with active metals to liberate hydrogen, reacts with carbonates to give carbon dioxide, and neutralises bases to form a salt and water. The deeper explanation came from Svante Arrhenius, who defined an **acid** as a substance that produces hydrogen ions, H^+, as the only positive ions when dissolved in water. In fact a bare proton does not exist alone in water; it attaches to a water molecule to form the **hydroxonium ion** (oxonium or hydronium ion), H_3O^+, so that, for hydrogen chloride,

$$HCl(g) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq).$$

For convenience, $H^+(aq)$ is used as shorthand for the hydroxonium ion.

Acids are classified by their **basicity**, the number of replaceable hydrogen atoms per molecule that can form H^+. Hydrochloric acid, nitric acid and ethanoic acid are monobasic (monoprotic), sulphuric acid and carbonic acid are dibasic, and phosphoric acid is tribasic. Note that ethanoic acid, CH_3COOH, has four hydrogen atoms but is monobasic, since only the hydrogen in the carboxyl group ionises. Acids are also classed as *mineral* (inorganic) acids, such as hydrochloric, nitric and sulphuric acids, and *organic* acids, such as ethanoic acid in vinegar, citric acid in oranges and lactic acid in sour milk.

Reactions of acids

- **With metals above hydrogen in the reactivity series**, to form a salt and hydrogen: $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$. (Nitric acid is an oxidising acid and usually gives oxides of nitrogen rather than hydrogen.)
- **With bases and alkalis**, to give a salt and water: $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$.
- **With carbonates and hydrogencarbonates**, to give a salt, water and carbon dioxide: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$. The effervescence is the standard test for a carbonate.

7.2 Bases and alkalis

A **base** is a substance that reacts with an acid to form a salt and water only. Most bases are oxides and hydroxides of metals, such as magnesium oxide, copper(II) oxide and iron(III) hydroxide. An **alkali** is a base that dissolves in water, giving hydroxide ions, OH^-, as the only negative ions. The common alkalis are sodium hydroxide, potassium hydroxide, calcium hydroxide (slightly soluble) and aqueous ammonia. All alkalis are bases, but not all bases are alkalis. Alkalis feel soapy, turn red litmus blue, and neutralise acids. Ammonia dissolves in water and ionises only slightly:

$NH_3(g) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$.

A **neutralisation** reaction is the reaction of an acid with a base to form a salt and water. In terms of ions, every neutralisation of a strong acid by a strong alkali reduces to a single ionic equation:

$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$.

Amphoteric oxides and hydroxides

An **amphoteric** substance reacts with both acids and bases. The principal examples are aluminium oxide and hydroxide, zinc oxide and hydroxide, and lead(II) oxide and hydroxide. For instance, zinc oxide reacts with hydrochloric acid and with sodium hydroxide:

$ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$
$ZnO + 2NaOH + H_2O \rightarrow Na_2Zn(OH)_4$ (sodium zincate).

Aluminium hydroxide similarly dissolves in excess sodium hydroxide to give a colourless solution of sodium aluminate; this behaviour is used in the tests for cations in Chapter 11.

7.3 The Brønsted-Lowry and Lewis ideas

Arrhenius's definitions apply only to aqueous solutions. Brønsted and Lowry proposed a wider idea: an **acid** is a proton (H^+) donor and a **base** is a proton acceptor. Every acid-base reaction therefore consists of a proton transfer. When hydrogen chloride dissolves in water, HCl donates a proton to H_2O, so HCl is the acid and water acts as a base. When ammonia dissolves, water donates a proton to NH_3, so water acts as the acid. A substance such as water that can behave both ways is *amphiprotic*.

When an acid gives up a proton, the species left is its **conjugate base**; when a base gains a proton, it becomes its **conjugate acid**. In $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$, NH_3 and NH_4^+ form a conjugate pair (base and acid), and so do H_2O and OH^- (acid and

base). A strong acid has a weak conjugate base, and a weak acid has a strong conjugate base.

Gilbert Lewis took the idea further: a **Lewis acid** is an electron-pair acceptor and a **Lewis base** is an electron-pair donor. The formation of the dative bond in the ammonium ion (Chapter 4) is an example, with H^+ as acceptor and ammonia as donor. Compounds like boron trifluoride, BF_3, are Lewis acids although they contain no hydrogen.

7.4 Strength and concentration

Two quite different ideas are often confused. The **concentration** of an acid says how much acid is dissolved in a given volume (in mol/dm^3), while the **strength** of an acid says how far it ionises in water. A **strong acid** ionises almost completely: HCl, HNO_3 and H_2SO_4 (the first ionisation). A **weak acid** ionises only partially, and a reversible equilibrium is established:

$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$.

Weak acids include ethanoic acid, carbonic acid, hydrogen sulphide and citric acid. Similarly, strong alkalis (NaOH, KOH, $Ba(OH)_2$) are fully ionised in solution, while ammonia solution is a weak base. Strength is measured by the ionisation (dissociation) constant K_a for an acid:

$K_a = [H^+][A^-]/[HA]$.

The larger the K_a, the stronger the acid. For ethanoic acid, $K_a = 1.8 \times 10^{-5}$ mol/dm^3 at 25 °C.

The two ideas are easily separated by experiment. A 0.1 mol/dm^3 solution of hydrochloric acid (strong) and one of ethanoic acid (weak) have equal concentrations, yet the former conducts electricity much better, has a lower pH, reacts more vigorously with magnesium and gives a faster reaction with marble chips. However, equal volumes of the two solutions need *the same* volume of

sodium hydroxide for neutralisation, because the weak acid keeps ionising as the hydrogen ions are used up, until all of it has reacted. A *dilute* solution contains little solute per unit volume and a *concentrated* solution a lot; a dilute solution of a strong acid can be much less acidic than a concentrated solution of a weak one.

> **Key point.** Strong and weak describe the degree of ionisation; concentrated and dilute describe the amount of solute per dm^3. Neutralisation calculations depend on the number of moles of acid and base, whatever their strength.

7.5 The pH scale

Pure water is very slightly ionised: $\mathrm{H_2O} \rightleftharpoons \mathrm{H^+} + \mathrm{OH^-}$. At 25 °C, the concentrations of $\mathrm{H^+}$ and $\mathrm{OH^-}$ in pure water are each 1.0×10^{-7} mol/dm^3. The product of the two concentrations in any aqueous solution at 25 °C is a constant, the **ionic product of water**:

$$K_w = [\mathrm{H^+}][\mathrm{OH^-}] = 1.0 \times 10^{-14}\ \mathrm{mol^2/dm^6}.$$

In an acid solution $[\mathrm{H^+}]$ exceeds 10^{-7}, and in an alkaline solution $[\mathrm{OH^-}]$ exceeds 10^{-7}. Since the concentrations vary over a huge range, Sørensen introduced a logarithmic scale. The **pH** of a solution is defined as

$$\mathrm{pH} = -\log_{10}[\mathrm{H^+}], \text{ so that } [\mathrm{H^+}] = 10^{-\mathrm{pH}},$$

and similarly $\mathrm{pOH} = -\log_{10}[\mathrm{OH^-}]$, with $\mathrm{pH} + \mathrm{pOH} = 14$ at 25 °C. A solution with pH less than 7 is acidic, pH equal to 7 is neutral, and pH greater than 7 is alkaline. Because the scale is logarithmic, a change of one pH unit is a tenfold change in $[\mathrm{H^+}]$: a solution of pH 3 is ten times as acidic as one of pH 4 and a hundred times as acidic as one of pH 5.

Substance	Approximate pH
Gastric juice (stomach acid)	1.5 to 2
Lime juice	2 to 2.5
Vinegar	about 3
Rainwater (unpolluted)	about 5.6
Pure water, sodium chloride solution	7
Blood	about 7.4
Baking soda solution	about 8.3
Milk of magnesia	about 10.5
Household bleach, oven cleaner	12 to 13

WORKED EXAMPLE 7.1

Calculate the pH of (a) 0.010 mol/dm^3 hydrochloric acid, (b) 0.050 mol/dm^3 sulphuric acid, (c) 0.010 mol/dm^3 sodium hydroxide.

Solution: (a) HCl is monobasic and strong, so $[H^+] = 0.010 = 10^{-2}$ and pH = 2. (b) H_2SO_4 is dibasic; assuming complete ionisation, $[H^+] = 2 \times 0.050 = 0.10 = 10^{-1}$, so pH = 1. (c) NaOH is fully ionised, so $[OH^-] = 10^{-2}$, pOH = 2 and pH = 14 − 2 = 12. Alternatively $[H^+] = 10^{-14}/10^{-2} = 10^{-12}$, pH = 12.

WORKED EXAMPLE 7.2

10.0 cm^3 of 0.10 mol/dm^3 hydrochloric acid is diluted to 1.00 dm^3 with distilled water. Find the new pH.

Solution: Moles of H^+ = 0.0100 dm^3 × 0.10 = 1.0×10^{-3} mol. New concentration = $1.0 \times 10^{-3}/1.00 = 1.0 \times 10^{-3}$ mol/dm^3, so pH = 3. The dilution factor of 100 raised the pH from 1 to 3, that is, by two units.

WORKED EXAMPLE 7.3

Calculate the pH of 0.010 mol/dm^3 ethanoic acid, given K_a = 1.0 × 10^{-5} mol/dm^3. (Assume that very little of the acid ionises.)

Solution: For HA ⇌ H$^+$ + A$^-$, [H$^+$] = [A$^-$] = x and [HA] ≈ 0.010. Then K_a = x^2/0.010, so x^2 = 1.0 × 10^{-5} × 0.010 = 1.0 × 10^{-7} and x = 3.16 × 10^{-4} mol/dm^3. pH = −log(3.16 × 10^{-4}) = 3.50. The ionised fraction is 3.16 × 10^{-4}/0.010 = 3.2%, which justifies the assumption.

7.6 Indicators

An **indicator** is a substance, usually a weak organic acid or base, that has different colours in acidic and alkaline solutions. The colour change occurs over a narrow pH range. The most useful ones are:

Indicator	Colour in acid	Colour in alkali	Approximate range
Litmus	Red	Blue	5 to 8
Methyl orange	Red	Yellow	3.1 to 4.4
Methyl red	Red	Yellow	4.4 to 6.2
Bromothymol blue	Yellow	Blue	6.0 to 7.6
Phenolphthalein	Colourless	Pink	8.3 to 10

Universal indicator is a mixture that changes through red, orange, yellow, green, blue and violet as the pH rises from 1 to 14, and so gives an approximate pH. Natural indicators can be made from the pigments of hibiscus (zobo) flowers, red cabbage and turmeric, which turn from red to green or yellow in alkali. In a titration, the indicator must change colour sharply at the pH of the equivalence point, as Section 7.7 shows.

7.7 Acid-base titrations

In a titration, a solution of known concentration (the *standard solution*) is added from a burette to a measured volume of the other solution (pipetted into a conical flask) until the reaction is exactly complete. The point at which the acid and base have reacted in exactly the proportions of the equation is the **equivalence point**, and the point at which the indicator changes colour is the **end point**. The method and the arithmetic were developed in Chapter 5; here we concentrate on the choice of indicator.

- **Strong acid with strong base** (HCl with NaOH): the pH at equivalence is 7 and the pH changes sharply from about 3 to 11 near the end point, so methyl orange, litmus or phenolphthalein can all be used.
- **Weak acid with strong base** (ethanoic acid with NaOH): the salt formed hydrolyses to give an alkaline solution at equivalence (pH above 7), so **phenolphthalein** is used; methyl orange would change colour too early.
- **Strong acid with weak base** (HCl with aqueous ammonia): the salt gives an acidic solution at equivalence (pH below 7), so **methyl orange** is used.
- **Weak acid with weak base**: there is no sharp change of pH, so no simple indicator is suitable.

WORKED EXAMPLE 7.4

25.0 cm^3 of sodium hydroxide solution required 20.0 cm^3 of 0.10 mol/dm^3 sulphuric acid for complete neutralisation. Calculate the concentration of the alkali in mol/dm^3 and in g/dm^3. (Na = 23, O = 16, H = 1)

Solution: $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$. Moles of acid = $0.0200 \times 0.10 = 2.0 \times 10^{-3}$ mol. Moles of NaOH = $2 \times 2.0 \times 10^{-3} = 4.0 \times 10^{-3}$ mol in 0.0250 dm^3. Concentration = $4.0 \times 10^{-3} / 0.0250 = 0.16$ mol/dm^3. In g/dm^3: $0.16 \times 40 = 6.4$ g/dm^3.

WORKED EXAMPLE 7.5

20.0 cm^3 of 0.10 mol/dm^3 HCl is mixed with 30.0 cm^3 of 0.10 mol/dm^3 NaOH. Find the pH of the mixture.

Solution: Moles HCl = $0.0200 \times 0.10 = 2.0 \times 10^{-3}$; moles NaOH = $0.0300 \times 0.10 = 3.0 \times 10^{-3}$. The acid is used up and 1.0×10^{-3} mol of NaOH remains in a total volume of 50.0 cm^3. $[OH^-] = 1.0 \times 10^{-3}/0.0500 = 0.020$ mol/dm^3. pOH = $-\log(0.020) = 1.70$, so pH = $14 - 1.70 = 12.30$.

7.8 Salts

A **salt** is a compound formed when the hydrogen of an acid is wholly or partly replaced by a metal ion or the ammonium ion. Salts are named from the metal and the acid they come from: sulphates from sulphuric acid, nitrates from nitric acid, chlorides from hydrochloric acid, carbonates from carbonic acid and ethanoates from ethanoic acid.

Classes of salts

- **Normal salts** result from the replacement of *all* the replaceable hydrogen of the acid: NaCl, Na_2SO_4, $CaCO_3$, K_3PO_4.
- **Acid salts** result from partial replacement and still contain ionisable hydrogen: sodium hydrogensulphate $NaHSO_4$, sodium hydrogencarbonate $NaHCO_3$. Their solutions can still react as acids ($NaHSO_4$ turns blue litmus red).
- **Basic salts** contain hydroxide or oxide ions as well as the acid anion, such as basic zinc carbonate or basic lead carbonate (white lead).
- **Double salts** crystallise from a solution of two simple salts and contain two different cations, for example alum, $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$, and Mohr's salt, $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$. In solution they behave as a mixture of their two component salts.

- **Complex salts** contain a complex ion that stays intact in solution, for example potassium hexacyanoferrate(II), $K_4Fe(CN)_6$, and tetraamminecopper(II) sulphate, $[Cu(NH_3)_4]SO_4$.
- **Hydrated salts** contain water of crystallisation, such as $CuSO_4 \cdot 5H_2O$ and $Na_2CO_3 \cdot 10H_2O$.

Solubility rules

The method used to prepare a salt depends on its solubility, so the following rules should be memorised:

- All sodium, potassium and ammonium salts are soluble.
- All nitrates are soluble.
- All common chlorides are soluble, except silver chloride and lead(II) chloride (soluble in hot water).
- All common sulphates are soluble, except barium sulphate, lead(II) sulphate and calcium sulphate (slightly soluble).
- All common carbonates are insoluble, except those of sodium, potassium and ammonium.
- All common hydroxides are insoluble, except those of sodium, potassium and barium; calcium hydroxide is slightly soluble.

7.9 Preparation of salts

Soluble salts from an acid are made by reacting the acid with an excess of (a) a metal (for example zinc and dilute sulphuric acid, giving zinc sulphate), (b) an insoluble base (copper(II) oxide with sulphuric acid), or (c) an insoluble carbonate (magnesium carbonate with hydrochloric acid). The excess solid is filtered off, the filtrate is evaporated to the point of crystallisation, and left to cool so that crystals separate. Sodium, potassium and ammonium salts cannot be made from the metals or insoluble bases, because sodium and potassium react dangerously and their

oxides and carbonates are soluble. They are prepared by **titration**, adding the acid to the alkali with an indicator until neutral, then repeating without indicator and crystallising.

Insoluble salts are made by **precipitation**: mixing two soluble salts whose ions combine to form the insoluble one. For example, lead(II) iodide: $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$, or ionically $Pb^{2+}(aq) + 2I^-(aq) \rightarrow PbI_2(s)$, a bright yellow solid. The precipitate is filtered, washed with distilled water and dried.

Anhydrous chlorides of iron(III) and aluminium are prepared by **direct combination** of the metal with dry chlorine, since they cannot be made in water: $2Fe + 3Cl_2 \rightarrow 2FeCl_3$. Other methods include the action of an acid on a salt of a volatile acid (for example, $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$).

WORKED EXAMPLE 7.6

What mass of hydrated copper(II) sulphate crystals, $CuSO_4·5H_2O$, can be obtained from 4.0 g of copper(II) oxide, assuming complete conversion? (Cu = 64, O = 16, S = 32, H = 1)

Solution: $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$. $M_r(CuO) = 80$, so moles of CuO = 4.0/80 = 0.050 mol, which gives 0.050 mol of $CuSO_4$, and hence 0.050 mol of $CuSO_4·5H_2O$. Its $M_r = 64 + 32 + 64 + 5 \times 18 = 250$. Mass = $0.050 \times 250 = 12.5$ g.

7.10 Hydrolysis of salts

Dissolving a salt in water does not always give a neutral solution. **Salt hydrolysis** is the reaction of the ions of a salt with water to give an acidic or alkaline solution. It depends on the strength of the acid and base from which the salt was derived:

- **Strong acid and strong base** (NaCl, KNO_3, Na_2SO_4): neither ion reacts with water, and the solution is neutral (pH 7).
- **Weak acid and strong base** (Na_2CO_3, CH_3COONa): the anion takes a proton from water, leaving excess OH^-, so the solution is alkaline: $CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$. This is why washing soda solution feels soapy.
- **Strong acid and weak base** (NH_4Cl, $CuSO_4$, $FeCl_3$, $AlCl_3$): the cation donates a proton to water, forming H_3O^+, and the solution is acidic: $NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$.

Salts of a weak acid and a weak base, such as ammonium ethanoate, give an almost neutral solution whose exact pH depends on the relative strength of the two. This explains why aqueous copper(II) sulphate turns blue litmus red although it is a salt.

7.11 Buffer solutions

A **buffer solution** resists changes in pH when small amounts of acid or alkali are added. It contains a weak acid together with its salt (for example ethanoic acid with sodium ethanoate, an *acidic buffer*) or a weak base together with its salt (ammonia with ammonium chloride, an *alkaline buffer*). If acid is added, the large reserve of the anion, CH_3COO^-, combines with the extra H^+; if alkali is added, the weak acid supplies H^+ to neutralise it. Blood is buffered by the carbonic acid-hydrogencarbonate system so that its pH stays near 7.4; a deviation of a few tenths would be fatal. Buffers are also important in the food industry and in the treatment of wastewater.

7.12 Uses of acids, bases and salts

Sulphuric acid is the most widely used industrial chemical, serving in fertilisers, detergents, batteries and paints. Hydrochloric acid cleans metals before galvanising, nitric acid is the starting material for fertilisers and explosives, and ethanoic acid is the acid in vinegar. Sodium hydroxide is used for soap making, paper, textiles and

the refining of bauxite. Magnesium hydroxide (milk of magnesia) and aluminium hydroxide relieve indigestion by neutralising excess stomach acid, and lime (calcium hydroxide) is used to reduce the acidity of farm soil. Among salts, sodium chloride preserves and seasons food, sodium hydrogencarbonate is baking powder, ammonium nitrate and ammonium sulphate are nitrogenous fertilisers, and calcium sulphate (plaster of Paris) sets fractures.

Exercise 7.1

1. Calculate the pH of 0.0010 mol/dm^3 nitric acid.
2. Calculate the pH of 0.050 mol/dm^3 sodium hydroxide. (log 5 = 0.70)
3. What is the hydrogen-ion concentration of a solution of pH 5?
4. Calculate the pH of 0.025 mol/dm^3 sulphuric acid, assuming complete ionisation. (log 5 = 0.70)
5. What volume of 0.10 mol/dm^3 hydrochloric acid will neutralise 25 cm^3 of 0.20 mol/dm^3 sodium hydroxide?
6. Calculate the mass of zinc sulphate, $ZnSO_4$, obtainable from 1.3 g of zinc and excess dilute sulphuric acid. (Zn = 65, S = 32, O = 16)
7. Find the pH of the solution formed by mixing 20 cm^3 of 0.10 mol/dm^3 HCl with 30 cm^3 of 0.10 mol/dm^3 NaOH solution. (log 2 = 0.30)
8. Calculate the mass of silver chloride precipitated when excess sodium chloride is added to 50 cm^3 of 0.10 mol/dm^3 silver nitrate. (Ag = 108, Cl = 35.5)
9. Calculate the pH of 0.10 mol/dm^3 of a weak monobasic acid with $K_a = 1.0 \times 10^{-5}$ mol/dm^3.
10. State whether aqueous solutions of (a) sodium carbonate, (b) ammonium chloride and (c) sodium chloride are acidic, alkaline or neutral, and explain the first two.
11. Write the ionic equation for the reaction between dilute hydrochloric acid and sodium carbonate.

12. Which indicator, methyl orange or phenolphthalein, would you use for the titration of ethanoic acid with sodium hydroxide, and why?

Answers to Exercise 7.1

1. $[H^+] = 10^{-3}$, so pH = 3.
2. $[OH^-] = 0.050$, pOH = $-\log(5 \times 10^{-2}) = 2 - 0.70 = 1.30$; pH = $14 - 1.30 = 12.70$.
3. $[H^+] = 10^{-5}$ mol/dm^3.
4. $[H^+] = 2 \times 0.025 = 0.050$; pH = $-\log(5 \times 10^{-2}) = 1.30$.
5. NaOH moles = $0.025 \times 0.20 = 5.0 \times 10^{-3}$; HCl needed = $5.0 \times 10^{-3}/0.10 = 0.050$ dm^3 = 50 cm^3.
6. Zn moles = 1.3/65 = 0.020 mol; $M_r(ZnSO_4) = 65 + 32 + 64 = 161$; mass = $0.020 \times 161 = 3.22$ g.
7. NaOH in excess: $3.0 \times 10^{-3} - 2.0 \times 10^{-3} = 1.0 \times 10^{-3}$ mol in 0.050 dm^3, so $[OH^-] = 0.020$, pOH = $2 - 0.30 = 1.70$, pH = 12.30.
8. $AgNO_3$ moles = $0.050 \times 0.10 = 5.0 \times 10^{-3}$; 1 : 1 with AgCl ($M_r = 143.5$); mass = $5.0 \times 10^{-3} \times 143.5 = 0.7175$ g, about 0.72 g.
9. $[H^+] = \sqrt{(K_a \times C)} = \sqrt{(1.0 \times 10^{-5} \times 0.10)} = \sqrt{(10^{-6})} = 10^{-3}$; pH = 3.
10. (a) Alkaline: $CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$ (weak acid, strong base). (b) Acidic: $NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$ (strong acid, weak base). (c) Neutral (strong acid, strong base).
11. $2H^+(aq) + CO_3^{2-}(aq) \rightarrow H_2O(l) + CO_2(g)$.
12. Phenolphthalein: the salt sodium ethanoate is alkaline at the equivalence point, and phenolphthalein changes colour in the pH range 8.3 to 10, which contains the steep part of the curve.

Chapter summary

- An Arrhenius acid gives H^+ (H_3O^+) in water; an alkali gives OH^-. Brønsted-Lowry: acids donate protons, bases accept them; conjugate pairs differ by one proton. Lewis: acids accept electron pairs, bases donate them.
- Amphoteric oxides and hydroxides (Al, Zn, Pb) react with both acids and alkalis.
- Strength is the degree of ionisation (measured by K_a); concentration is the amount per dm^3. Equal volumes of equally concentrated strong and weak monobasic acids need equal volumes of alkali for neutralisation.
- $pH = -\log[H^+]$; $K_w = [H^+][OH^-] = 10^{-14}$ and $pH + pOH = 14$ at 25 °C.
- Choose the indicator to match the equivalence point: methyl orange for strong acid with weak base, phenolphthalein for weak acid with strong base, any of these for strong with strong.
- Salts may be normal, acid, basic, double, complex or hydrated. Use solubility rules to choose the method of preparation: acid with metal, base or carbonate; titration for Na, K and ammonium salts; precipitation for insoluble salts; direct combination for anhydrous $FeCl_3$ and $AlCl_3$.
- Hydrolysis makes solutions of salts of weak acids with strong bases alkaline and those of strong acids with weak bases acidic. Buffers resist pH change because they contain a weak acid (or base) and its salt.

Chapter 8: Oxidation, Reduction and Electrochemistry

Every time a generator engine burns petrol in a Lagos neighbourhood during a power cut, every time a nail rusts on a zinc roof in Port Harcourt, every time a phone battery is charged and every time a jeweller coats a bracelet with gold, electrons are being transferred from one substance to another. Reactions of this kind, in which electrons move, are called oxidation-reduction or **redox** reactions, and they form the basis of electrochemistry, the study of the relationship between chemical change and electrical energy. In this chapter you will learn how to recognise oxidation and reduction, how to assign oxidation numbers, how to balance redox equations, how electricity can be used to bring about chemical change (electrolysis), how chemical change can in turn be used to produce electricity (electrochemical cells), and how the same ideas explain corrosion. Calculations based on Faraday's laws are common in the UTME, and they become routine once the underlying logic is understood.

8.1 The meaning of oxidation and reduction

The terms oxidation and reduction have been defined in three successively broader ways, and a candidate should be comfortable with all three.

- **In terms of oxygen:** oxidation is the gain of oxygen by a substance, and reduction is the loss of oxygen. When magnesium burns, $2Mg + O_2 \rightarrow 2MgO$, magnesium is oxidised; when copper(II) oxide is heated in a stream of hydrogen, $CuO + H_2 \rightarrow Cu + H_2O$, the copper oxide is reduced.
- **In terms of hydrogen:** oxidation is the loss of hydrogen, and reduction is the gain of hydrogen. When hydrogen sulphide reacts with chlorine, $H_2S + Cl_2 \rightarrow 2HCl + S$, the hydrogen sulphide has lost hydrogen and is oxidised, while chlorine has gained hydrogen and is reduced.

- **In terms of electrons:** oxidation is the loss of electrons, and reduction is the gain of electrons. This is the most general definition and applies even where neither oxygen nor hydrogen is involved. The memory aid *OIL RIG* (Oxidation Is Loss, Reduction Is Gain) is widely used.

Consider zinc dropped into copper(II) sulphate solution. The blue colour fades and a reddish-brown deposit of copper appears. The ionic equation is $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$. It can be split into two **half-equations**:

$Zn \rightarrow Zn^{2+} + 2e^{-}$ (oxidation: zinc loses electrons)
$Cu^{2+} + 2e^{-} \rightarrow Cu$ (reduction: copper(II) ions gain electrons)

Oxidation and reduction always occur together; one cannot happen without the other, because electrons lost by one species must be gained by another. The overall process is therefore called a redox reaction.

> **Key point.** An *oxidising agent* is the substance that accepts electrons and is itself reduced. A *reducing agent* is the substance that donates electrons and is itself oxidised. In the reaction above, Cu^{2+} is the oxidising agent and Zn is the reducing agent. Candidates frequently confuse the agent with the substance oxidised; remember that the agent does the opposite to itself.

8.2 Oxidation numbers

When electrons are transferred only partially, as in covalent compounds, it is hard to say how many electrons an atom has lost or gained. The **oxidation number** (oxidation state) is a bookkeeping device that solves this problem. It is the charge an atom would have if all the shared electrons in its bonds were assigned to the more electronegative atom. The following rules are applied in order.

1. The oxidation number of an atom in an uncombined element is zero (Na, O_2, S_8, P_4).

2. The oxidation number of a simple monatomic ion equals its charge (Na^+ is +1, Cl^- is −1, Fe^{3+} is +3).
3. The sum of the oxidation numbers in a neutral compound is zero; in a polyatomic ion it equals the charge on the ion.
4. Group 1 metals are always +1, Group 2 metals always +2, and aluminium is +3.
5. Hydrogen is +1 in most compounds, but −1 in metal hydrides such as NaH.
6. Oxygen is −2 in most compounds, but −1 in peroxides (H_2O_2, Na_2O_2) and +2 in OF_2.
7. Fluorine is always −1; other halogens are −1 except when combined with oxygen or a more electronegative halogen.

WORKED EXAMPLE 8.1

Find the oxidation number of (a) manganese in $KMnO_4$, (b) chromium in $K_2Cr_2O_7$, (c) sulphur in the sulphate ion SO_4^{2-}, (d) nitrogen in NH_4^+.

Solution: (a) K is +1 and four oxygen atoms contribute 4(−2) = −8. Let Mn be x: +1 + x − 8 = 0, so x = +7.

(b) Two K contribute +2 and seven O contribute −14. With two Cr atoms: +2 + 2x − 14 = 0, so 2x = 12 and x = +6.

(c) Four O contribute −8, and the total charge is −2: x − 8 = −2, so x = +6.

(d) Four H contribute +4, and the total charge is +1: x + 4 = +1, so x = −3.

Using oxidation numbers to identify redox

A reaction is a redox reaction if, and only if, the oxidation number of at least one element changes. An *increase* in oxidation number is oxidation; a *decrease* is reduction. In $2FeCl_2 + Cl_2 \rightarrow 2FeCl_3$, iron rises from +2 to +3 (oxidised) and chlorine falls from 0 to −1 (reduced). By contrast, in $NaOH + HCl \rightarrow NaCl + H_2O$ no element changes oxidation number, so neutralisation is not a redox process. A

reaction in which the same element is both oxidised and reduced is called **disproportionation**; for instance, chlorine in cold dilute sodium hydroxide, $Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$, goes from 0 to −1 and to +1 at the same time.

Oxidation numbers also explain the Roman numerals in names. Iron(II) sulphate contains Fe^{2+}, and iron(III) chloride contains Fe^{3+}; the numeral is the oxidation number of the metal. Names such as manganate(VII) for MnO_4^- and chromate(VI) for CrO_4^{2-} follow the same pattern.

8.3 Common oxidising and reducing agents

Good oxidising agents are substances whose atoms readily accept electrons or contain an element in a high oxidation state. Common examples are oxygen, the halogens (especially fluorine and chlorine), acidified potassium manganate(VII) (purple, turning almost colourless as Mn^{2+} forms), acidified potassium dichromate(VI) (orange, turning green as Cr^{3+} forms), concentrated nitric acid, hydrogen peroxide and concentrated sulphuric acid. Good reducing agents readily give up electrons or contain an element in a low oxidation state. Examples are the reactive metals (sodium, magnesium, zinc), hydrogen, carbon and carbon monoxide, hydrogen sulphide, sulphur dioxide, iron(II) salts and the iodide ion.

Several colour changes serve as qualitative tests. Sulphur dioxide turns acidified dichromate from orange to green; iodide ions are oxidised by chlorine to brown iodine, which gives a blue-black colour with starch; and iron(II) ions (pale green) are oxidised to iron(III) (yellow-brown) by manganate(VII) or by exposure to air.

Balancing redox equations by the half-equation method

For reactions in acidic solution the procedure is: write each half-equation; balance atoms other than oxygen and hydrogen; balance oxygen by adding H_2O; balance hydrogen by adding H^+; balance charge by adding electrons; multiply the half-equations so the electrons cancel; and add them. For the reaction between manganate(VII) and iron(II):

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

$Fe^{2+} \rightarrow Fe^{3+} + e^-$ (multiply by 5)

Adding gives $MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$. Check: the charge on the left is −1 + 8 + 10 = +17 and on the right is +2 + 15 = +17, so charge is balanced.

WORKED EXAMPLE 8.2

A 25.0 cm^3 portion of an iron(II) sulphate solution required 20.0 cm^3 of 0.0200 mol dm^{-3} potassium manganate(VII) for complete oxidation in acid solution. Calculate the concentration of the iron(II) solution.

Solution: Amount of MnO_4^- = 0.0200 × 20.0/1000 = 4.00 × 10^{-4} mol. From the equation, 1 mol MnO_4^- reacts with 5 mol Fe^{2+}, so Fe^{2+} = 5 × 4.00 × 10^{-4} = 2.00 × 10^{-3} mol. This is contained in 25.0 cm^3, so concentration = 2.00 × 10^{-3} × 1000/25.0 = 0.0800 mol dm^{-3}.

Manganate(VII) is its own indicator in this titration: the first permanent faint pink colour marks the end point, since the purple ion is colourless once reduced to Mn^{2+}. Hydrochloric acid must not be used to acidify, because manganate(VII) would oxidise chloride ions to chlorine and give a falsely large titre; dilute sulphuric acid is used.

8.4 Electrolysis: basic ideas

A substance conducts electricity if it contains freely moving charged particles. Metals conduct through mobile electrons, and conduction produces no chemical change. Ionic compounds conduct only when molten or dissolved, because then the ions are free to move; in this case conduction is accompanied by chemical change at

the electrodes. **Electrolysis** is the decomposition of a molten or dissolved ionic compound by the passage of an electric current. The compound decomposed is the **electrolyte**; the conductors that carry current into and out of it are the **electrodes**.

The **anode** is the positive electrode, connected to the positive terminal of the supply; it attracts anions, and oxidation occurs there. The **cathode** is the negative electrode; it attracts cations, and reduction occurs there. (*AnOx* and *RedCat*: anode-oxidation, reduction-cathode.) Electrons travel through the external wire from anode to cathode. Electrolytes are classed as **strong** (almost fully ionised: salts, strong acids, strong alkalis) or **weak** (partly ionised: ethanoic acid, aqueous ammonia). Non-electrolytes such as sugar solution, ethanol and pure water in practice do not conduct. Inert electrodes (graphite or platinum) merely carry current, while active electrodes (such as copper) may take part in the reaction.

Electrolysis of molten compounds

Only two kinds of ions are present, so the products are easy to predict. In molten lead(II) bromide the cathode half-equation is $Pb^{2+} + 2e^- \rightarrow Pb$, giving a bead of grey molten lead, and the anode half-equation is $2Br^- \rightarrow Br_2 + 2e^-$, giving brown bromine vapour. In molten sodium chloride, sodium metal forms at the cathode and chlorine gas at the anode; this is how sodium is manufactured in the Downs cell.

8.5 Electrolysis of aqueous solutions: selective discharge

An aqueous solution contains, besides the ions of the dissolved compound, a small number of H^+ and OH^- ions from the ionisation of water. At each electrode there is therefore competition, and only one ion is discharged; this is **selective (preferential) discharge**. Three factors decide the outcome.

1. **Position in the electrochemical series.** At the cathode, the metal lower in the series (less reactive) is discharged in preference to hydrogen, so ions such as Cu^{2+}, Ag^+ and Pb^{2+} are discharged before H^+, whereas H^+ is discharged in

preference to Na^+, K^+, Ca^{2+}, Mg^{2+} and Al^{3+}. At the anode, the usual order of preference is I^-, Br^-, Cl^-, then OH^-, with the sulphate, nitrate and similar oxo-anions practically never discharged.

2. **Concentration.** A high concentration favours discharge of an ion. In concentrated sodium chloride solution (brine), chloride is discharged at the anode in preference to hydroxide even though OH^- is nominally lower in the series; in very dilute chloride solution, oxygen is evolved instead.
3. **Nature of the electrode.** Active electrodes can themselves dissolve or be deposited, changing the products.

Important cases

- **Dilute sulphuric acid (platinum or graphite electrodes).** Cathode: $2H^+ + 2e^- \rightarrow H_2$. Anode: $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$. Hydrogen and oxygen are evolved in the volume ratio 2 : 1; effectively water is decomposed and the acid becomes more concentrated.
- **Copper(II) sulphate solution with inert electrodes.** Cathode: $Cu^{2+} + 2e^- \rightarrow Cu$, a pink-brown deposit. Anode: oxygen is evolved. The blue colour fades and the solution becomes acidic because H^+ and SO_4^{2-} remain.
- **Copper(II) sulphate with copper electrodes.** The anode dissolves, $Cu \rightarrow Cu^{2+} + 2e^-$, and copper is deposited at the cathode. The blue colour stays constant, the anode loses mass and the cathode gains the same mass. This is the principle of copper purification and electroplating.
- **Concentrated sodium chloride solution (brine).** Chlorine is evolved at the anode and hydrogen at the cathode; the solution left behind contains Na^+ and OH^-, that is, sodium hydroxide. This is the basis of the chlor-alkali industry.
- **Dilute sodium chloride solution or sodium sulphate solution.** Hydrogen at the cathode and oxygen at the anode; again water is decomposed.

Key point. Always list every ion present, including H^+ and OH^- from water, before deciding what is discharged. Then apply the three factors in turn. Examination questions often change the concentration or the electrode to catch candidates who rely on memorised products.

8.6 Faraday's laws of electrolysis

Michael Faraday showed that the extent of chemical change in electrolysis is strictly proportional to the quantity of electricity passed. His two laws are as follows.

- **First law:** the mass of a substance liberated at an electrode is directly proportional to the quantity of electricity (charge) that passes through the electrolyte.
- **Second law:** when the same quantity of electricity passes through different electrolytes, the masses of substances liberated are proportional to their equivalent masses (relative atomic mass divided by the charge on the ion).

The quantity of electricity is measured in coulombs: $Q = I \times t$, where I is current in amperes and t is time in seconds. One **faraday** (1 F) is the quantity of electricity carried by one mole of electrons, and it equals approximately 96 500 C (precisely 96 485 C mol^{-1}; the value 96 500 is used in the UTME). It follows that 1 mol of a metal ion M^{n+} needs n faradays for discharge: 1 F deposits 1 mol of Ag (108 g) from Ag^+, 2 F deposit 1 mol of Cu (63.5 g) from Cu^{2+}, and 3 F deposit 1 mol of Al (27 g) from Al^{3+}. For gases, 4 F liberate 1 mol of O_2, 2 F liberate 1 mol of H_2 or Cl_2, and each mole of gas occupies 22.4 dm^3 at s.t.p.

A reliable method for every calculation is: (1) find $Q = It$; (2) find moles of electrons $= Q/96\,500$; (3) use the half-equation to convert moles of electrons into moles of product; (4) convert to mass or volume.

WORKED EXAMPLE 8.3

A current of 2.0 A is passed through copper(II) sulphate solution for 30 minutes using inert electrodes. Calculate the mass of copper deposited. (Cu = 64, 1 F = 96 500 C)

Solution: t = 30 × 60 = 1800 s, so Q = 2.0 × 1800 = 3600 C. Moles of electrons = 3600/96 500 = 0.03731 mol. $Cu^{2+} + 2e^{-} \rightarrow Cu$, so moles of Cu = 0.03731/2 = 0.01865 mol. Mass = 0.01865 × 64 = 1.19 g.

WORKED EXAMPLE 8.4

How long will it take a current of 3.0 A to deposit 5.4 g of silver from silver nitrate solution? (Ag = 108)

Solution: Moles of Ag = 5.4/108 = 0.050 mol. $Ag^{+} + e^{-} \rightarrow Ag$, so 0.050 mol of electrons is needed. Q = 0.050 × 96 500 = 4825 C. Time = Q/I = 4825/3.0 = 1608 s, which is about 26.8 minutes (26 min 48 s).

WORKED EXAMPLE 8.5

The same quantity of electricity is passed through copper(II) sulphate solution and silver nitrate solution connected in series. If 0.635 g of copper is deposited in the first cell, what mass of silver is deposited in the second? (Cu = 63.5, Ag = 108)

Solution: Moles of Cu = 0.635/63.5 = 0.0100 mol, which required 2 × 0.0100 = 0.0200 mol of electrons. The same 0.0200 mol of electrons passes through the silver cell, and since one electron deposits one Ag atom, moles of Ag = 0.0200 mol. Mass = 0.0200 × 108 = 2.16 g.

WORKED EXAMPLE 8.6

Dilute sulphuric acid is electrolysed with platinum electrodes using a current of 2.0 A for 965 s. Calculate the volumes of hydrogen and oxygen liberated at s.t.p. (molar volume 22.4 dm^3)

Solution: Q = 2.0 × 965 = 1930 C; moles of electrons = 1930/96 500 = 0.0200 mol. Cathode: $2H^+ + 2e^- \rightarrow H_2$, so moles of H_2 = 0.0100 mol, volume = 0.0100 × 22.4 = 0.224 dm^3 = 224 cm^3. Anode: $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$, so moles of O_2 = 0.0200/4 = 0.00500 mol, volume = 0.00500 × 22.4 = 0.112 dm^3 = 112 cm^3. The ratio 224 : 112 = 2 : 1 agrees with the formula of water.

8.7 Applications of electrolysis

- **Electroplating.** The article to be plated is made the cathode, the plating metal is made the anode, and the electrolyte is a salt of the plating metal. Chromium plating of car bumpers, silver plating of cutlery and tin plating of steel food cans are typical. A thin, even, adherent coat is obtained at low current density.
- **Purification of copper.** Impure copper is the anode and a thin sheet of pure copper is the cathode in acidified copper(II) sulphate solution. Copper dissolves from the anode and is deposited as pure metal on the cathode, while less reactive impurities such as silver and gold fall as anode sludge and more reactive ones such as zinc and iron stay in solution. Copper for electrical wiring must be 99.9 % pure because impurities greatly increase resistance.
- **Extraction of reactive metals.** Sodium, potassium, calcium and magnesium are obtained by electrolysis of their molten chlorides. Aluminium is obtained by electrolysing purified alumina (Al_2O_3, from bauxite) dissolved in molten cryolite (Na_3AlF_6) at about 900 °C; the cryolite lowers the melting point from over 2000 °C and improves conductivity. Aluminium forms at the cathode

(carbon lining) and oxygen at the carbon anodes, which burn away and must be replaced.

- **Chlor-alkali industry.** Electrolysis of brine in the diaphragm or membrane cell yields chlorine, hydrogen and sodium hydroxide, each of great industrial value. The mercury cell has been largely abandoned because of mercury pollution.
- **Anodising** of aluminium, which thickens its protective oxide layer, and manufacture of hydrogen and oxygen from water are further uses.

8.8 Electrochemical (galvanic) cells

Electrolysis uses electricity to drive a non-spontaneous reaction. An **electrochemical cell** works the other way round: a spontaneous redox reaction is arranged so that the electrons pass through an external circuit and do useful work. In the Daniell cell, a zinc rod dips into zinc sulphate solution and a copper rod into copper(II) sulphate solution; the two solutions are linked by a **salt bridge** (filter paper soaked in potassium nitrate, or a U-tube of agar gel) which completes the circuit by allowing ions to migrate and keeps each half-cell electrically neutral. Zinc is the more reactive metal and has the greater tendency to lose electrons, so it forms the negative terminal. Anode (negative): $\mathrm{Zn \rightarrow Zn^{2+} + 2e^-}$. Cathode (positive): $\mathrm{Cu^{2+} + 2e^- \rightarrow Cu}$. Notice that in a galvanic cell the anode is the negative electrode, the opposite of the electrolytic cell, though oxidation still occurs at the anode in both. The cell is written in shorthand as $\mathrm{Zn(s) \mid Zn^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s)}$, with the anode on the left and the double bar representing the salt bridge.

Standard electrode potentials

The tendency of a half-cell to gain electrons is measured by its **standard electrode potential**, E°, in volts. It is the potential of the half-cell measured against the standard hydrogen electrode (platinum coated with platinum black, immersed in 1.0

mol dm^{-3} H^+, hydrogen gas at 1 atmosphere pressure and 25 °C), which is arbitrarily assigned 0.00 V. Metals arranged in order of E° form the **electrochemical series**.

Half-reaction (reduction)	E° / V
$K^+ + e^- \rightleftharpoons K$	−2.92
$Na^+ + e^- \rightleftharpoons Na$	−2.71
$Mg^{2+} + 2e^- \rightleftharpoons Mg$	−2.37
$Al^{3+} + 3e^- \rightleftharpoons Al$	−1.66
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	−0.76
$Fe^{2+} + 2e^- \rightleftharpoons Fe$	−0.44
$Pb^{2+} + 2e^- \rightleftharpoons Pb$	−0.13
$2H^+ + 2e^- \rightleftharpoons H_2$	0.00
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	+0.34
$Ag^+ + e^- \rightleftharpoons Ag$	+0.80
$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	+1.36

The more negative the E°, the stronger the reducing agent (the metal readily gives up electrons); the more positive the E°, the stronger the oxidising agent (the ion readily gains electrons). The e.m.f. of a cell under standard conditions is $E^\circ_{cell} = E^\circ(\text{cathode}) - E^\circ(\text{anode})$, where the cathode is the half-cell with the more positive potential. A positive E°_{cell} signals a spontaneous reaction. For the Daniell cell, $E^\circ_{cell} = +0.34 - (-0.76) = +1.10$ V.

The series explains displacement reactions. A metal displaces from solution any metal below it in the series, because its half-cell has the more negative potential. Zinc displaces copper and silver; copper displaces silver but not zinc or iron; magnesium displaces hydrogen from dilute acid, whereas copper and silver do not, since their E° values lie above that of hydrogen.

WORKED EXAMPLE 8.7

A cell is made from a magnesium half-cell and a silver half-cell. Write the overall equation and calculate E°_{cell}.

Solution: Magnesium (−2.37 V) is more negative, so it is the anode and silver (+0.80 V) is the cathode. Anode: $Mg \rightarrow Mg^{2+} + 2e^-$. Cathode: $Ag^+ + e^- \rightarrow Ag$, which is multiplied by 2 so that electrons balance. Overall: $Mg + 2Ag^+ \rightarrow Mg^{2+} + 2Ag$. $E^\circ_{cell} = +0.80 - (-2.37) = +3.17$ V. Note that the potential is not multiplied by 2; electrode potential is an intensive property independent of the number of electrons.

Practical cells

The **dry (Leclanché) cell** used in torches and radios has a zinc case as anode, a carbon rod as cathode surrounded by a paste of manganese(IV) oxide and ammonium chloride, and gives about 1.5 V; it is not rechargeable. The **lead-acid accumulator** in motor vehicles has a lead anode and a lead(IV) oxide cathode in dilute sulphuric acid, gives about 2 V per cell (six cells in series for a 12 V battery), and is rechargeable: during discharge both electrodes become coated with lead(II) sulphate and the acid becomes more dilute, so its density falls, which allows the state of charge to be checked with a hydrometer. Charging reverses the reactions. The nickel-cadmium and lithium-ion cells in phones and laptops are further rechargeable examples, and **fuel cells** convert the energy of reacting hydrogen and oxygen directly into electricity, with water as the only product.

8.9 Corrosion and its prevention

Corrosion is the gradual destruction of a metal by chemical reaction with its environment. The rusting of iron is the best-known example and costs economies enormous sums every year. Rust is hydrated iron(III) oxide, $Fe_2O_3 \cdot xH_2O$. Iron rusts only when *both oxygen and water* are present; salt, acidic gases such as carbon

dioxide and sulphur dioxide, and contact with a less reactive metal speed the process. This is why vehicles and roofing sheets near the Lagos lagoon and the coast corrode faster than those in dry northern towns. Rusting is an electrochemical process: a drop of water on iron forms tiny cells in which iron acts as the anode, $Fe \rightarrow Fe^{2+} + 2e^{-}$, and oxygen is reduced at the cathode, $O_2 + 2H_2O + 4e^{-} \rightarrow 4OH^{-}$. The iron(II) ions are then oxidised further by air to rust.

Methods of prevention include the following.

- **Barrier methods** keep out air and water: painting, oiling and greasing, plastic coating, enamelling and plating with tin or chromium. Tin plating protects only while the coat is unbroken; once scratched, iron (more reactive) corrodes preferentially and rusts faster than unplated iron.
- **Galvanising** coats iron with zinc. Zinc is more reactive than iron, so even where the coat is scratched the zinc corrodes in preference and the iron is protected. This is **sacrificial protection**, and the common corrugated roofing sheets are galvanised.
- **Sacrificial anodes:** blocks of magnesium or zinc are bolted to ship hulls and buried against pipelines, and are replaced when consumed.
- **Alloying:** stainless steel contains chromium and nickel and resists rusting.
- **Cathodic protection** connects the iron object to the negative terminal of a power supply so that it cannot lose electrons.

Aluminium, though a reactive metal, resists corrosion because it quickly forms a thin, tough, self-healing layer of aluminium oxide that seals the surface. Copper slowly develops a green patina of basic copper carbonate on roofs and statues.

Exercise 8.1

1. State the oxidation number of (a) sulphur in $Na_2S_2O_3$ and (b) nitrogen in HNO_3.

2. In the reaction $Zn + 2HCl \rightarrow ZnCl_2 + H_2$, identify the oxidising agent and the reducing agent.
3. What mass of zinc is deposited from zinc sulphate solution by a current of 0.50 A flowing for 3860 s? (Zn = 65)
4. What volume of chlorine at s.t.p. is liberated at the anode when 0.20 mol of electrons passes through brine? (molar volume = 22.4 dm^3)
5. Calculate the mass of sodium produced when 4825 C of electricity passes through molten sodium chloride. (Na = 23)
6. Calculate the mass of aluminium deposited by 2.0 A flowing for 4825 s. (Al = 27)
7. Will iron displace copper from copper(II) sulphate solution? Support your answer with E° values and calculate $E°_{cell}$.
8. A 25.0 cm^3 sample of iron(II) solution is oxidised by 22.5 cm^3 of 0.0200 mol dm^{-3} $KMnO_4$ in acid. Calculate the iron(II) concentration.
9. How long must a current of 0.50 A flow to deposit 1.27 g of copper from copper(II) sulphate? (Cu = 63.5)
10. Calculate $E°_{cell}$ for the cell $Zn \mid Zn^{2+} \parallel Ag^+ \mid Ag$.
11. Name the products at each electrode when concentrated brine is electrolysed with graphite electrodes, and say what is left in solution.
12. Explain why galvanised iron resists rusting even when scratched, whereas tin-plated iron does not.

Answers to Exercise 8.1

1. (a) +2: $2(+1) + 2x + 3(-2) = 0$ gives $2x = 4$, $x = +2$. (b) +5: $+1 + x - 6 = 0$, $x = +5$.
2. Oxidising agent: HCl (H^+, reduced from +1 to 0). Reducing agent: Zn (oxidised from 0 to +2).

3. Q = 0.50 × 3860 = 1930 C = 0.0200 mol e^-; Zn = 0.0100 mol; mass = 0.0100 × 65 = 0.65 g.
4. $2Cl^- \rightarrow Cl_2 + 2e^-$: 0.10 mol Cl_2; volume = 0.10 × 22.4 = 2.24 dm^3.
5. 4825/96 500 = 0.050 mol e^-, so 0.050 mol Na; mass = 0.050 × 23 = 1.15 g.
6. Q = 2.0 × 4825 = 9650 C = 0.100 mol e^-; Al = 0.100/3 = 0.0333 mol; mass = 0.0333 × 27 = 0.90 g.
7. Yes. $E°(Fe^{2+}/Fe) = -0.44$ V is more negative than $E°(Cu^{2+}/Cu) = +0.34$ V; $E°_{cell} = 0.34 - (-0.44) = +0.78$ V, positive, so the reaction is spontaneous.
8. MnO_4^- = 0.0200 × 22.5/1000 = 4.50 × 10^{-4} mol; Fe^{2+} = 5 × 4.50 × 10^{-4} = 2.25 × 10^{-3} mol in 25.0 cm^3; concentration = 2.25 × 10^{-3} × 1000/25.0 = 0.0900 mol dm^{-3}.
9. Cu = 1.27/63.5 = 0.0200 mol; electrons = 0.0400 mol; Q = 0.0400 × 96 500 = 3860 C; t = 3860/0.50 = 7720 s (2 h 8 min 40 s).
10. $E°_{cell} = +0.80 - (-0.76) = +1.56$ V.
11. Anode: chlorine gas; cathode: hydrogen gas; the solution left contains Na^+ and OH^-, that is, sodium hydroxide solution.
12. Zinc is more reactive than iron, so it corrodes in preference and protects the iron (sacrificial protection). Tin is less reactive than iron, so once the coat is broken iron becomes the anode of a cell with tin and rusts faster.

Chapter summary

- Oxidation is loss of electrons (increase in oxidation number, gain of oxygen, loss of hydrogen); reduction is gain of electrons (decrease in oxidation number). Both occur together, and the oxidising agent is the species reduced.
- Oxidation numbers follow fixed rules; a reaction is redox only if some oxidation number changes. Disproportionation is simultaneous oxidation and reduction of one element.

- Redox equations are balanced with half-equations; $MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$ is the key titration equation.
- In electrolysis the anode is positive and oxidation occurs there, while reduction occurs at the negative cathode. In aqueous solutions, selective discharge depends on the electrochemical series, concentration and electrode material.
- Faraday's laws: mass liberated is proportional to charge; 1 F = 96 500 C is the charge on one mole of electrons; use Q = It, then moles of electrons, then moles of product.
- Electrolysis is used for plating, purifying copper, extracting Na, Mg, Ca and Al, and manufacturing NaOH, Cl_2 and H_2.
- In galvanic cells the anode is negative and the cathode positive; $E°_{cell}$ = E°(cathode) − E°(anode), and a positive value means a spontaneous reaction. Common cells include the Leclanché cell and the lead-acid accumulator.
- Rusting needs oxygen and water and is electrochemical; it is prevented by barrier coats, galvanising, sacrificial anodes, alloying and cathodic protection.

Chapter 9: Energy Changes, Rates of Reaction and Chemical Equilibrium

Every chemical change is accompanied by a change in energy, and every chemical change takes place at some speed. A cook in Kano who lights a charcoal fire is releasing stored chemical energy as heat; a farmer in Ibadan who notices that a bag of cement hardens faster in warm weather is observing the effect of temperature on rate; a manufacturer who makes ammonia for fertiliser must decide how to coax a reaction that never goes to completion into giving as much product as possible. These three ideas, energy, speed and the extent of reaction, are the subject of this chapter. They answer three questions that a chemist always asks: how much heat is given out or taken in, how fast does the reaction go, and how far does it go? The UTME tests all three, mostly through short calculations and through conceptual questions on the effect of changing conditions.

9.1 Exothermic and endothermic reactions

Chemical substances store energy in their bonds and in the arrangement of their particles. This stored energy is called the **enthalpy** (heat content) of the substance and is given the symbol H. We cannot measure the absolute enthalpy of a substance, but we can measure the change in enthalpy, ΔH, when a reaction occurs at constant pressure. By definition,

ΔH = H(products) − H(reactants)

A reaction in which heat is given out to the surroundings is **exothermic**. The products have less enthalpy than the reactants, so ΔH is negative. The temperature of the surroundings rises. Combustion, neutralisation, respiration, the reaction of sodium with water and the setting of cement are all exothermic. A reaction in which heat is absorbed from the surroundings is **endothermic**. The products have more

enthalpy than the reactants, so ΔH is positive, and the surroundings become cooler. Photosynthesis, the thermal decomposition of calcium carbonate, the dissolving of ammonium nitrate in water and the cracking of hydrocarbons are endothermic.

> **Key point.** Exothermic: ΔH negative, heat released, temperature of surroundings rises, products are more stable. Endothermic: ΔH positive, heat absorbed, temperature of surroundings falls. The sign of ΔH is always given from the point of view of the reacting system, not the surroundings.

Energy changes are conveniently shown on an **enthalpy level diagram**. For an exothermic reaction the products sit lower than the reactants and the difference is |ΔH|. For an endothermic reaction the products sit higher. Even an exothermic reaction usually needs a small initial supply of energy to start, for example the flame that lights a gas burner. This minimum energy that colliding particles must have before they can react is the **activation energy**, E_a. On the diagram it appears as a hump between reactants and products; the larger the hump, the slower the reaction at a given temperature. Activation energy is a kinetic barrier and has nothing to do with the sign of ΔH.

Chemically, the explanation lies in bonds. Breaking bonds always requires energy (endothermic) and forming bonds always releases energy (exothermic). The overall ΔH is the balance between the two: if more energy is released in forming the new bonds than was absorbed in breaking the old ones, the reaction is exothermic.

9.2 Standard enthalpy changes

Because ΔH depends on temperature, pressure and the physical states of the substances, chemists quote values under **standard conditions**: a pressure of 101.3 kPa (1 atmosphere), a stated temperature, normally 298 K (25 °C), and every

substance in its normal physical state at those conditions. A standard value carries the symbol ΔH° and the unit kJ mol^{-1}. Several named enthalpy changes appear regularly in examinations.

Term	Definition	Example
Enthalpy of combustion, ΔH°_c	Heat change when one mole of a substance burns completely in oxygen	$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$; $\Delta H^\circ = -890$ kJ mol^{-1}
Enthalpy of formation, ΔH°_f	Heat change when one mole of a compound is formed from its elements in their standard states	$C(s) + 2H_2(g) \rightarrow CH_4(g)$; $\Delta H^\circ = -76$ kJ mol^{-1}
Enthalpy of neutralisation, ΔH°_n	Heat change when an acid and an alkali react to form one mole of water	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$; $\Delta H^\circ \approx -57$ kJ mol^{-1}
Enthalpy of solution, ΔH°_s	Heat change when one mole of a substance dissolves in so much water that further dilution produces no further change	$NH_4NO_3(s) \rightarrow NH_4^+(aq) + NO_3^-(aq)$; ΔH° is positive
Enthalpy of atomisation	Heat change when one mole of gaseous atoms is formed from the element in its standard state	$\tfrac{1}{2}Cl_2(g) \rightarrow Cl(g)$

Two points deserve emphasis. First, the enthalpy of formation of any element in its standard state, such as $O_2(g)$, C(s) as graphite, or Na(s), is defined as zero. Second, the strong acid, strong base neutralisation value is almost constant at about −57 kJ mol^{-1} because the real change is always the same: H^+ and OH^- combine to form water. When a weak acid such as ethanoic acid is used, the value is smaller in magnitude because some energy is absorbed in ionising the acid.

9.3 Measuring heat: calorimetry

The heat transferred in a reaction in solution or in a burning fuel can be found by letting it warm or cool a known mass of water. The relation is

$$q = m \times c \times \Delta T$$

where q is the heat in joules, m the mass of the water (or solution) in grams, c the specific heat capacity (4.2 J g^{-1} K^{-1} for water) and ΔT the temperature change in kelvin or degrees Celsius (the size of the change is the same on both scales). One cm^3 of dilute aqueous solution is taken to have a mass of 1 g. The molar enthalpy change is then

ΔH = −q / n (exothermic, temperature rise) in kJ mol^{-1}, where n is the number of moles of the limiting reactant or of fuel burnt, and q is converted from joules to kilojoules by dividing by 1000.

WORKED EXAMPLE 9.1

50 cm^3 of 1.0 mol dm^{-3} hydrochloric acid is mixed with 50 cm^3 of 1.0 mol dm^{-3} sodium hydroxide in an insulated cup. The temperature rises by 6.8 °C. Calculate the enthalpy of neutralisation. (Specific heat capacity of the solution = 4.2 J g^{-1} K^{-1}; density = 1 g cm^{-3}.)

Solution: Mass of solution = 100 cm^3 × 1 g cm^{-3} = 100 g. q = 100 × 4.2 × 6.8 = 2856 J = 2.856 kJ. Moles of water formed = 0.050 dm^3 × 1.0 mol dm^{-3} = 0.050 mol. ΔH = −2.856 / 0.050 = −57.1 kJ mol^{-1}.

WORKED EXAMPLE 9.2

1.15 g of ethanol (C_2H_5OH) is burnt and the heat produced raises the temperature of 200 g of water by 40.0 °C. Calculate the enthalpy of combustion of ethanol. (C = 12, H = 1, O = 16.)

Solution: Molar mass of C_2H_5OH = 24 + 6 + 16 = 46 g mol^{-1}. Moles burnt = 1.15 / 46 = 0.025 mol. q = 200 × 4.2 × 40.0 = 33 600 J = 33.6 kJ. ΔH_c = −33.6 / 0.025 = −1344 kJ mol^{-1}. The experimental value is lower than the accepted figure (about −1367 kJ mol^{-1}) because some heat escapes to the air and combustion may be incomplete.

9.4 Hess's law

Many enthalpy changes cannot be measured directly. The formation of methane from carbon and hydrogen, for instance, does not take place cleanly in a laboratory. **Hess's law** solves the difficulty: *the total enthalpy change for a chemical reaction is the same whichever route is followed from reactants to products, provided the initial and final conditions are the same.* It is a consequence of the conservation of energy; if different routes gave different totals, energy could be created simply by going round a cycle.

Two practical forms of the law are very useful. Using enthalpies of formation:

$$\Delta H^\circ(\text{reaction}) = \Sigma \Delta H^\circ_f(\text{products}) - \Sigma \Delta H^\circ_f(\text{reactants})$$

Using enthalpies of combustion the order reverses:

$$\Delta H^\circ(\text{reaction}) = \Sigma \Delta H^\circ_c(\text{reactants}) - \Sigma \Delta H^\circ_c(\text{products})$$

Each term must be multiplied by the number of moles in the balanced equation. If an equation is reversed, the sign of ΔH is reversed; if it is multiplied by a factor, ΔH is multiplied by the same factor.

WORKED EXAMPLE 9.3

Calculate the enthalpy of formation of methane from these combustion data: $C(s) + O_2(g) \rightarrow CO_2(g)$, $\Delta H = -394$ kJ; $H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l)$, $\Delta H = -286$ kJ; $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$, $\Delta H = -890$ kJ.

Solution: Required: $C(s) + 2H_2(g) \rightarrow CH_4(g)$. Reactants burnt: 1 mol C and 2 mol H_2, so $\Sigma \Delta H_c(\text{reactants}) = (-394) + 2(-286) = -394 - 572 = -966$ kJ. Product burnt: 1 mol CH_4, $\Delta H_c = -890$ kJ. $\Delta H_f = -966 - (-890) = -966 + 890 = -76$ kJ mol^{-1}.

WORKED EXAMPLE 9.4

Given ΔH°_f values (kJ mol^{-1}): $Fe_2O_3(s)$ = −824; CO(g) = −111; $CO_2(g)$ = −394. Find ΔH for $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$.

Solution: Products: 2(0) + 3(−394) = −1182. Reactants: (−824) + 3(−111) = −824 − 333 = −1157. ΔH = −1182 − (−1157) = −1182 + 1157 = −25 kJ. The reaction is mildly exothermic.

9.5 Bond energies

The **bond energy** (bond enthalpy) is the energy required to break one mole of a particular covalent bond in gaseous molecules. Because it is an average taken over many compounds, it gives only an estimate of ΔH. The calculation is simple:

$\Delta H \approx \Sigma$(bond energies of bonds broken) − Σ(bond energies of bonds formed)

WORKED EXAMPLE 9.5

Estimate ΔH for $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ (all gases), using these bond energies in kJ mol^{-1}: C–H 413; O=O 498; C=O 805; O–H 464.

Solution: Bonds broken: 4 C–H = 4 × 413 = 1652; 2 O=O = 2 × 498 = 996; total = 2648 kJ. Bonds formed: 2 C=O = 2 × 805 = 1610; 4 O–H = 4 × 464 = 1856; total = 3466 kJ. ΔH = 2648 − 3466 = −818 kJ. The negative sign confirms that combustion is exothermic. The figure differs from −890 kJ because water is here a gas, not a liquid, and bond energies are averages.

9.6 Rate of reaction

The **rate of a reaction** is the change in concentration of a reactant or product per unit time. Its usual unit is mol dm^{-3} s^{-1}. In the laboratory, rate may be followed by measuring the volume of gas given off at intervals, the loss of mass of the reaction

flask, the time taken for a cross under the flask to disappear as a precipitate forms (as in the reaction of sodium thiosulphate with acid), or a colour change. On a graph of amount of product against time, the rate at any instant is the gradient of the tangent to the curve. The curve is steepest at the start, when reactant concentrations are highest, and flattens to a horizontal line when one reactant is used up.

WORKED EXAMPLE 9.6

In a reaction the concentration of reactant A falls from 0.50 mol dm^{-3} to 0.30 mol dm^{-3} in 40 seconds. Find the average rate of reaction.

Solution: Change in concentration = 0.50 − 0.30 = 0.20 mol dm^{-3}. Average rate = 0.20 / 40 = 5.0 × 10^{-3} mol dm^{-3} s^{-1}.

9.7 Collision theory and the factors affecting rate

The **collision theory** says that a reaction can occur only when particles collide, and then only if two conditions are met: the colliding particles must possess at least the activation energy, and they must be correctly oriented. Most collisions fail one or both tests and are unproductive. Anything that increases the number of effective collisions per second increases the rate.

- **Concentration.** At higher concentration there are more particles in a given volume, so collisions are more frequent. For gases, raising the pressure has the same effect. Doubling the concentration of a reactant often doubles the rate, though the exact relationship must be found by experiment.
- **Surface area.** In a reaction involving a solid, only the surface particles can collide with the other reactant. Powdered marble reacts with hydrochloric acid far faster than the same mass of marble chips. Fine flour dust and charcoal powder can burn explosively for the same reason.
- **Temperature.** Raising the temperature increases the average kinetic energy of the particles. Collisions are somewhat more frequent, but the dominant effect is

that a much larger fraction of particles have energy equal to or greater than E_a. As a rough rule, a rise of 10 °C approximately doubles the rate of many reactions. This is why food spoils faster in the heat of a Lagos afternoon and why refrigeration preserves it.

- **Catalyst.** A catalyst speeds up a reaction without being used up. It provides an alternative reaction pathway with a lower activation energy, so a larger proportion of collisions are effective. It does not change ΔH, does not alter the position of equilibrium, and is chemically unchanged at the end, though it may change physically. Examples are manganese(IV) oxide in the decomposition of hydrogen peroxide, iron in the Haber process, vanadium(V) oxide in the Contact process and platinum in the oxidation of ammonia. Biological catalysts are called enzymes.

> **Key point.** To explain any change in rate in an examination, always say the same thing: the number of effective collisions per unit time has increased. Then state why: more crowded particles (concentration, pressure), more exposed surface (area), more particles above E_a (temperature) or lower E_a (catalyst).

9.8 Reversible reactions and chemical equilibrium

Many reactions do not go to completion. In a **reversible reaction** the products can react to re-form the reactants, and the reaction is written with a double arrow, ⇌. Heating blue hydrated copper(II) sulphate drives off water to give the white anhydrous salt; adding water reverses the change. In a closed vessel, a reversible reaction eventually reaches a state of **dynamic equilibrium**, in which the rate of the forward reaction equals the rate of the reverse reaction. The word *dynamic* matters: both reactions continue, but there is no overall change in the concentrations of reactants and products, which remain constant. Equilibrium can be reached only in a closed system and can be approached from either side.

For the general reaction aA + bB ⇌ cC + dD at a fixed temperature, the **equilibrium constant** in terms of concentrations is

$K_c = [C]^c[D]^d / ([A]^a[B]^b)$

where square brackets denote equilibrium concentrations in mol dm^{-3}. Pure solids and pure liquids are left out of the expression because their concentrations are constant. A large K_c means the equilibrium lies to the right (mostly products); a small K_c means it lies to the left. K_c changes only with temperature. Its units are obtained by substituting the units of concentration into the expression; if the total number of moles is the same on both sides, K_c has no units.

WORKED EXAMPLE 9.7

For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, the equilibrium concentrations are $[N_2] = 2.0$, $[H_2] = 0.50$ and $[NH_3] = 1.0$ mol dm^{-3}. Calculate K_c with its units.

Solution: $K_c = [NH_3]^2 / ([N_2][H_2]^3) = (1.0)^2 / (2.0 \times 0.50^3) = 1.0 / (2.0 \times 0.125) = 1.0 / 0.25 = 4.0$. Units: $mol^2\ dm^{-6} / (mol\ dm^{-3} \times mol^3\ dm^{-9}) = mol^{-2}\ dm^6$. So $K_c = 4.0\ mol^{-2}\ dm^6$.

WORKED EXAMPLE 9.8

1.0 mol of hydrogen and 1.0 mol of iodine are placed in a 1.0 dm^3 vessel and heated until $H_2 + I_2 \rightleftharpoons 2HI$ is at equilibrium, when 1.6 mol of HI is present. Calculate K_c.

Solution: HI formed = 1.6 mol, which needs 0.8 mol H_2 and 0.8 mol I_2. At equilibrium: $[H_2] = 1.0 - 0.8 = 0.2$ and $[I_2] = 0.2$ mol dm^{-3}; $[HI] = 1.6$ mol dm^{-3}. $K_c = (1.6)^2 / (0.2 \times 0.2) = 2.56 / 0.04 = 64$. There are two moles on each side, so there are no units.

9.9 Le Chatelier's principle

Le Chatelier's principle states that if a system at equilibrium is disturbed by a change of concentration, pressure or temperature, the position of equilibrium shifts in the direction that tends to oppose the change. It lets us predict qualitatively how conditions affect the yield.

- **Concentration.** Adding a reactant, or removing a product, shifts the equilibrium to the right; adding a product shifts it to the left.
- **Pressure** (gaseous systems only). Raising the pressure favours the side with fewer gas molecules; lowering it favours the side with more. If the number of gas molecules is equal on both sides, pressure has no effect.
- **Temperature.** Raising the temperature favours the endothermic direction (it absorbs the added heat); lowering the temperature favours the exothermic direction. This is the only change that alters the value of K_c.
- **Catalyst.** A catalyst speeds up the forward and reverse reactions equally. It brings the system to equilibrium sooner but does not change the equilibrium composition or K_c.

The nitrogen dioxide system shows the ideas clearly: $2NO_2(g) \rightleftharpoons N_2O_4(g)$, ΔH negative for the forward reaction. NO_2 is brown and N_2O_4 is colourless. Compressing the mixture shifts the equilibrium to the right (2 moles of gas become 1), so the colour fades after an initial darkening caused by the higher concentration. Heating the mixture shifts it to the left, the endothermic direction, so the brown colour deepens.

9.10 Equilibrium in industry

The **Haber process** makes ammonia: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H = -92$ kJ mol^{-1}. The forward reaction is exothermic and reduces the number of gas molecules from 4 to 2. In theory, therefore, high pressure and low temperature maximise the yield. In practice a compromise is used: a pressure of about 200

atmospheres, a temperature of about 450 °C, and an iron catalyst. A lower temperature would give a better equilibrium yield but the rate would be too slow; the high pressure raises both yield and rate but is limited by the cost and safety of the plant. Unreacted gases are recycled and the ammonia is liquefied and removed so that the equilibrium keeps moving to the right. Most of the ammonia is converted to fertilisers such as ammonium sulphate and urea, on which Nigerian agriculture depends.

The **Contact process** makes sulphuric acid, and its key step is $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $\Delta H = -197$ kJ mol^{-1}. It uses vanadium(V) oxide at about 450 °C and a pressure close to atmospheric, because the yield is already high (over 95 per cent) and high pressure would add cost without much benefit. The SO_3 is absorbed in concentrated sulphuric acid to form oleum, which is diluted to give the acid; it is not added directly to water because the reaction is violently exothermic and produces a corrosive mist.

Exercise 9.1

1. State whether each is exothermic or endothermic: (a) burning of kerosene, (b) dissolving ammonium chloride in water, (c) decomposition of limestone, (d) neutralisation of hydrochloric acid by sodium hydroxide.
2. Calculate q when 250 g of water is warmed from 24.0 °C to 31.0 °C. (c = 4.2 J g^{-1} K^{-1})
3. When 0.10 mol of a solid dissolves in 100 g of water, the temperature falls by 5.0 °C. Calculate the enthalpy of solution in kJ mol^{-1}. (c = 4.2 J g^{-1} K^{-1})
4. Given $\Delta H°_f$ (kJ mol^{-1}): $CO_2(g) = -394$, $H_2O(l) = -286$, $C_2H_5OH(l) = -277$. Calculate $\Delta H°_c$ of ethanol.
5. The enthalpy of combustion of carbon is −394 kJ mol^{-1}. How much heat is released when 6.0 g of carbon burns completely? (C = 12)

6. Use bond energies (H–H 436, Cl–Cl 242, H–Cl 431 kJ mol^{-1}) to estimate ΔH for $H_2 + Cl_2 \rightarrow 2HCl$.
7. A reactant concentration falls from 0.80 to 0.20 mol dm^{-3} in 2.0 minutes. Calculate the average rate in mol dm^{-3} s^{-1}.
8. Give two reasons why powdered zinc reacts faster with dilute acid than zinc granules of the same mass.
9. Write the expression for K_c for $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$.
10. For $A + B \rightleftharpoons C + D$, the equilibrium amounts in a 1.0 dm^3 vessel are A 0.20, B 0.30, C 0.60 and D 0.50 mol. Calculate K_c.
11. For $2NO_2 \rightleftharpoons N_2O_4$ (exothermic forward), state the effect of (a) raising the pressure, (b) raising the temperature, (c) adding a catalyst.

Answers to Exercise 9.1

1. (a) exothermic; (b) endothermic (the solution cools); (c) endothermic (heat is needed to drive off CO_2); (d) exothermic.
2. $\Delta T = 7.0\ °C$; $q = 250 \times 4.2 \times 7.0 = 7350\ J = 7.35\ kJ$.
3. $q = 100 \times 4.2 \times 5.0 = 2100\ J = 2.1\ kJ$ absorbed; $\Delta H = +2.1 / 0.10 = +21$ kJ mol^{-1}.
4. $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$. Products: $2(-394) + 3(-286) = -788 - 858 = -1646$. Reactants: $-277 + 0 = -277$. $\Delta H = -1646 - (-277) = -1369$ kJ mol^{-1}.
5. $6.0 / 12 = 0.50$ mol; heat $= 0.50 \times 394 = 197$ kJ.
6. Broken: $436 + 242 = 678$; formed: $2 \times 431 = 862$; $\Delta H = 678 - 862 = -184$ kJ.
7. Change $= 0.60$ mol dm^{-3}; time $= 120$ s; rate $= 0.60 / 120 = 5.0 \times 10^{-3}$ mol dm^{-3} s^{-1}.
8. Powder has a much larger exposed surface, so more acid particles collide with zinc atoms per second; hence more effective collisions per unit time and a higher rate.
9. $K_c = [CO_2]$, because the solids are omitted.

10. $K_c = (0.60 \times 0.50) / (0.20 \times 0.30) = 0.30 / 0.06 = 5.0$ (no units).
11. (a) shifts right to the side with fewer gas molecules (N_2O_4); (b) shifts left (endothermic direction), K_c decreases; (c) no change in position, equilibrium is reached faster.

Chapter summary

- ΔH = H(products) − H(reactants). Exothermic reactions have negative ΔH; endothermic reactions have positive ΔH. Activation energy is the barrier to reaction and is separate from ΔH.
- Standard enthalpy changes (combustion, formation, neutralisation, solution) are quoted in kJ mol^{-1}; ΔH_f of an element is zero; strong acid with strong base gives about −57 kJ mol^{-1}.
- Calorimetry: $q = mc\Delta T$, then $\Delta H = -q/n$ with q in kJ and n in moles.
- Hess's law: ΔH depends only on initial and final states. Use $\Sigma\Delta H_f$(products) − $\Sigma\Delta H_f$(reactants), or $\Sigma\Delta H_c$(reactants) − $\Sigma\Delta H_c$(products); bond energies give $\Delta H \approx \Sigma$(broken) − Σ(formed).
- Rate = change in concentration / time. Rate increases with concentration (pressure), surface area, temperature, catalyst and light, always because effective collisions become more frequent.
- At dynamic equilibrium forward and reverse rates are equal and concentrations are constant. K_c = [products]/[reactants] with powers equal to coefficients, solids omitted; only temperature changes K_c.
- Le Chatelier: the system shifts to oppose any change. Pressure favours fewer gas molecules, heat favours the endothermic direction, and a catalyst only speeds attainment of equilibrium.
- Industrial compromises: Haber process (about 200 atm, 450 °C, iron) and Contact process (V_2O_5, 450 °C, about 1 atm) balance yield, rate and cost.

Chapter 10: Organic Chemistry

Organic chemistry is the chemistry of the compounds of carbon. The name is historical: early chemists believed such substances could be made only by living things. That belief fell away in 1828, when urea was prepared from inorganic materials, but the name stayed, and today more than ten million organic compounds are known, from the methane in cooking gas to the proteins in the beans on a Nigerian plate. The variety is possible because of the special behaviour of carbon. Its atoms form four covalent bonds, they join to one another in long chains, branched chains and rings, and they form single, double and triple bonds. Add the capacity to combine with hydrogen, oxygen, nitrogen and the halogens, and an almost limitless family of compounds results. This chapter organises that family into homologous series, shows how to name and draw its members, and describes the reactions the UTME syllabus requires.

10.1 Sources, classification and homologous series

The main source of organic compounds is **petroleum** (crude oil), together with natural gas, coal and living matter. Nigeria's crude oil from the Niger Delta is separated by **fractional distillation** into fractions with different boiling ranges: refinery gas, petrol (gasoline), naphtha, kerosene, diesel (gas oil), lubricating oils and finally bitumen. Each fraction is a mixture of hydrocarbons of similar molecular size; the larger the molecules, the stronger the van der Waals forces, the higher the boiling point and the more viscous the fraction. The heavier fractions are in surplus, so they are **cracked**, broken into smaller, more useful molecules (alkanes and alkenes) by heat with a catalyst. For example, $C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$.

A **hydrocarbon** contains carbon and hydrogen only. Hydrocarbons are **saturated** if they contain only single bonds and **unsaturated** if they contain at least one double or triple bond. Organic compounds are also grouped as **aliphatic** (open chains, or non-aromatic rings) and **aromatic** (containing the benzene ring).

A **homologous series** is a family of organic compounds with the same general formula and the same functional group, in which successive members differ by a CH_2 group. A **functional group** is the atom or group of atoms that determines the characteristic reactions of the family. Members of a series show a gradual change in physical properties, such as a rise in boiling point as the chain lengthens, and have similar chemical properties and similar methods of preparation.

Series	General formula	Functional group	Example
Alkanes	C_nH_{2n+2}	C–C (single bonds)	Ethane, C_2H_6
Alkenes	C_nH_{2n}	C=C	Ethene, C_2H_4
Alkynes	C_nH_{2n-2}	C≡C	Ethyne, C_2H_2
Alkanols (alcohols)	$C_nH_{2n+1}OH$	–OH	Ethanol, C_2H_5OH
Alkanals (aldehydes)	$C_nH_{2n+1}CHO$	–CHO	Ethanal, CH_3CHO
Alkanones (ketones)	$C_nH_{2n+1}COC_mH_{2m+1}$	C=O between two carbons	Propanone, CH_3COCH_3
Alkanoic acids	$C_nH_{2n+1}COOH$	–COOH	Ethanoic acid, CH_3COOH
Alkyl alkanoates (esters)	$RCOOR'$	–COO–	Ethyl ethanoate, $CH_3COOC_2H_5$
Haloalkanes	$C_nH_{2n+1}X$	–Cl, –Br, –I	Chloromethane, CH_3Cl

10.2 Naming organic compounds (IUPAC rules)

The names of the first four alkanes are historical: **meth**ane (1 carbon), **eth**ane (2), **prop**ane (3) and **but**ane (4). After that the stems come from Greek numbers: pent (5), hex (6), hept (7), oct (8), non (9) and dec (10). To name a compound by the IUPAC system:

1. Find the longest continuous carbon chain that contains the functional group; this gives the stem.
2. Add the ending for the series: -ane, -ene, -yne, -ol, -al, -one, -oic acid.
3. Number the chain from the end that gives the lowest number to the functional group (or, for alkanes, to the first branch).
4. Name each side chain as an alkyl group (methyl, ethyl, propyl) with its position number; use di-, tri- for repeats and list the groups alphabetically.

Examples: $CH_3CH(CH_3)CH_3$ is 2-methylpropane; $CH_3CH{=}CHCH_3$ is but-2-ene; $CH_3CH_2CH(OH)CH_3$ is butan-2-ol; $CH_3CH(CH_3)CH_2CH_2CH_3$ is 2-methylpentane. Note that "propan-1-ol" means the –OH is on carbon 1, and that the final "e" of the stem is dropped before a vowel (ethanol) but kept before a consonant (propane-1,2,3-triol).

10.3 Isomerism

Isomers are compounds with the same molecular formula but different structural formulae. *Chain isomerism* arises from different arrangements of the carbon skeleton: butane ($CH_3CH_2CH_2CH_3$) and 2-methylpropane. Pentane, C_5H_{12}, has three isomers: pentane, 2-methylbutane and 2,2-dimethylpropane. *Position isomerism* occurs when the functional group occupies different positions: propan-1-ol and propan-2-ol, or but-1-ene and but-2-ene. *Functional group isomerism* occurs when the same atoms form different functional groups: ethanol (C_2H_5OH) and methoxymethane (CH_3OCH_3); propanal and propanone; ethanoic acid and methyl methanoate. A further kind, *geometric (cis-trans) isomerism*, arises in alkenes

because there is no rotation about a C=C bond: but-2-ene exists as a cis form, with the two methyl groups on the same side of the double bond, and a trans form with them on opposite sides. This requires two different groups on each doubly bonded carbon.

> **Key point.** The number of isomers grows quickly: C_4H_{10} has 2, C_5H_{12} has 3, C_6H_{14} has 5. C_4H_8 alkenes: but-1-ene, cis-but-2-ene, trans-but-2-ene and 2-methylpropene.

10.4 Alkanes

Alkanes are saturated hydrocarbons with the general formula C_nH_{2n+2}. Methane to butane are gases, pentane to about C_{16} are liquids, and higher members are waxy solids. They are non-polar, insoluble in water and burn readily. Natural gas is mostly methane, and "cooking gas" (LPG) is propane and butane.

Reactions. Alkanes are rather unreactive, because C–C and C–H bonds are strong and non-polar. (1) *Combustion*: in plenty of air alkanes give carbon dioxide and water, e.g. $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$. In limited air, poisonous carbon monoxide or soot (carbon) is formed. (2) *Substitution*: in ultraviolet light, chlorine replaces hydrogen atoms in a chain reaction: $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$, and further substitution gives CH_2Cl_2, $CHCl_3$ (trichloromethane, chloroform) and CCl_4 (tetrachloromethane). Substitution, not addition, is the characteristic reaction of alkanes. Bromine water is not decolourised by alkanes in the dark, which distinguishes them from alkenes.

Preparation of methane in the laboratory: heating anhydrous sodium ethanoate with soda lime, $CH_3COONa + NaOH \rightarrow CH_4 + Na_2CO_3$. It is collected over water because it is insoluble. Methane is also produced by the decay of vegetation in swamps and from biogas digesters.

10.5 Alkenes

Alkenes, C_nH_{2n}, contain a C=C double bond made of one sigma and one pi bond. The pi bond is weaker and electron-rich, so alkenes are far more reactive than alkanes. Ethene is prepared in the laboratory by dehydrating ethanol with excess concentrated sulphuric acid at about 170 °C (or by passing ethanol vapour over heated aluminium oxide): $C_2H_5OH \rightarrow C_2H_4 + H_2O$. Industrially it is obtained by cracking.

The characteristic reaction of alkenes is **addition**, in which the double bond opens and a small molecule joins on:

- *Hydrogenation* (nickel, about 150 °C): $C_2H_4 + H_2 \rightarrow C_2H_6$. Used to harden vegetable oils into margarine.
- *Halogenation*: $C_2H_4 + Br_2 \rightarrow CH_2BrCH_2Br$ (1,2-dibromoethane). Orange-brown bromine water is decolourised rapidly; this is the standard test for unsaturation.
- *Hydrohalogenation*: $C_2H_4 + HBr \rightarrow C_2H_5Br$.
- *Hydration* (steam, phosphoric acid catalyst, high pressure): $C_2H_4 + H_2O \rightarrow C_2H_5OH$, an industrial route to ethanol.
- *Oxidation*: cold, dilute, acidified potassium manganate(VII) is decolourised (purple to colourless) as the alkene is converted to a diol. This is a second test for unsaturation.

With an unsymmetrical alkene such as propene, HBr adds so that the hydrogen goes to the carbon that already has more hydrogens (Markovnikov's rule), giving mainly 2-bromopropane. Alkenes also undergo **polymerisation**, treated in section 10.10.

10.6 Alkynes

Alkynes, C_nH_{2n-2}, contain a C≡C triple bond. Ethyne (acetylene) is made by adding water to calcium carbide: $CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$. It burns with a very hot flame (oxy-acetylene welding) and, in air, a smoky flame because of its high carbon content. Like alkenes, alkynes undergo addition and decolourise bromine water, but they can add two molecules of reagent: $C_2H_2 + 2H_2 \rightarrow C_2H_6$. Terminal alkynes also form an explosive reddish precipitate with ammoniacal copper(I) chloride, a test that distinguishes ethyne from ethene.

10.7 Alkanols

Alkanols contain the hydroxyl group. Their general formula is $C_nH_{2n+1}OH$. Hydrogen bonding between –OH groups gives them much higher boiling points than alkanes of similar mass, and the lower members dissolve in water. Alkanols are classified as **primary** (the carbon bearing –OH is joined to one other carbon, e.g. ethanol), **secondary** (two carbons, e.g. propan-2-ol) or **tertiary** (three carbons, e.g. 2-methylpropan-2-ol).

Ethanol is made industrially by hydration of ethene and, traditionally, by **fermentation** of sugar: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$, catalysed by zymase, an enzyme in yeast, at 25–35 °C. Palm wine, burukutu and pito are made by fermentation, and the product is concentrated by fractional distillation because ethanol (78 °C) boils lower than water. The reactions of ethanol are:

- *Combustion*: $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$.
- *Reaction with sodium*: $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$, giving sodium ethoxide and hydrogen.
- *Oxidation* with acidified potassium dichromate(VI) or potassium manganate(VII): a primary alkanol gives an alkanal and then an alkanoic acid ($CH_3CH_2OH \rightarrow CH_3CHO \rightarrow CH_3COOH$); a secondary alkanol gives an

alkanone; a tertiary alkanol resists oxidation. The orange dichromate turns green. This is the principle of the breathalyser test.

- *Esterification* with an alkanoic acid (see below).
- *Dehydration* to ethene (concentrated H_2SO_4, 170 °C), or to ethoxyethane at 140 °C with excess alcohol.

Ethanol is a solvent, a fuel (as a blend in petrol), an antiseptic and the intoxicant in alcoholic drinks. Methanol is poisonous, causing blindness and death.

10.8 Alkanoic acids and esters

Alkanoic (carboxylic) acids contain the –COOH group and are weak acids; they turn litmus red, liberate carbon dioxide from carbonates, and react with metals and bases to form salts: $2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$. Ethanoic acid is the acid in vinegar. It is prepared by oxidising ethanol (as palm wine turns sour on standing). Pure ethanoic acid freezes at 17 °C and is called glacial ethanoic acid. Methanoic acid is the irritant in ant and bee stings.

Esterification is the reversible reaction of an acid with an alcohol, catalysed by concentrated sulphuric acid, forming an ester and water: $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$. Esters are sweet-smelling liquids responsible for the scent of fruits and are used in perfumes and flavourings. Reaction of an ester with aqueous alkali is **hydrolysis** (saponification) and gives an alcohol and the salt of the acid. The reaction that produces soap is the same: a fat or oil (an ester of propane-1,2,3-triol, glycerol, with long-chain fatty acids) is boiled with sodium hydroxide to give glycerol and sodium salts of the fatty acids, which are soap. The cleaning action of soap depends on its structure: a long non-polar hydrocarbon tail dissolves in grease, and a polar carboxylate head dissolves in water, so grease is lifted into the water as droplets. In hard water soap forms insoluble scum with Ca^{2+} and Mg^{2+}, whereas detergents (for example sodium alkylbenzenesulphonates) do not, but non-biodegradable ones pollute rivers.

10.9 Alkanals and alkanones

Both contain the carbonyl group, C=O. In an alkanal it is at the end of the chain (–CHO); in an alkanone it is inside the chain. Alkanals are readily oxidised, alkanones are not. This difference gives the standard tests: with **Fehling's solution** (blue) on warming an alkanal gives a red-brown precipitate of copper(I) oxide; with **Tollens' reagent** it gives a silver mirror. Alkanones give no reaction. Methanal in solution (formalin) is used to preserve biological specimens.

10.10 Benzene, polymers and natural products

Benzene, C_6H_6, is a planar ring of six carbon atoms with delocalised electrons shared equally, so all the C–C bonds have the same length, between a single and a double bond. This stability means it behaves unlike an alkene: it does not decolourise bromine water and resists addition. Instead it undergoes **substitution**: with concentrated nitric and sulphuric acids at about 55 °C it forms nitrobenzene, $C_6H_5NO_2$; with bromine and an iron(III) bromide catalyst it forms bromobenzene. It will add hydrogen over a nickel catalyst only under forcing conditions to form cyclohexane. It burns with a smoky, sooty flame.

Polymers are giant molecules built from many small **monomers**. In *addition polymerisation*, alkene monomers join with no other product: n $CH_2{=}CH_2$ → $-(CH_2CH_2)_n-$, poly(ethene), used for sachets ("pure water" bags) and buckets; chloroethene gives PVC for pipes; propene gives polypropene. In *condensation polymerisation* monomers join with the loss of a small molecule such as water. Nylon, a polyamide, is made from a diamine and a dicarboxylic acid, and terylene (polyester) from a diol and a dicarboxylic acid. Plastics that are not biodegradable create waste-disposal problems in cities such as Lagos, where they block drains.

Carbohydrates are compounds of carbon, hydrogen and oxygen with the general formula $C_x(H_2O)_y$. Monosaccharides (glucose, fructose, $C_6H_{12}O_6$), disaccharides (sucrose, maltose, $C_{12}H_{22}O_{11}$) and polysaccharides (starch, cellulose, glycogen) are

related by hydrolysis: sucrose + H_2O → glucose + fructose. Starch gives a blue-black colour with iodine; glucose reduces Fehling's solution. **Proteins** are polymers of amino acids joined by peptide bonds (–CONH–); they are hydrolysed to amino acids and detected by the biuret (violet) test. **Fats and oils** are triesters of glycerol; fats are saturated and solid, oils unsaturated and liquid.

10.11 Calculations in organic chemistry

WORKED EXAMPLE 10.1

A hydrocarbon contains 85.7% carbon and 14.3% hydrogen by mass and has a relative molecular mass of 56. Find its molecular formula. (C = 12, H = 1.)

Solution: Moles C = 85.7 / 12 = 7.14; moles H = 14.3 / 1 = 14.3. Dividing by the smaller (7.14): C = 1, H = 2.0. Empirical formula = CH_2, mass 14. n = 56 / 14 = 4. Molecular formula = C_4H_8, a butene.

WORKED EXAMPLE 10.2

10 cm³ of a gaseous hydrocarbon is burnt in excess oxygen and produces 40 cm³ of carbon dioxide and 50 cm³ of steam, all volumes at the same temperature and pressure. Find its formula.

Solution: $C_xH_y + (x + y/4)O_2 \rightarrow xCO_2 + (y/2)H_2O$. By Avogadro's law, volumes are in mole ratio. 1 volume gives 40/10 = 4 volumes of CO_2, so x = 4; it gives 50/10 = 5 volumes of steam, so y/2 = 5 and y = 10. Formula = C_4H_{10} (butane). Check: oxygen needed = 4 + 10/4 = 6.5 volumes, and the carbon and hydrogen balance.

WORKED EXAMPLE 10.3

90 g of glucose ($C_6H_{12}O_6$) undergoes fermentation. Calculate the mass of ethanol obtained if the yield is 80%. (C = 12, H = 1, O = 16.)

Solution: M(glucose) = 72 + 12 + 96 = 180 g mol^{-1}; moles = 90 / 180 = 0.50 mol. One mole of glucose gives two of ethanol, so the theoretical amount is 1.0 mol. M(C_2H_5OH) = 46; theoretical mass = 46 g. Actual mass = 46 × 80/100 = 36.8 g.

WORKED EXAMPLE 10.4

A polythene sample has a relative molecular mass of 28 000. How many ethene units does each molecule contain? What is the maximum mass of ethyl ethanoate (M = 88) from 6.0 g of ethanoic acid (M = 60) and excess ethanol?

Solution: The ethene unit has mass 28, so n = 28 000 / 28 = 1000. For the ester: moles of acid = 6.0 / 60 = 0.10 mol; the equation is 1 : 1, so maximum ester = 0.10 mol × 88 = 8.8 g.

WORKED EXAMPLE 10.5

A compound has the composition 54.5% carbon, 9.1% hydrogen and 36.4% oxygen by mass. Its relative molecular mass is 88. Find the molecular formula and suggest the functional group it could contain. (C = 12, H = 1, O = 16.)

Solution: Moles: C = 54.5 / 12 = 4.54; H = 9.1 / 1 = 9.1; O = 36.4 / 16 = 2.275. Dividing by 2.275: C = 2.0, H = 4.0, O = 1.0. Empirical formula = C_2H_4O, mass = 24 + 4 + 16 = 44. n = 88 / 44 = 2, so the molecular formula is $C_4H_8O_2$. This fits a carboxylic acid (butanoic acid) or an ester (ethyl ethanoate), which are functional group isomers; a test with sodium carbonate, which liberates carbon dioxide only from the acid, would decide between them.

Exercise 10.1

1. Name (a) $CH_3CH_2CH_2CHO$, (b) $CH_3CH(CH_3)CH_2CH_3$, (c) $CH_3CH_2CH{=}CH_2$.
2. Draw or write the structural formulae of all the isomers of C_4H_{10} and name them.
3. Write the equation for the complete combustion of butane and find the volume of oxygen needed to burn 20 cm^3 of butane completely.
4. State one test that distinguishes ethene from ethane and give the observation.
5. Write the equation for the reaction of ethene with bromine and name the product.
6. Calculate the mass of ethanol, in grams, produced when 3.0 mol of glucose ferments completely. (C = 12, H = 1, O = 16)
7. Name the products when ethanol is oxidised first partially, then fully, by acidified dichromate(VI).

8. What is formed when ethanoic acid is warmed with propan-1-ol and concentrated sulphuric acid? Name the ester.
9. A hydrocarbon has an empirical formula CH and a relative molecular mass of 78. Give its molecular formula and name one hydrocarbon with that formula.
10. What volume of carbon dioxide is produced at s.t.p. when 4.4 g of propane (M = 44) burns completely? (Molar volume = 22.4 dm³.)
11. Explain why soap is less effective than a detergent in hard water.

Answers to Exercise 10.1

1. (a) butanal; (b) 2-methylbutane; (c) but-1-ene.
2. $CH_3CH_2CH_2CH_3$ is butane; $CH_3CH(CH_3)CH_3$ is 2-methylpropane. There are two isomers.
3. $2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$. Ratio butane : O_2 = 2 : 13, so 20 cm³ needs 20 × 13/2 = 130 cm³ of oxygen.
4. Shake with bromine water: ethene decolourises it (orange to colourless); ethane does not. Acidified $KMnO_4$ also works (purple to colourless).
5. $CH_2{=}CH_2 + Br_2 \rightarrow CH_2BrCH_2Br$, 1,2-dibromoethane.
6. 3.0 mol glucose gives 6.0 mol ethanol; mass = 6.0 × 46 = 276 g.
7. Ethanal (CH_3CHO) then ethanoic acid (CH_3COOH); the orange dichromate turns green.
8. Propyl ethanoate, $CH_3COOCH_2CH_2CH_3$, and water.
9. n = 78 / 13 = 6, so C_6H_6, benzene.
10. Moles propane = 4.4 / 44 = 0.10; $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ gives 0.30 mol CO_2; volume = 0.30 × 22.4 = 6.72 dm³.
11. Soap reacts with Ca^{2+} and Mg^{2+} to form insoluble scum, which wastes soap and prevents lather; detergent calcium salts are soluble.

Chapter summary

- Carbon forms four covalent bonds and chains, giving millions of compounds. Petroleum is separated by fractional distillation and heavy fractions are cracked.
- A homologous series has one general formula, one functional group and members differing by CH_2. Naming follows the longest chain, the series ending and the lowest position numbers.
- Isomers share a molecular formula: chain, position, functional group and cis-trans types.
- Alkanes (C_nH_{2n+2}) burn and undergo substitution; alkenes (C_nH_{2n}) and alkynes (C_nH_{2n-2}) undergo addition and decolourise bromine water.
- Ethanol is made by fermentation or hydration; it oxidises to ethanal then ethanoic acid, forms esters with acids and is dehydrated to ethene. Esters are hydrolysed by alkali, and this is saponification (soap-making).
- Alkanals reduce Fehling's and Tollens' reagents but alkanones do not.
- Benzene undergoes substitution because of delocalised electrons. Polymers form by addition or condensation; carbohydrates, proteins and fats are natural polymers or esters.
- For empirical formulae divide percentage by atomic mass; for gas combustion use volume ratios equal to mole ratios.

Chapter 11: Metals, Non-metals, Environmental Chemistry and Qualitative Analysis

The earlier chapters of this book developed the principles of chemistry. This chapter applies them to the elements themselves, to the materials a country such as Nigeria extracts, manufactures and consumes, to the effects of those activities on the air, water and soil, and to the laboratory methods by which an unknown substance is identified. The UTME draws many questions from this area, and they reward candidates who have learnt properties systematically and linked them to underlying ideas such as the reactivity series, redox and equilibrium. The chapter is organised in four parts: metals and their extraction, important non-metals and their compounds, environmental chemistry, and qualitative analysis.

11.1 General properties of metals and the reactivity series

Metals are good conductors of heat and electricity, are malleable and ductile, have lustre, and are usually dense with high melting points (the Group 1 metals are exceptions). They form positive ions by losing electrons, and their oxides are basic or amphoteric. These properties follow from metallic bonding, a lattice of positive ions held together by a sea of delocalised electrons.

The **reactivity series** arranges metals in order of their tendency to lose electrons: potassium, sodium, calcium, magnesium, aluminium, (carbon), zinc, iron, tin, lead, (hydrogen), copper, mercury, silver, gold. It explains most of metal chemistry. Potassium, sodium and calcium react with cold water giving hydroxides and hydrogen ($2Na + 2H_2O \rightarrow 2NaOH + H_2$); magnesium reacts with steam, giving the oxide and hydrogen; zinc and iron react with steam and are reversible; metals above

hydrogen displace hydrogen from dilute acids; and a more reactive metal displaces a less reactive one from solution, e.g. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$. The more reactive the metal, the more stable its compounds (the carbonates and nitrates of Group 1 are the most stable to heat) and the more difficult it is to extract from its ore.

11.2 Occurrence and extraction of metals

Few metals occur free; gold, silver and platinum do, because they are so unreactive. Most are found as oxides, sulphides or carbonates in ores. The method of extraction depends on the position in the series:

Metals	Method	Reason
K, Na, Ca, Mg, Al	Electrolysis of the molten compound	Too reactive to be reduced by carbon
Zn, Fe, Sn, Pb	Reduction of the oxide by carbon or carbon monoxide	Less reactive than carbon
Cu (from sulphide, then purified electrolytically); Hg, Ag	Roasting or reduction; displacement	Low reactivity

Iron. Iron is extracted from haematite, Fe_2O_3, in a blast furnace charged with ore, coke and limestone, while a blast of hot air enters near the bottom. Coke burns to give carbon dioxide and heat: $C + O_2 \rightarrow CO_2$. The carbon dioxide is reduced by hot coke to carbon monoxide: $CO_2 + C \rightarrow 2CO$. Carbon monoxide is the main reducing agent: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$. Limestone decomposes, $CaCO_3 \rightarrow CaO + CO_2$, and the calcium oxide removes acidic impurities such as silica as slag: $CaO + SiO_2 \rightarrow CaSiO_3$. Molten iron and slag collect at the bottom, the slag floating on the iron; the iron is run off as pig iron, about 4% carbon and brittle. Removing most of the carbon by blowing oxygen through it gives **steel**.

Aluminium. Aluminium is extracted from bauxite, $Al_2O_3{\cdot}2H_2O$. The ore is purified to give pure alumina, which has a melting point of over 2000 °C. It is dissolved in molten cryolite (Na_3AlF_6), which lowers the working temperature to about 900 °C, and electrolysed using carbon electrodes. At the cathode, $Al^{3+} + 3e^- \rightarrow Al$; at the anode, $2O^{2-} \rightarrow O_2 + 4e^-$, and the oxygen burns away the carbon anodes, which must be replaced regularly. The process uses huge amounts of electricity.

Sodium is obtained by electrolysis of molten sodium chloride in the Downs cell, with calcium chloride added to lower the melting point. **Zinc** is obtained by roasting the sulphide ore, zinc blende (ZnS), to the oxide, which is then reduced with carbon. **Copper** is usually obtained from copper pyrites; the crude metal is refined by electrolysis, with impure copper as anode and pure copper as cathode in copper(II) sulphate solution. **Tin**, important in the history of Jos Plateau, comes from cassiterite, SnO_2, reduced by carbon. **Lead** comes from galena, PbS, roasted to oxide and reduced by carbon.

WORKED EXAMPLE 11.1

Calculate the mass of iron obtainable from 320 kg of pure Fe_2O_3. (Fe = 56, O = 16.)

Solution: Molar mass of Fe_2O_3 = 112 + 48 = 160 g mol^{-1}. Moles = 320 000 / 160 = 2000 mol. Each mole gives 2 mol Fe, so 4000 mol Fe. Mass = 4000 × 56 = 224 000 g = 224 kg. Check: the oxide is 112/160 = 70% iron, and 0.70 × 320 = 224 kg.

WORKED EXAMPLE 11.2

Calculate the quantity of charge needed to deposit 5.4 g of aluminium at the cathode. (Al = 27; 1 F = 96 500 C.)

Solution: Moles Al = 5.4 / 27 = 0.20 mol. Each Al^{3+} needs 3 electrons, so 0.20 × 3 = 0.60 mol of electrons. Q = 0.60 × 96 500 = 57 900 C.

Alloys and uses

An **alloy** is a mixture of a metal with other elements, which is usually harder, stronger or more resistant to corrosion than the pure metal, because atoms of different sizes disrupt the regular layers so that they cannot slide easily. *Brass* (copper and zinc) is used for door fittings and musical instruments; *bronze* (copper and tin) for statues and bells, as the famous Benin bronzes show; *solder* (tin and lead) for joining wires; *stainless steel* (iron with chromium and nickel) for cutlery and sinks; *duralumin* (aluminium with copper and magnesium) in aircraft. Aluminium is used for cooking pots, cables and roofing sheets because it is light, a good conductor and protected by a thin coherent layer of oxide. Copper is used for electric wiring, iron for construction and zinc for galvanising.

Important compounds of metals

Sodium hydroxide is made by the electrolysis of brine and is used in soap-making and the paper industry. **Sodium carbonate** is made by the Solvay process and used in glass and as washing soda. **Calcium carbonate** (limestone, marble, chalk) decomposes on heating to quicklime, CaO, which reacts with water to give slaked lime, $Ca(OH)_2$, used to neutralise acidic soil and in making cement. Limewater turns milky with carbon dioxide because insoluble calcium carbonate forms, but clears with excess CO_2 as soluble calcium hydrogencarbonate forms. **Aluminium oxide and hydroxide** are amphoteric: they react with both acids and alkalis, $Al(OH)_3 + NaOH \rightarrow NaAl(OH)_4$.

WORKED EXAMPLE 11.3

250 g of an impure limestone sample is strongly heated and 88 g of carbon dioxide is evolved. Find the percentage of calcium carbonate in the sample. (Ca = 40, C = 12, O = 16.)

Solution: $CaCO_3 \rightarrow CaO + CO_2$. $M(CO_2) = 44$, so moles = 88 / 44 = 2.0 mol. Moles $CaCO_3$ = 2.0, with M = 100, so mass = 200 g. Percentage purity = 200 / 250 × 100 = 80%.

11.3 Non-metals and their compounds

Hydrogen. The lightest element is prepared in the laboratory by reacting zinc with dilute hydrochloric or sulphuric acid: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$. It is collected over water or by upward displacement of air (it is less dense than air), and burns with a pop; the product of burning is water. Industrially it comes from steam and natural gas and from the electrolysis of brine, and is used to make ammonia and margarine and as a rocket fuel.

Oxygen. The laboratory method is the catalytic decomposition of hydrogen peroxide with manganese(IV) oxide: $2H_2O_2 \rightarrow 2H_2O + O_2$. Industrially it is obtained by fractional distillation of liquid air. It relights a glowing splint. It forms oxides with most elements; acidic oxides (CO_2, SO_2, P_4O_{10}) are formed by non-metals, basic oxides (Na_2O, CaO) by metals, amphoteric oxides (Al_2O_3, ZnO) react with both acids and bases, and neutral oxides include water, CO and N_2O. Ozone, O_3, is an allotrope of oxygen.

Nitrogen and its compounds. Nitrogen forms about 78% of air and is unreactive because of the strong triple bond. It is used for fertilisers and as an inert atmosphere. *Ammonia* is prepared in the laboratory by heating an ammonium salt with a base: $2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2NH_3 + 2H_2O$; it is dried with quicklime (not with concentrated sulphuric acid or calcium chloride, which react

with it) and collected by downward delivery because it is lighter than air and extremely soluble in water. It is the only common alkaline gas: it turns damp red litmus blue and forms dense white fumes with hydrogen chloride ($NH_3 + HCl \rightarrow NH_4Cl$). *Nitric acid* is made industrially by the Ostwald process: ammonia is oxidised over a hot platinum catalyst to nitrogen monoxide, which is oxidised to nitrogen dioxide and dissolved in water. Nitric acid is a strong oxidising acid and its reactions with metals give oxides of nitrogen, not hydrogen. Nitrates decompose on heating: potassium and sodium nitrates give the nitrite and oxygen, most other nitrates give the metal oxide, brown NO_2 and oxygen, and silver nitrate gives metallic silver.

Sulphur and its compounds. Sulphur occurs as free sulphur and in sulphide ores. It has two crystalline allotropes, rhombic (stable below 96 °C) and monoclinic. Sulphur dioxide is prepared by burning sulphur or by the action of acid on a sulphite; it is a colourless, choking gas, soluble in water to give sulphurous acid, a reducing agent that turns acidified potassium dichromate(VI) from orange to green and decolourises potassium manganate(VII); it also bleaches. *Sulphuric acid* (Contact process) has three characteristic properties: as an acid it reacts with metals, bases and carbonates; as a dehydrating agent it removes water from sugar, leaving black carbon, and from copper(II) sulphate crystals; as an oxidising agent (hot, concentrated) it oxidises copper and carbon. Hydrogen sulphide, H_2S, smells of rotten eggs, is made by reacting iron(II) sulphide with dilute acid, is poisonous and blackens lead(II) ethanoate paper.

Chlorine and the halogens. Chlorine is prepared in the laboratory by oxidising concentrated hydrochloric acid with manganese(IV) oxide: $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$. It is a greenish-yellow poisonous gas, collected by upward delivery after being washed with water (to remove HCl) and dried with concentrated sulphuric acid. It bleaches damp litmus paper and is used to sterilise water. With sodium hydroxide it forms bleach (sodium chlorate(I)). Hydrogen chloride is made from sodium chloride and concentrated sulphuric acid, is dried by

concentrated sulphuric acid and dissolves in water to give hydrochloric acid. The halogens become less reactive down the group, so chlorine displaces bromine and iodine from their salts.

Carbon and its compounds. Carbon has crystalline allotropes: diamond, in which each atom bonds tetrahedrally to four others, giving the hardest natural substance; graphite, in which atoms form layers of hexagons that slide, giving a soft solid which conducts electricity because of delocalised electrons; and fullerenes. Amorphous forms include charcoal, coke and soot. Carbon dioxide is prepared by the action of dilute hydrochloric acid on marble or any carbonate: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$. It is denser than air, mildly acidic, and turns limewater milky; it is used in fire extinguishers and fizzy drinks, and as dry ice. Carbon monoxide, formed in incomplete combustion, is a colourless, odourless, poisonous gas that combines with haemoglobin more strongly than oxygen does. It is a neutral oxide and a good reducing agent.

WORKED EXAMPLE 11.4

Calculate the volume of ammonia produced at s.t.p. by heating 10.7 g of ammonium chloride with excess calcium hydroxide. (N = 14, H = 1, Cl = 35.5; molar volume = 22.4 dm³ at s.t.p.)

Solution: $M(NH_4Cl) = 14 + 4 + 35.5 = 53.5$. Moles = 10.7 / 53.5 = 0.20 mol. From the equation, 2 mol NH_4Cl gives 2 mol NH_3, so 0.20 mol NH_3. Volume = $0.20 \times 22.4 = 4.48$ dm³.

11.4 Environmental chemistry

Air pollution. Dry air is about 78% nitrogen, 21% oxygen, 0.9% argon and 0.04% carbon dioxide, with traces of other noble gases. The main pollutants are: *carbon monoxide* from vehicle exhausts, generators and bush burning; *sulphur dioxide* from burning fuels containing sulphur and from smelting; *oxides of nitrogen* from

engines and lightning; *hydrocarbons* and *particulates* (soot, dust); and *lead compounds* from leaded petrol. Gas flaring in the Niger Delta releases carbon dioxide, methane, soot and sulphur compounds, and affects health and farmland.

Acid rain forms when SO_2 and nitrogen oxides dissolve in rainwater to make sulphurous, sulphuric and nitric acids (pH below 5.6). It corrodes buildings and zinc roofing sheets, damages crops and leaches nutrients from soils, and kills aquatic life. **The greenhouse effect** arises because carbon dioxide, methane, nitrous oxide and water vapour absorb infrared radiation re-emitted by the Earth and keep it warm. An increase in these gases because of fossil fuel burning and deforestation causes global warming, melting ice, rising sea level and flooding of coastal cities such as Lagos. **Ozone layer depletion** is caused by chlorofluorocarbons (CFCs) from refrigerators and aerosols, which release chlorine atoms in the stratosphere that destroy ozone, allowing more harmful ultraviolet radiation to reach the ground. **Smog** is a mixture of smoke and fog formed in polluted urban air.

Water pollution. Sources include sewage, oil spills, industrial effluent containing heavy metals such as lead and mercury, agricultural run-off of fertilisers, and refuse. Excess nitrates and phosphates cause *eutrophication*: algae grow rapidly, then die and decay, and bacteria consume the dissolved oxygen so that fish die. Crude oil spills coat the surface, cut off oxygen and destroy fishing in the creeks. Water for drinking is purified by sedimentation, coagulation with alum (aluminium sulphate), filtration through sand and chlorination.

Solid waste and soil pollution. Plastics, scrap metal and electronic waste accumulate, and non-biodegradable materials last for centuries. The remedies are reduction of waste, recycling (e.g. of aluminium cans and glass), proper landfills, composting of organic waste and biogas generation. Pollution can be controlled by catalytic converters on vehicles (which convert CO and NO to CO_2 and N_2), the use of unleaded petrol, scrubbing sulphur dioxide from flue gases with lime, the use of renewable energy and the planting of trees.

11.5 Qualitative analysis

Qualitative analysis identifies the ions or gases present in a sample. Observations must be stated precisely: colour of flame, colour of precipitate, and whether the precipitate dissolves in excess.

Flame tests

A clean wire dipped in the sample and held in a non-luminous flame gives: sodium, persistent golden-yellow; potassium, lilac; calcium, brick-red; copper, blue-green; barium, apple-green; lithium, crimson.

Tests for cations

Ion	With NaOH(aq)	With NH_3(aq)
Ca^{2+}	White precipitate, insoluble in excess	No precipitate or very slight
Al^{3+}	White precipitate, soluble in excess	White precipitate, insoluble in excess
Zn^{2+}	White precipitate, soluble in excess	White precipitate, soluble in excess
Pb^{2+}	White precipitate, soluble in excess	White precipitate, insoluble in excess
Fe^{2+}	Dirty green precipitate, insoluble; turns brown in air	Dirty green precipitate, insoluble
Fe^{3+}	Reddish-brown precipitate, insoluble	Reddish-brown precipitate, insoluble
Cu^{2+}	Light blue precipitate, insoluble	Light blue precipitate, dissolving in excess to a deep blue solution
NH_4^+	On warming, gives ammonia, which turns damp red litmus blue	(not applicable)

Zn^{2+} and Al^{3+} are distinguished by excess ammonia, and Al^{3+} and Pb^{2+} by the effect of adding potassium iodide (yellow precipitate for lead) or dilute hydrochloric acid (white precipitate of $PbCl_2$, soluble on heating).

Tests for anions and gases

- **Carbonate (CO_3^{2-}):** dilute acid gives effervescence of CO_2, which turns limewater milky.
- **Chloride (Cl^-):** add dilute nitric acid, then silver nitrate: a white precipitate soluble in dilute ammonia. Bromide gives a cream precipitate and iodide a yellow one.
- **Sulphate (SO_4^{2-}):** add dilute hydrochloric acid, then barium chloride: a white precipitate insoluble in acid.
- **Sulphite (SO_3^{2-}):** dilute acid gives SO_2, which turns acidified dichromate(VI) from orange to green.
- **Nitrate (NO_3^-):** the brown ring test: add iron(II) sulphate, then carefully run concentrated sulphuric acid down the side of the tube; a brown ring forms at the junction.
- **Gases:** hydrogen burns with a pop; oxygen relights a glowing splint; carbon dioxide turns limewater milky; ammonia turns damp red litmus blue; chlorine bleaches damp litmus; hydrogen chloride gives white fumes with ammonia; hydrogen sulphide blackens lead ethanoate paper; sulphur dioxide turns acidified dichromate green.

Key point. Two tests must always be done in the correct order: acidify with nitric acid before testing for halides, and with hydrochloric acid before testing for sulphates. The acid removes carbonates and sulphites, which would also form precipitates and give a false positive.

Exercise 11.1

1. Name the ore from which each metal is mainly obtained: (a) aluminium, (b) iron, (c) tin.
2. Why is cryolite added to alumina before electrolysis?

3. Write the equation for the reduction of iron(III) oxide by carbon monoxide.
4. Calculate the percentage by mass of iron in Fe_3O_4. (Fe = 56, O = 16)
5. Calculate the mass of calcium oxide formed when 500 g of calcium carbonate is heated strongly. (Ca = 40, C = 12, O = 16)
6. Calculate the mass of aluminium deposited by 9650 C of charge. (Al = 27; 1 F = 96 500 C)
7. What volume of carbon dioxide at s.t.p. is liberated when 10.6 g of sodium carbonate reacts with excess dilute acid? (Na = 23, C = 12, O = 16; 22.4 $dm^3\ mol^{-1}$)
8. Calculate the volume of chlorine at s.t.p. obtained from 8.7 g of manganese(IV) oxide and excess concentrated hydrochloric acid. (Mn = 55, O = 16; $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$)
9. A solution gives a white precipitate with sodium hydroxide that dissolves in excess, and a white precipitate with ammonia that dissolves in excess. Name the cation.
10. State two ways in which burning fossil fuels harms the environment, and name the gas responsible for each.
11. Explain why ammonia cannot be dried with concentrated sulphuric acid.

Answers to Exercise 11.1

1. (a) bauxite; (b) haematite (or magnetite); (c) cassiterite.
2. Cryolite dissolves alumina and lowers the melting point from over 2000 °C to about 900 °C, saving energy; the melt also conducts electricity.
3. $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$.
4. $M(Fe_3O_4) = 168 + 64 = 232$; percentage $= 168 / 232 \times 100 = 72.4\%$.
5. Moles $CaCO_3 = 500 / 100 = 5.0$; $CaO = 5.0 \times 56 = 280$ g.

6. Moles of electrons = 9650 / 96 500 = 0.10; moles Al = 0.10 / 3 = 0.0333; mass = 0.0333 × 27 = 0.90 g.
7. Moles Na_2CO_3 = 10.6 / 106 = 0.10; CO_2 = 0.10 mol; volume = 2.24 dm^3.
8. Moles MnO_2 = 8.7 / 87 = 0.10; Cl_2 = 0.10 mol; volume = 2.24 dm^3.
9. Zn^{2+}, because only zinc hydroxide dissolves in excess of both reagents.
10. Carbon dioxide causes global warming; sulphur dioxide causes acid rain (carbon monoxide is poisonous).
11. Ammonia is a base and reacts with the acid to form ammonium sulphate; quicklime is used instead.

Chapter summary

- Metals are conductors, malleable and form basic or amphoteric oxides. The reactivity series governs reactions with water, acids and salts, and the method of extraction: electrolysis for Na, Ca, Mg, Al; carbon reduction for Zn, Fe, Sn, Pb.
- Iron is extracted in the blast furnace (coke, limestone, hot air, CO as reductant); aluminium by electrolysis of alumina in molten cryolite.
- Alloys are harder and more resistant than pure metals: brass, bronze, solder, steel, duralumin.
- Know the laboratory preparation, collection and tests of H_2, O_2, NH_3, Cl_2, CO_2, SO_2 and H_2S, and the industrial processes (Haber, Ostwald, Contact).
- Pollution arises from CO, SO_2, NO_x, CFCs, CO_2 and CH_4, oil spills, fertilisers and plastics. The main effects are acid rain, global warming, ozone depletion and eutrophication.

- Qualitative analysis uses flame tests, NaOH and NH_3 for cations, and specific reagents for anions and gases. Report colours and solubility in excess exactly, and acidify before testing for halides and sulphates.